NIALL CASSIDY

TESTING

A2 WJEC COMPUTER SCIENCE

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Section 8: Testing (10/100) marks

- **produced a comprehensive plan for testing all areas of the system**
- **made use of a wide range of appropriate testing methods**
- **made use of test data including typical, extreme and invalid data where appropriate**
- **presented all results with detailed and informed commentaries and included specific suggestions to refine the system**

8.0Test Plan (100 Tests)

Testing Methods: Functional, Boundary, Negative,
Test Data: Typical (T), Extreme (E), Invalid (I)

1. Login & Account Creation (10 Tests)

Form: LoginPage.py

TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	Success/Fail
1	Valid Admin Login	Username: ncass, Password: admin	Functional	T	Redirect to Manager Home	SUCCESS
2	Valid Sales Login	Username: sales, Password: 12	Functional	T	Redirect to Sales Home	SUCCESS
3	Invalid Password	Username: alice_admin, Password: wrong	Negative	I	Error: "Invalid username or password."	SUCCESS
4	Empty Username/Password	Username: "", Password: ""	Negative	I	Error: "All fields are required."	SUCCESS
5	Login with incorrect username but correct password	Username invalid Password: admin	Negative	I	Error: Login failed	SUCCESS
6	Create Account: Duplicate Username	StaffID=1, Username=alice_admin	Negative	I	Error: "Username already taken."	SUCCESS
7	Create Account: Invalid StaffID (Non-Integer)	StaffID=abc	Negative	I	Error: "StaffID must be a number."	SUCCESS
8	Create Account: Mismatched Passwords	Password=pass123, Confirm=pass456	Negative	I	Error: "Passwords do not match."	SUCCESS
9	Create Account: Valid Admin Account	StaffID=1, Username=new_admin, Password=pass123	Functional	T	Success message.	SUCCESS
10	Create Account: Extreme Password Length (100 chars)	Password=A" * 100	Boundary	E	Success .	SUCCESS

8.1 WHITEBOX TESTING

TEST 1



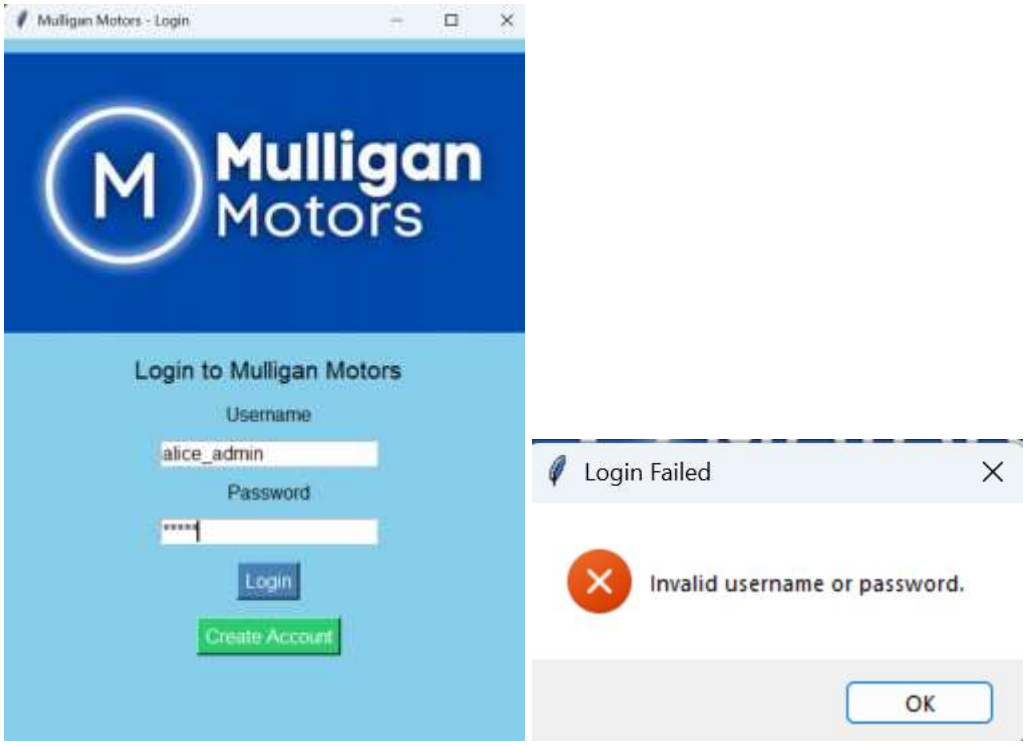


TEST 2


Mulligan Motors
Login to Mulligan Motors
Username
sales
Password
Login
Create Account

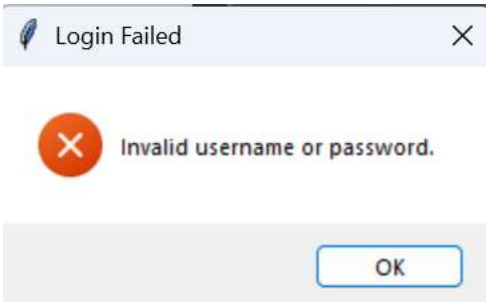

Mulligan Motors - Sales
Mulligan Motors
Mulligan Motors (SALES ACCESS)
Manage Vehicle Stock
Manage Customers
Manage Orders
Manage Test Drives
Generate Report
Backup Database
Help
Log Off

TEST 3


Mulligan Motors - Login
Mulligan
Motors
Login to Mulligan Motors
Username
alice_admin
Password
Login
Create Account
Login Failed
Invalid username or password.
OK

TEST 4


Mulligan Motors - Login
Mulligan
Motors
Login to Mulligan Motors
Username
Password
Login
Create Account

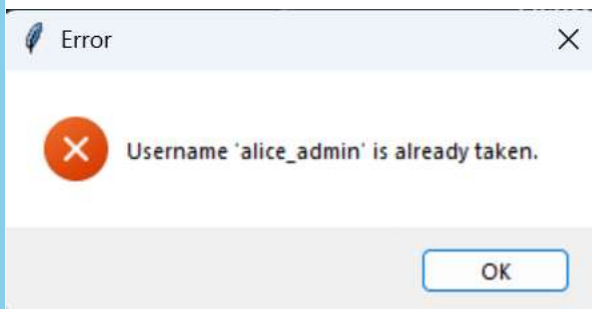

Login Failed
Invalid username or password.
OK

TEST 5

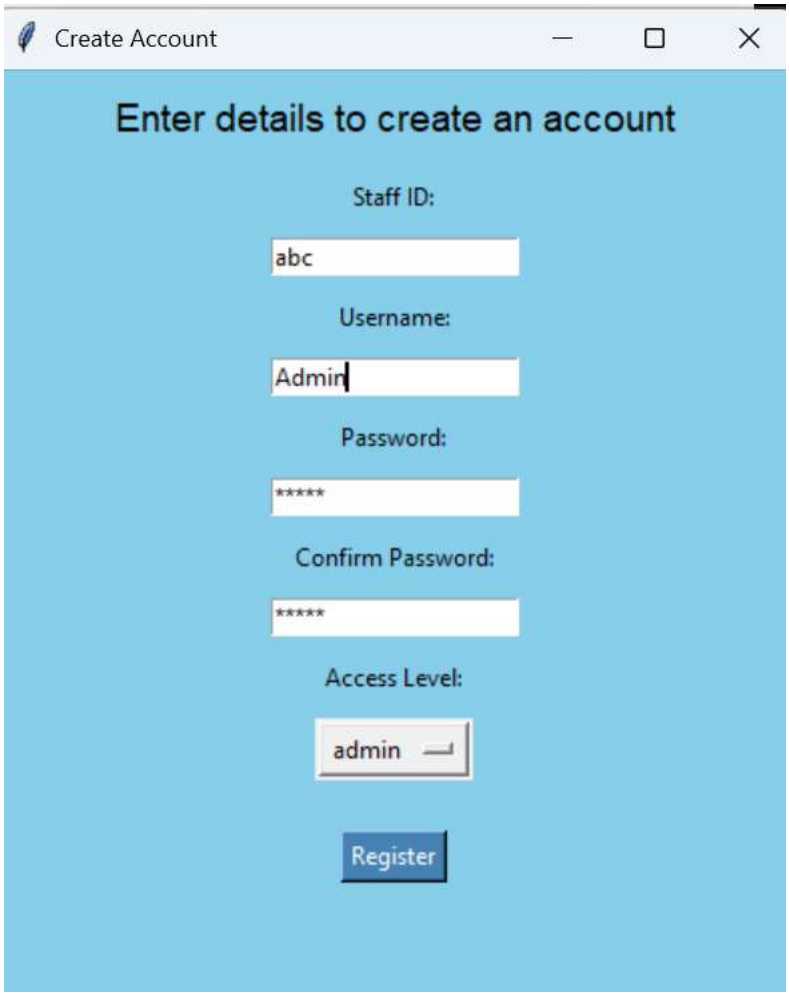

Mulligan Motors - Login
Mulligan
Motors
Login to Mulligan Motors
Username
invalid
Password
*****
Login
Create Account
Login Failed
Invalid username or password.
OK

TEST 6


Create Account
Enter details to create an account
Staff ID:
1
Username:
alice_admin
Password:
Confirm Password:
Access Level:
admin
Register

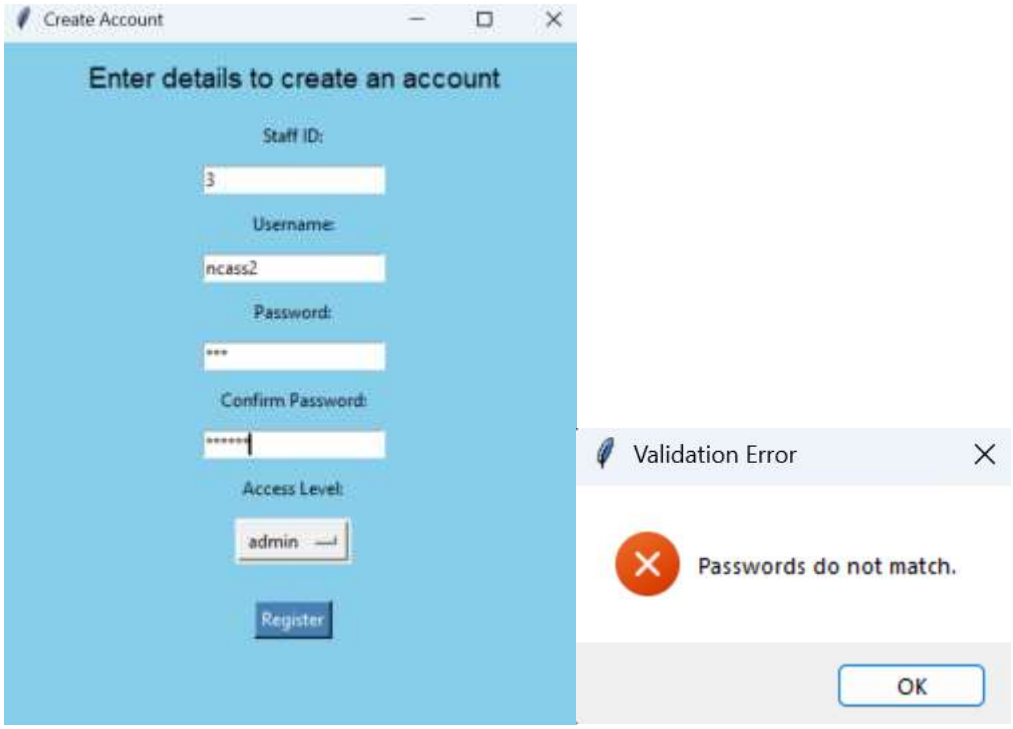

Error
Username 'alice_admin' is already taken.
OK

TEST 7


Create Account
Enter details to create an account
Staff ID:
abc
Username:
Admin
Password:
*****
Confirm Password:
*****
Access Level:
admin
Register

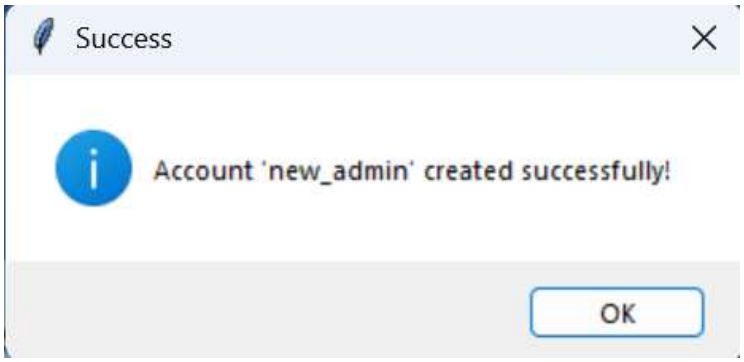

Validation Error
StaffID must be a number.
OK

TEST 8

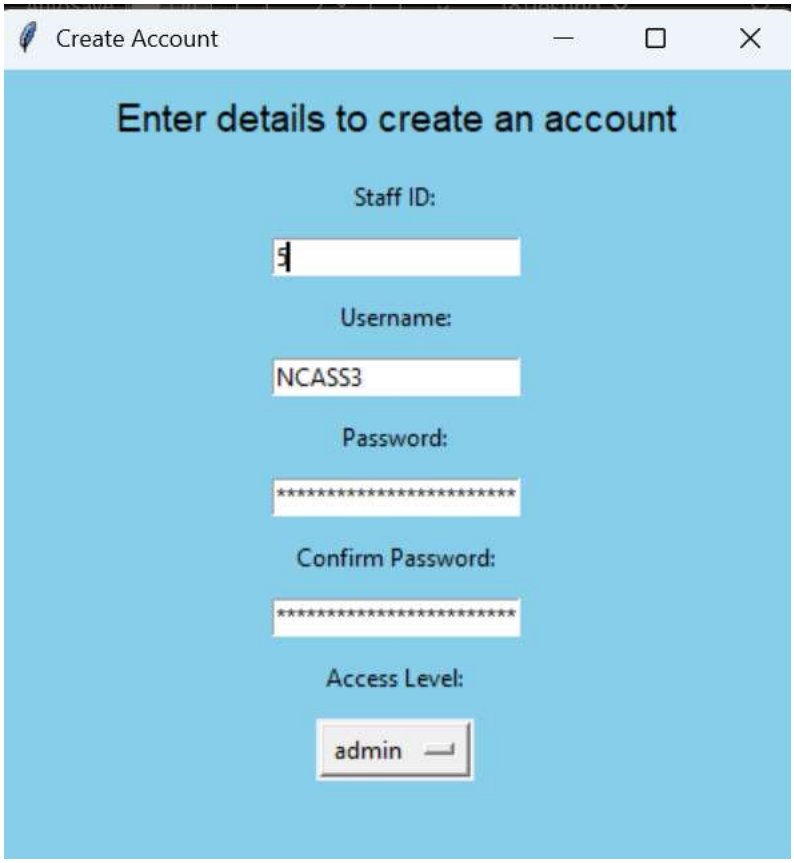

Create Account
Enter details to create an account
Staff ID:
3
Username:
ncass2
Password:
Confirm Password:
Access Level:
admin
Register
Validation Error
Passwords do not match.
OK

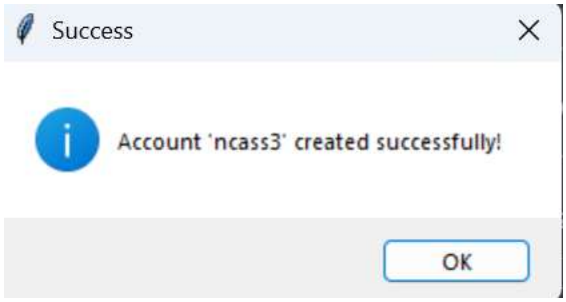
TEST 9


Create Account
Enter details to create an account
Staff ID:
5
Username:
new_admin
Password:
Confirm Password:
Access Level:
admin
Register


Success
Account 'new_admin' created successfully!
OK

TEST 10


Create Account
Enter details to create an account
Staff ID:
5
Username:
NCASS3
Password:
****
Confirm Password:
****
Access Level:
admin

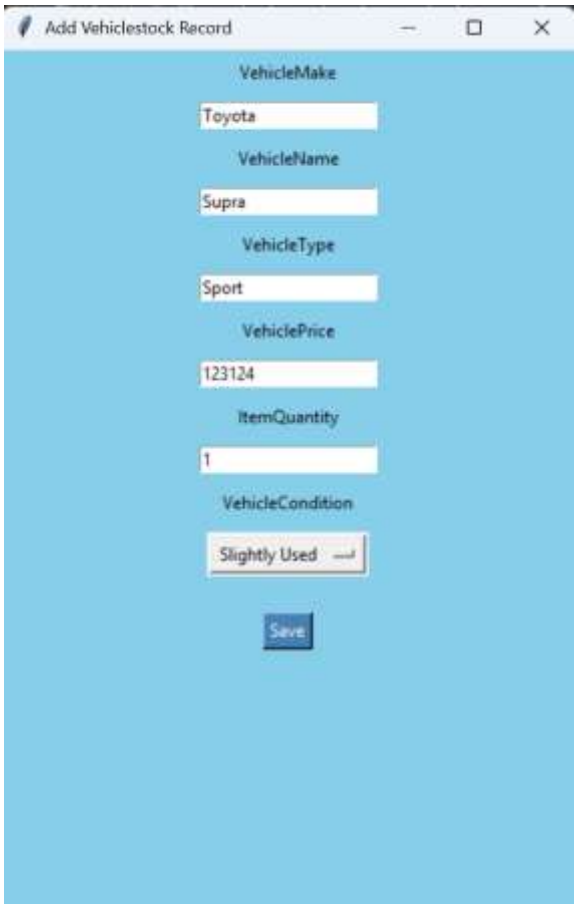

Success
Account 'ncass3' created successfully!
OK

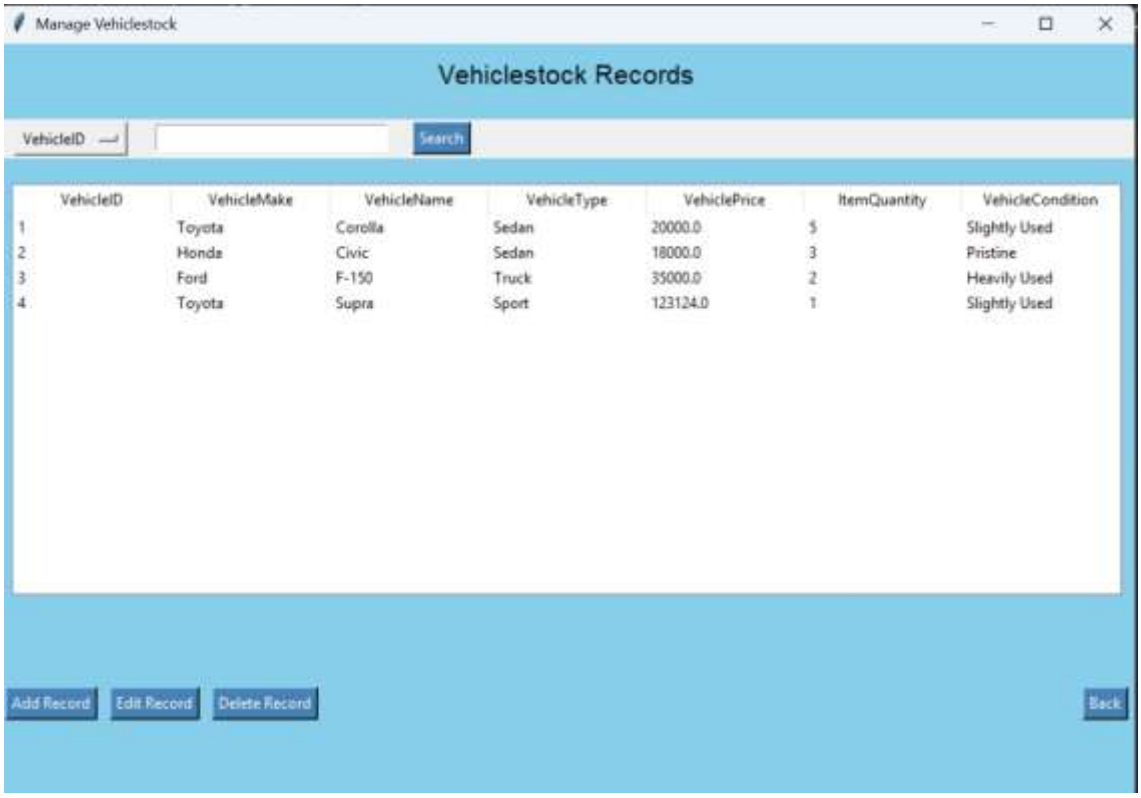
2. VehicleStock Management (10 Tests)

Form: manageTable.py (Vehicle Stock)

TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
11	Add Vehicle: Valid Data	Make=Toyota, Price=20000, Quantity=5	Functional	T	Record added.	SUCCESS
12	Add Vehicle: Negative Price	Price=-1000	Negative	I	Error: "VehiclePrice must be numeric."	SUCCESS
13	Add Vehicle: Extreme Quantity (999999)	Quantity=999999	Boundary	E	Added .	SUCCESS
14	Add Vehicle: Missing VehicleMake	Make=""	Negative	I	Error: "VehicleMake cannot be empty."	SUCCESS
15	Edit Vehicle: Update Quantity to 0	Quantity=0	Functional	E	Record updated.	SUCCESS
16	Delete Vehicle: Valid ID	Select existing VehicleID	Functional	T	Record deleted.	SUCCESS
17	Search Vehicle: By Make=Honda	Search Term=Honda	Functional	T	Matching records.	SUCCESS
18	Search by type =Sports	Search == Sports	Functional	I	Matching records	SUCCESS
19	Edit Vehicle: Invalid Price (Text)	Price=twenty	Negative	I	Error: "Numeric required."	SUCCESS
20	Search Vehicle: Non-Existent Make	Make=Tesla	Functional	T	No results.	SUCCESS

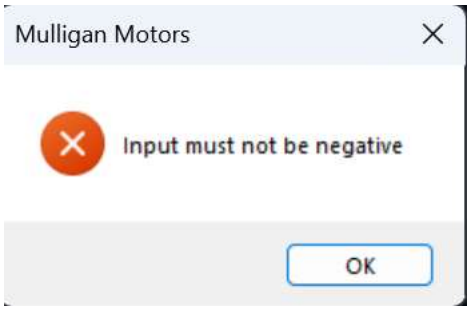
TEST 11


Add Vehiclestock Record
VehicleMake
Toyota
VehicleName
Supra
VehicleType
Sport
VehiclePrice
123124
ItemQuantity
1
VehicleCondition
Slightly Used
Save

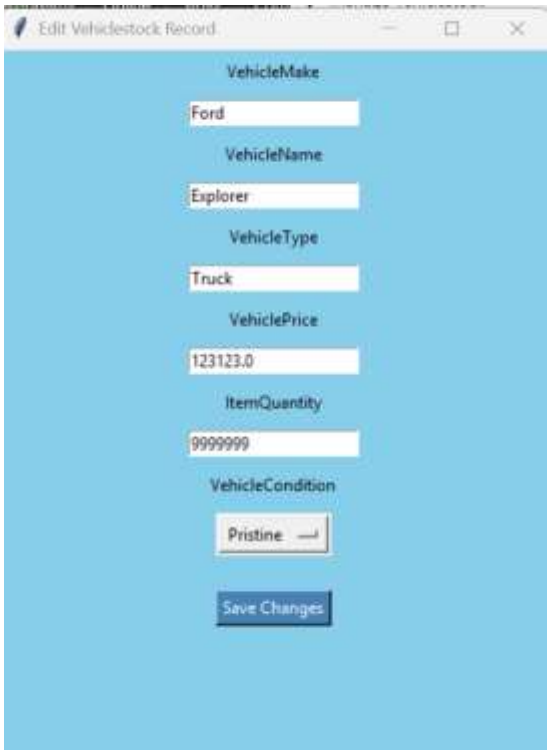

Manage Vehiclestock
Vehiclestock Records
VehicleID
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
1 Toyota Corolla Sedan 20000.0 5 Slightly Used
2 Honda Civic Sedan 18000.0 3 Pristine
3 Ford F-150 Truck 35000.0 2 Heavily Used
4 Toyota Supra Sport 123124.0 1 Slightly Used
Add Record
Edit Record
Delete Record
Back

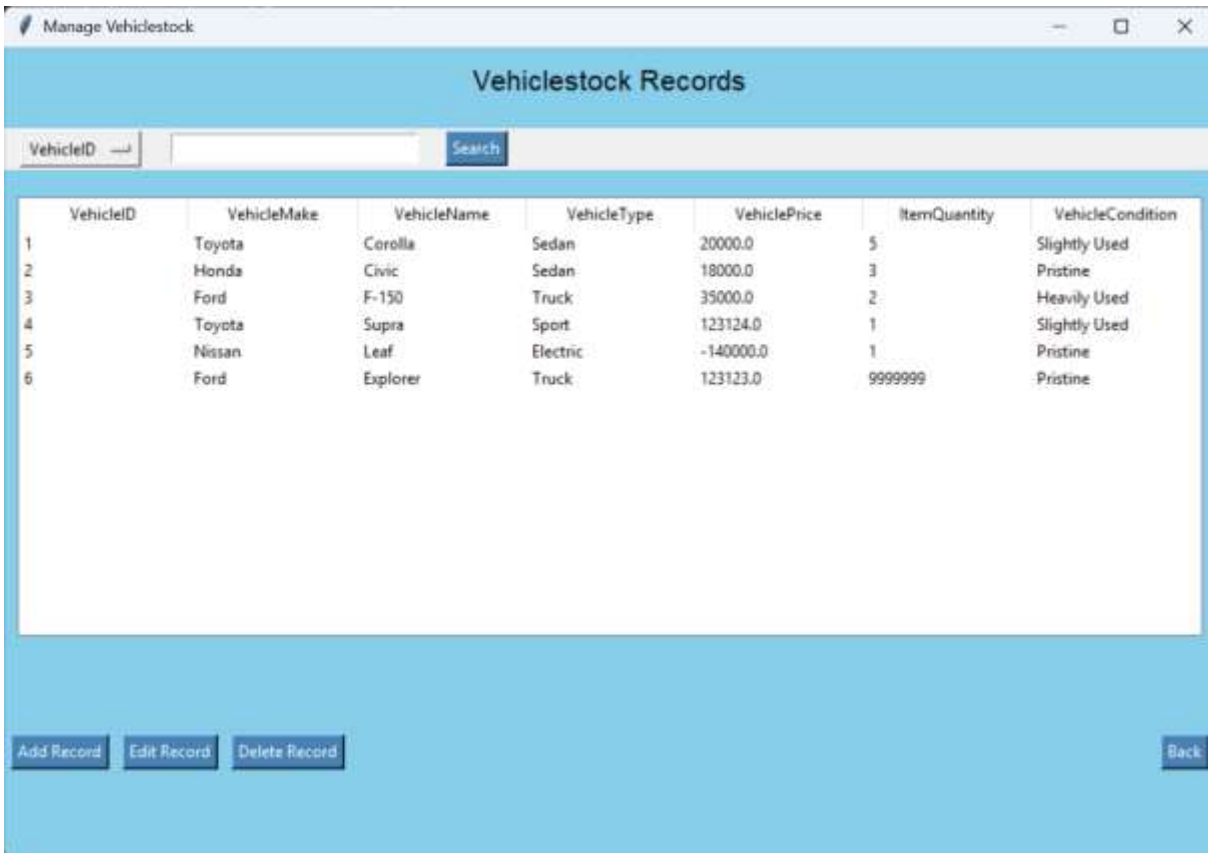
TEST 12


Edit Vehiclestock Record
VehicleMake
Nissan
VehicleName
Leaf
VehicleType
Electric
VehiclePrice
-140000.0
ItemQuantity
1
VehicleCondition
Pristine
Save Changes

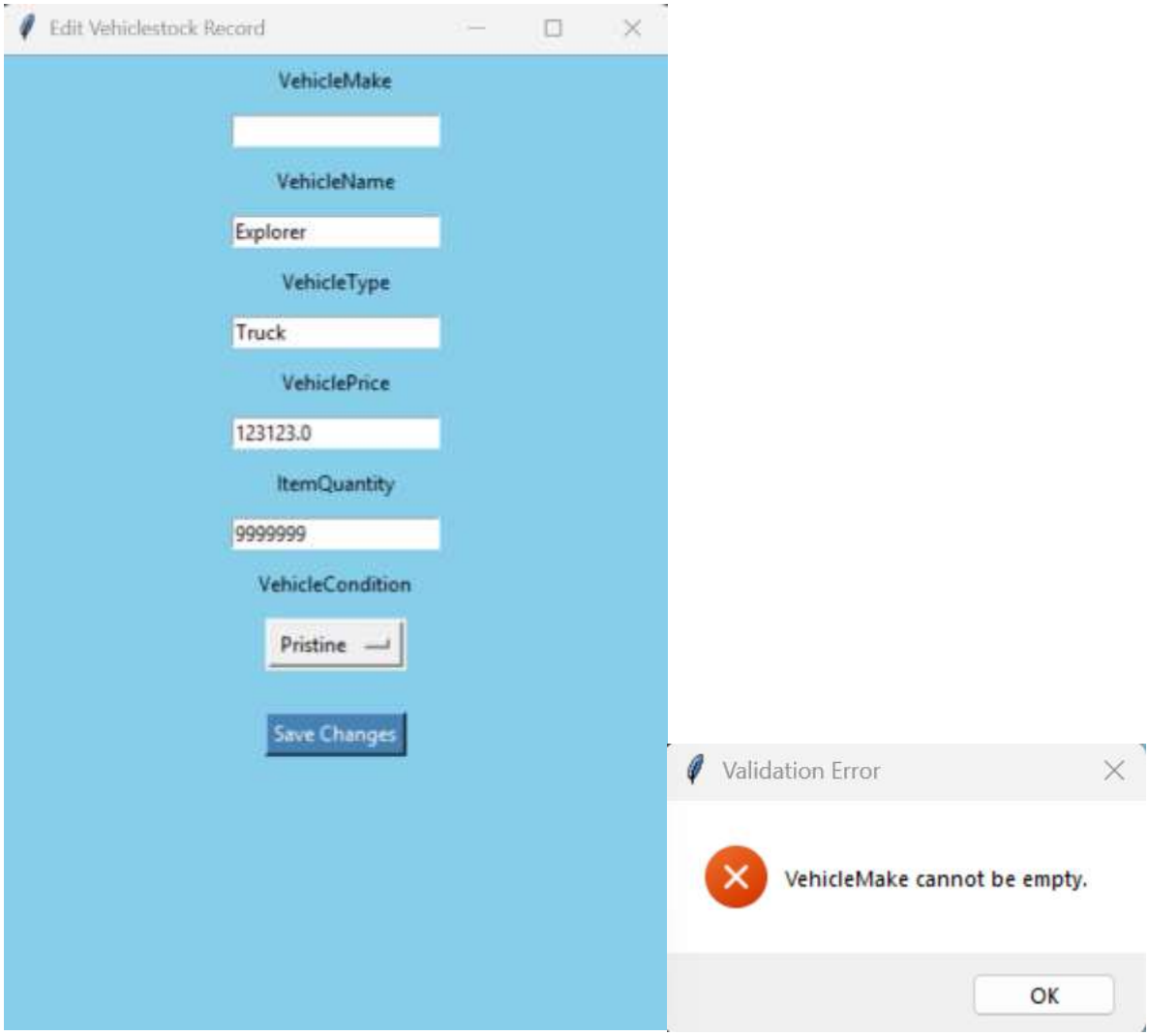

Mulligan Motors
Input must not be negative
OK

TEST 13

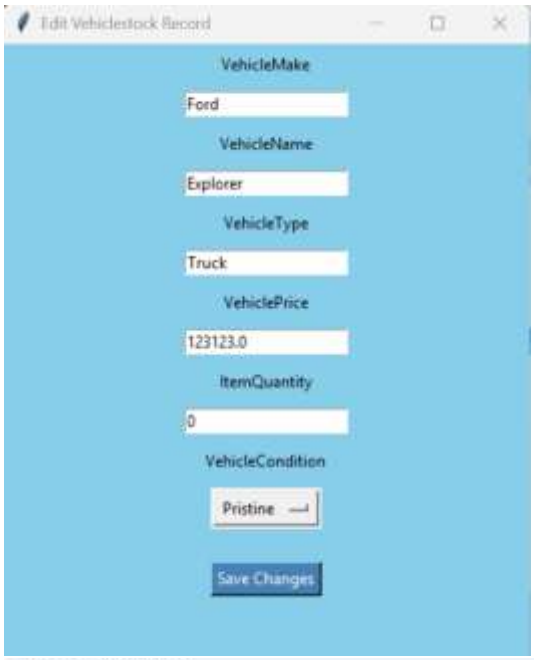

Edit Vehiclestock Record
VehicleMake
Ford
VehicleName
Explorer
VehicleType
Truck
VehiclePrice
123123.0
ItemQuantity
9999999
VehicleCondition
Pristine
Save Changes

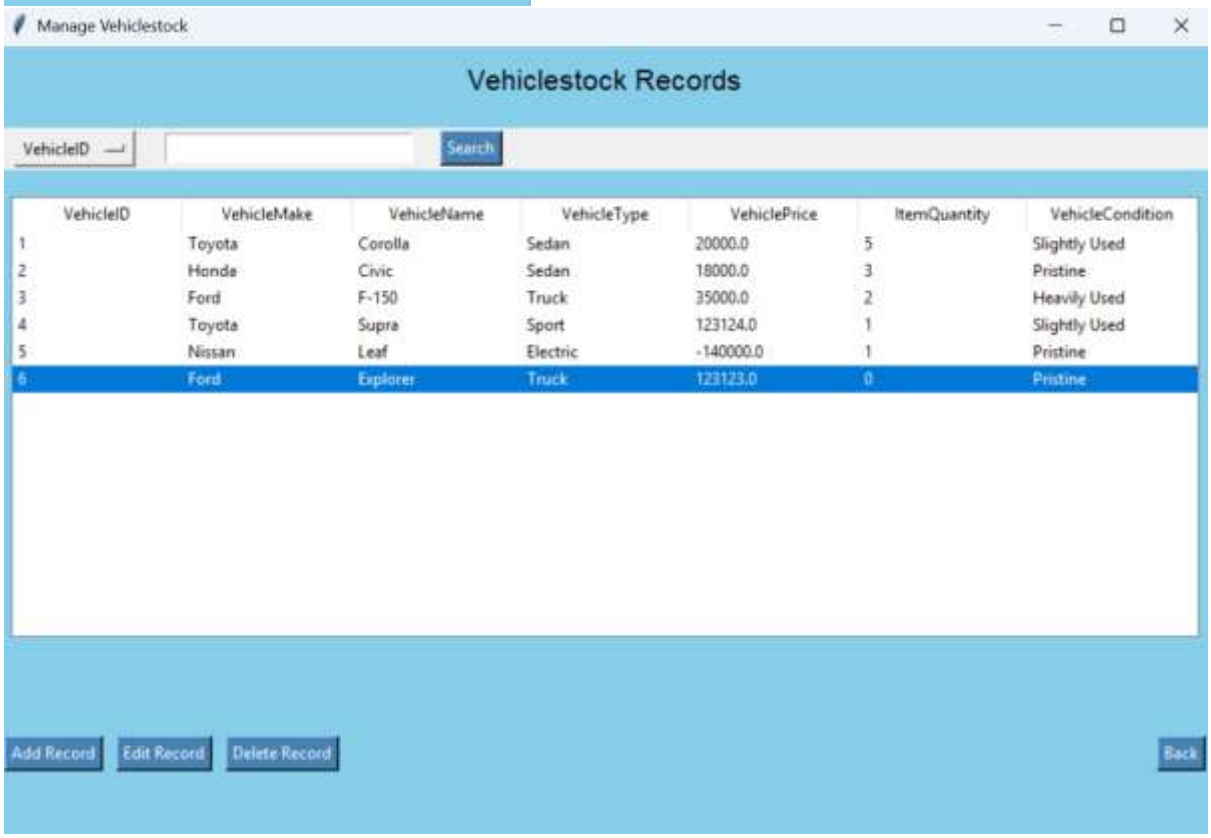

Manage Vehiclestock
Vehiclestock Records
VehicleID
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
1 Toyota Corolla Sedan 20000.0 5 Slightly Used
2 Honda Civic Sedan 18000.0 3 Pristine
3 Ford F-150 Truck 35000.0 2 Heavily Used
4 Toyota Supra Sport 123124.0 1 Slightly Used
5 Nissan Leaf Electric -140000.0 1 Pristine
6 Ford Explorer Truck 123123.0 9999999 Pristine
Add Record
Edit Record
Delete Record
Back

TEST 14

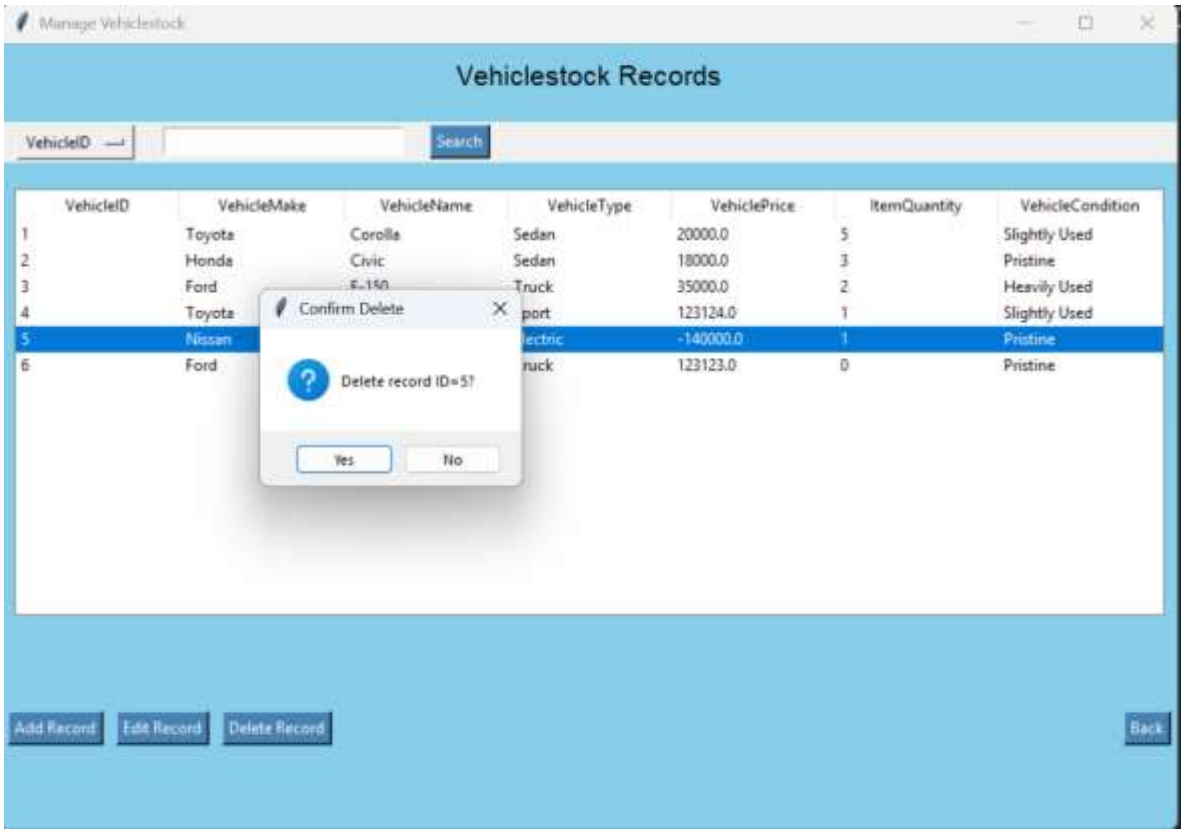

Edit Vehiclestock Record
VehicleMake
VehicleName
Explorer
VehicleType
Truck
VehiclePrice
123123.0
ItemQuantity
9999999
VehicleCondition
Pristine
Save Changes
Validation Error
VehicleMake cannot be empty.
OK

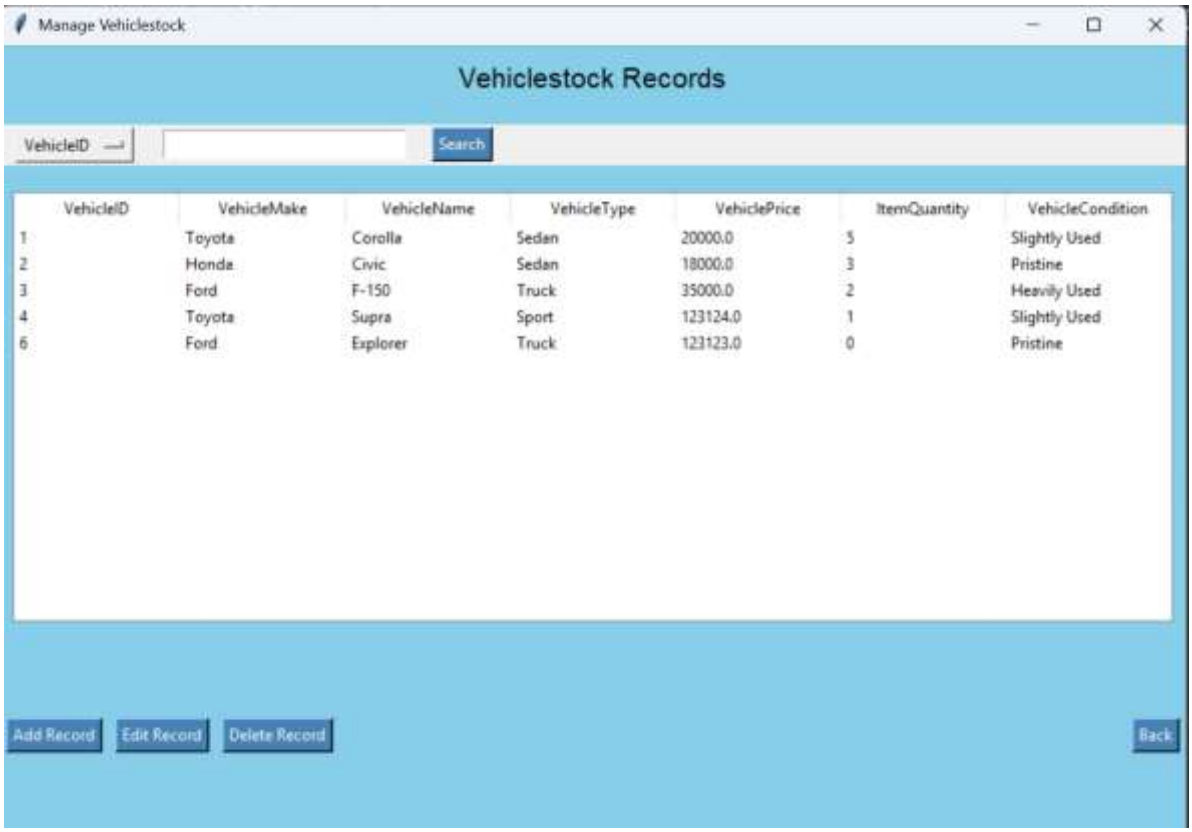
TEST 15


Edit Vehiclestock Record
VehicleMake
Ford
VehicleName
Explorer
VehicleType
Truck
VehiclePrice
123123.0
ItemQuantity
0
VehicleCondition
Pristine
Save Changes


Manage Vehiclestock
Vehiclestock Records
VehicleID
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
1 Toyota Corolla Sedan 20000.0 5 Slightly Used
2 Honda Civic Sedan 18000.0 3 Pristine
3 Ford F-150 Truck 35000.0 2 Heavily Used
4 Toyota Supra Sport 123124.0 1 Slightly Used
5 Nissan Leaf Electric -140000.0 1 Pristine
6 Ford Explorer Truck 123123.0 0 Pristine
Add Record
Edit Record
Delete Record
Back

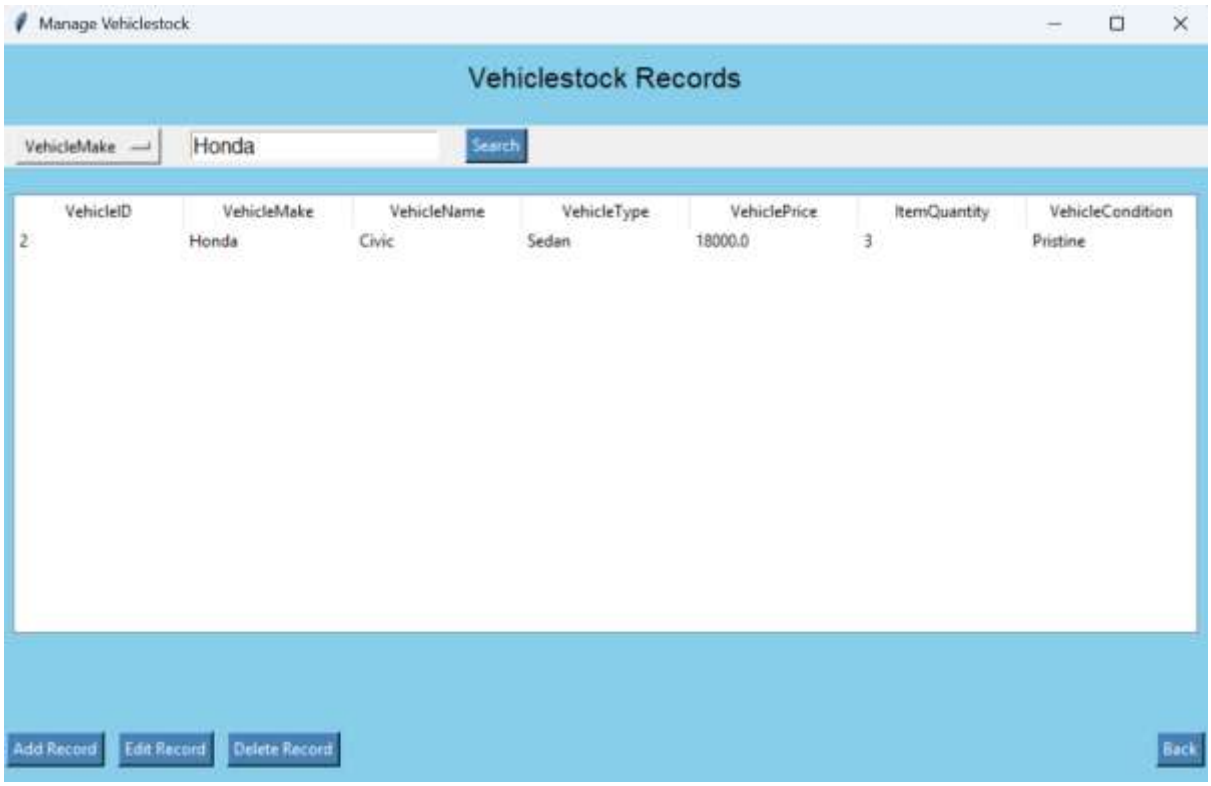
TEST 16


Manage Vehiclestock
Vehiclestock Records
VehicleID
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
1 Toyota Corolla Sedan 20000.0 5 Slightly Used
2 Honda Civic Sedan 18000.0 3 Pristine
3 Ford F-150 Truck 35000.0 2 Heavily Used
4 Toyota port 123124.0 1 Slightly Used
5 Nissan lectric -140000.0 1 Pristine
6 Ford ruck 123123.0 0 Pristine
Confirm Delete
Delete record ID=5?
Yes
No
Add Record
Edit Record
Delete Record
Back

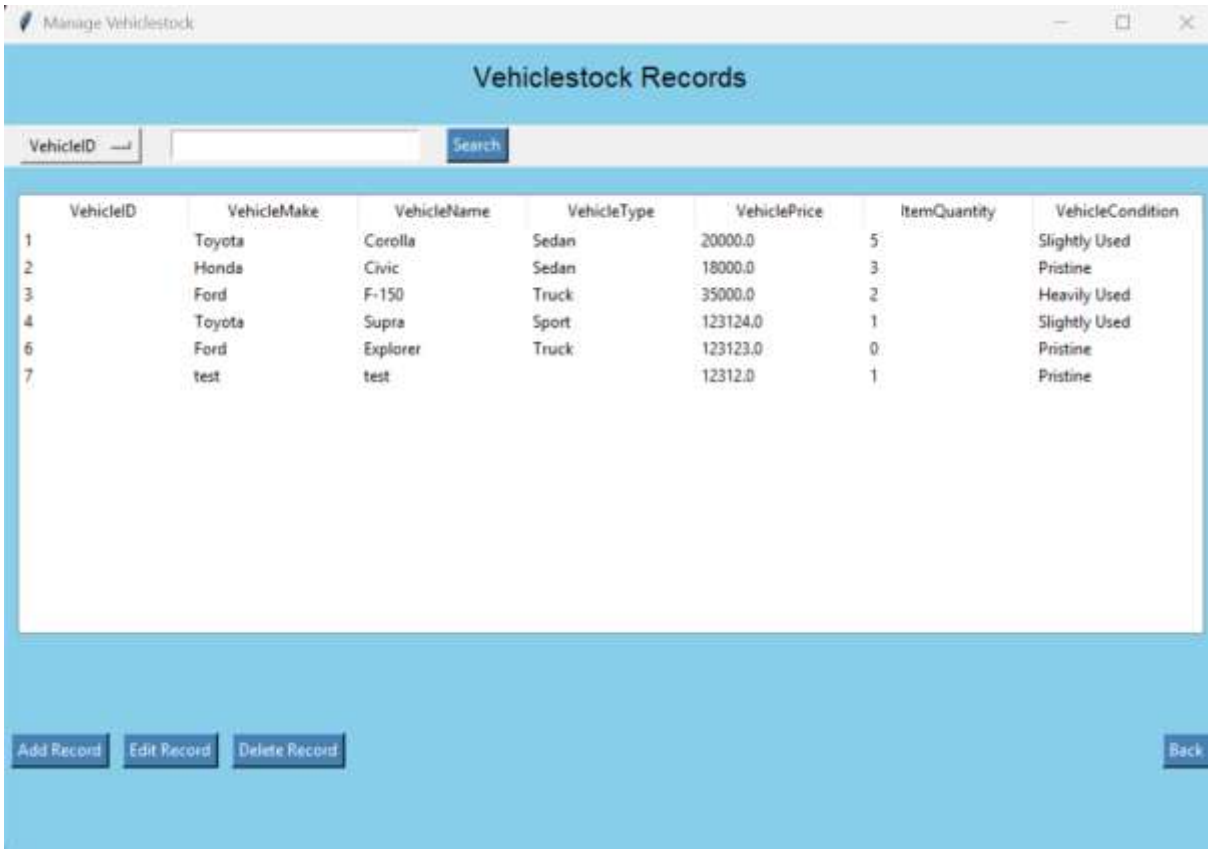

Manage Vehiclestock
Vehiclestock Records
VehicleID
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
1 Toyota Corolla Sedan 20000.0 5 Slightly Used
2 Honda Civic Sedan 18000.0 3 Pristine
3 Ford F-150 Truck 35000.0 2 Heavily Used
4 Toyota Supra Sport 123124.0 1 Slightly Used
6 Ford Explorer Truck 123123.0 0 Pristine
Add Record
Edit Record
Delete Record
Back

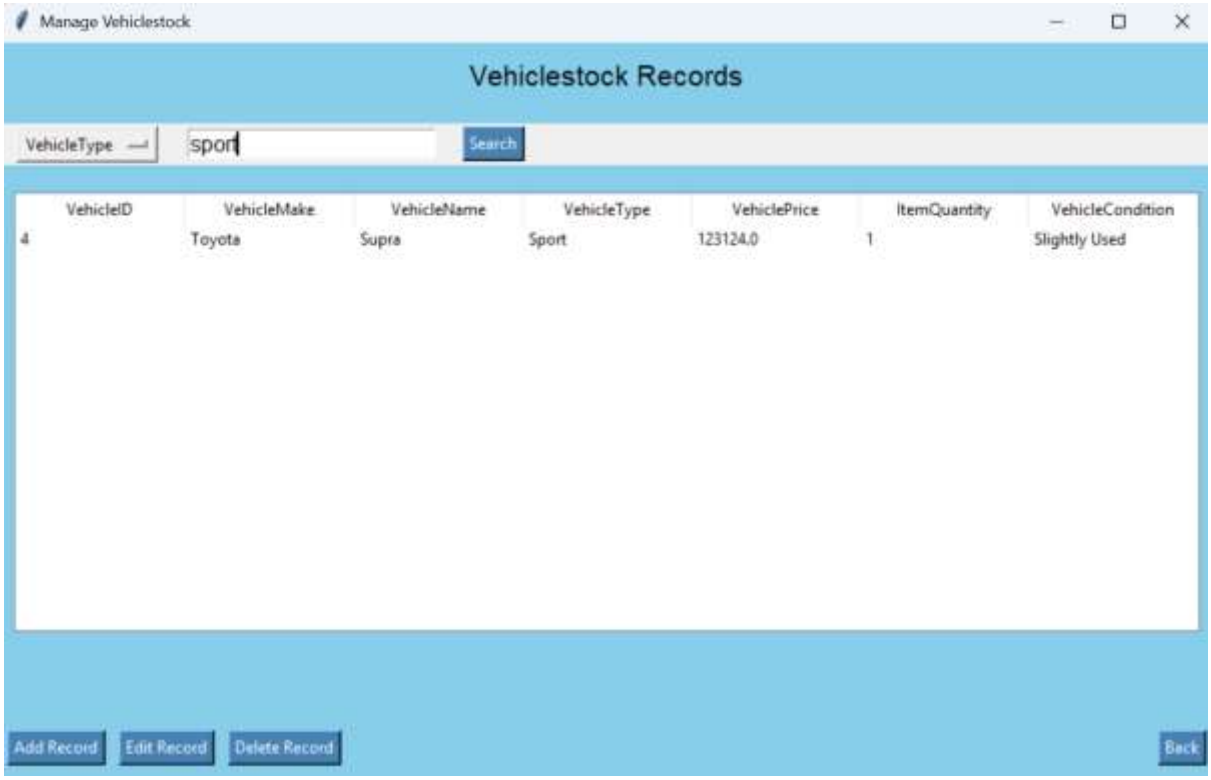
TEST 17


Manage Vehiclestock
Vehiclestock Records
VehicleID
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
1 Toyota Corolla Sedan 20000.0 5 Slightly Used
2 Honda Civic Sedan 18000.0 3 Pristine
3 Ford F-150 Truck 35000.0 2 Heavily Used
4 Toyota Supra Sport 123124.0 1 Slightly Used
6 Ford Explorer Truck 123123.0 0 Pristine
Add Record
Edit Record
Delete Record
Back

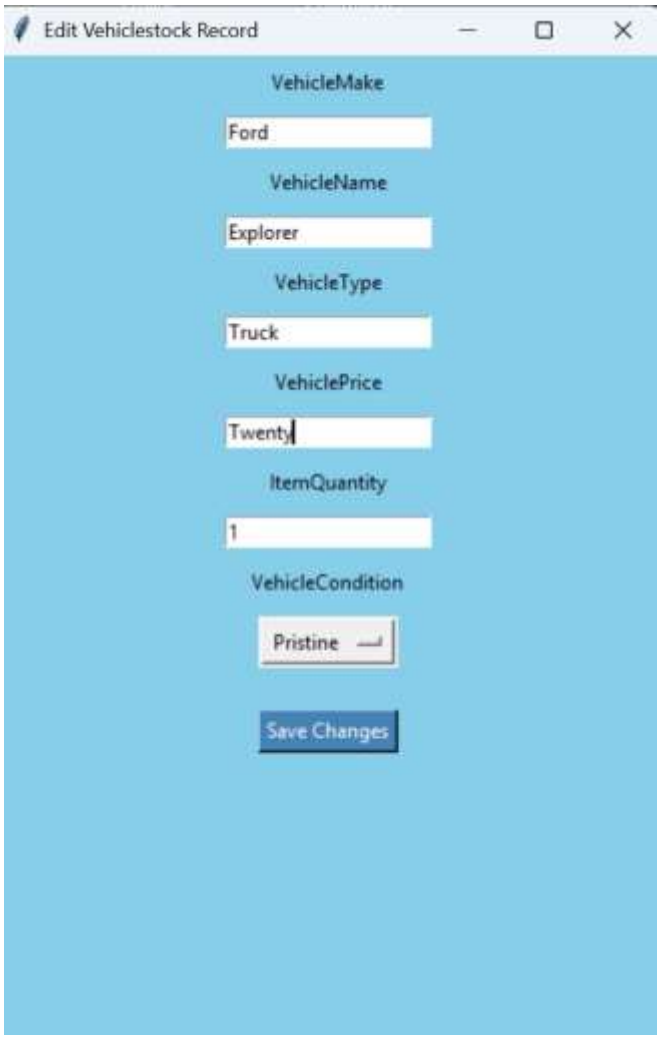

Manage Vehiclestock
Vehiclestock Records
VehicleMake
Honda
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
2 Honda Civic Sedan 18000.0 3 Pristine
Add Record
Edit Record
Delete Record
Back

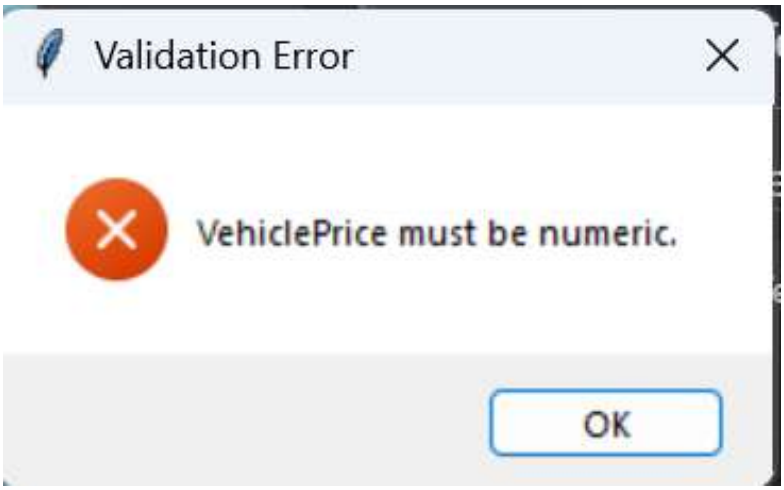
TEST 18


Manage Vehiclestock
Vehiclestock Records
VehicleID
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
1 Toyota Corolla Sedan 20000.0 5 Slightly Used
2 Honda Civic Sedan 18000.0 3 Pristine
3 Ford F-150 Truck 35000.0 2 Heavily Used
4 Toyota Supra Sport 123124.0 1 Slightly Used
6 Ford Explorer Truck 123123.0 0 Pristine
7 test test 12312.0 1 Pristine
Add Record
Edit Record
Delete Record
Back

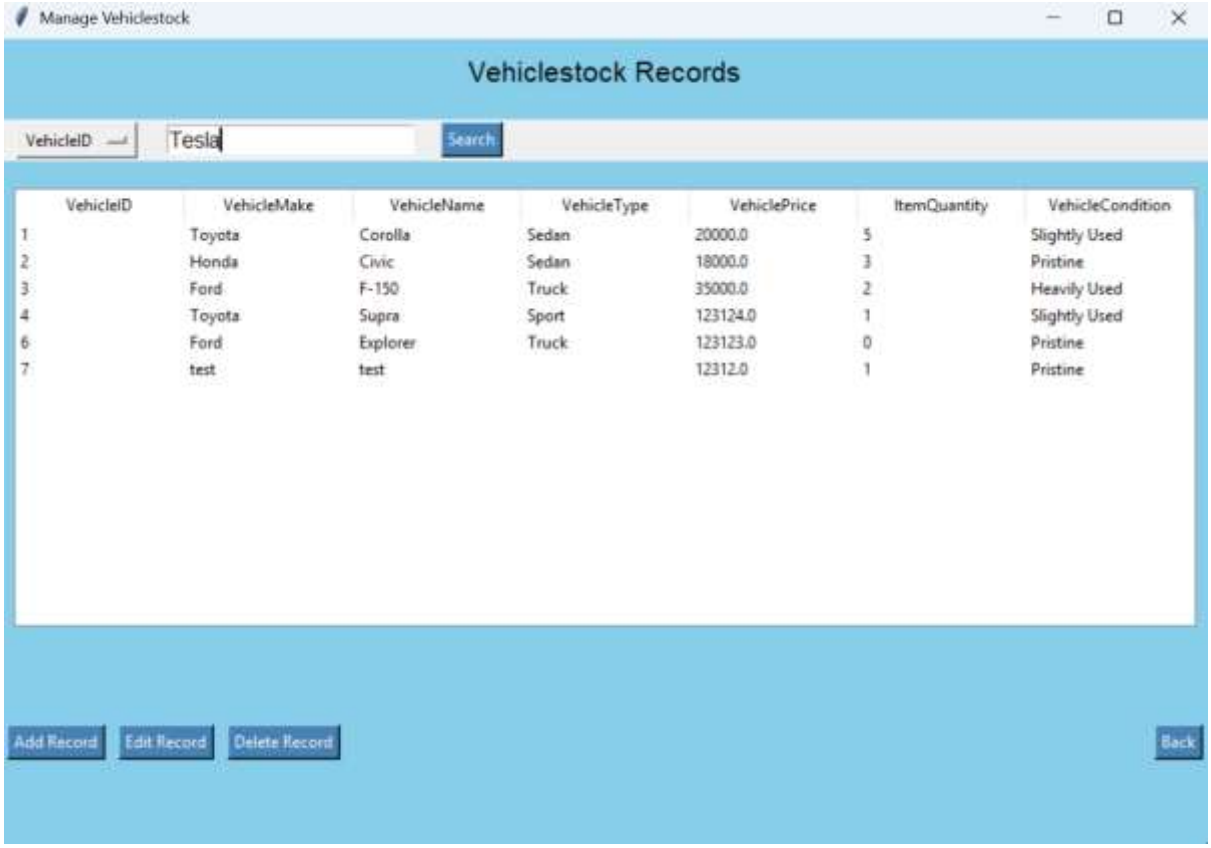

Manage Vehiclestock
Vehiclestock Records
VehicleType
sport
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
4 Toyota Supra Sport 123124.0 1 Slightly Used
Add Record
Edit Record
Delete Record
Back

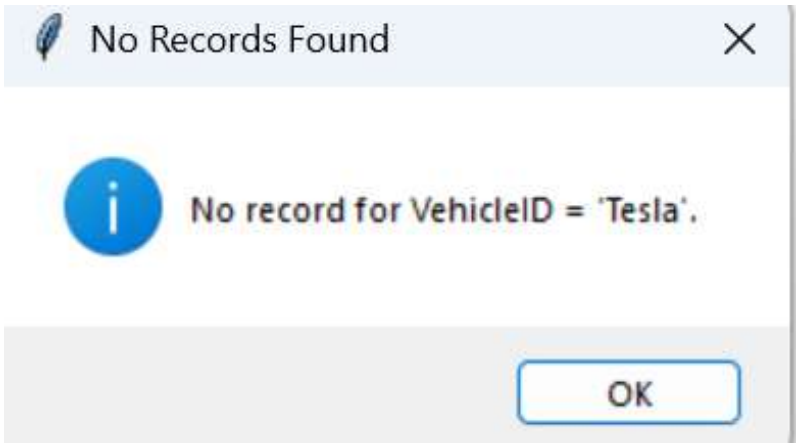
TEST 19


Edit Vehiclestock Record
VehicleMake
Ford
VehicleName
Explorer
VehicleType
Truck
VehiclePrice
Twenty
ItemQuantity
1
VehicleCondition
Pristine
Save Changes


Validation Error
VehiclePrice must be numeric.
OK

TEST 20


Manage Vehiclestock
Vehiclestock Records
VehicleID
Tesla
Search
VehicleID VehicleMake VehicleName VehicleType VehiclePrice ItemQuantity VehicleCondition
1 Toyota Corolla Sedan 20000.0 5 Slightly Used
2 Honda Civic Sedan 18000.0 3 Pristine
3 Ford F-150 Truck 35000.0 2 Heavily Used
4 Toyota Supra Sport 123124.0 1 Slightly Used
6 Ford Explorer Truck 123123.0 0 Pristine
7 test test 12312.0 1 Pristine
Add Record
Edit Record
Delete Record
Back

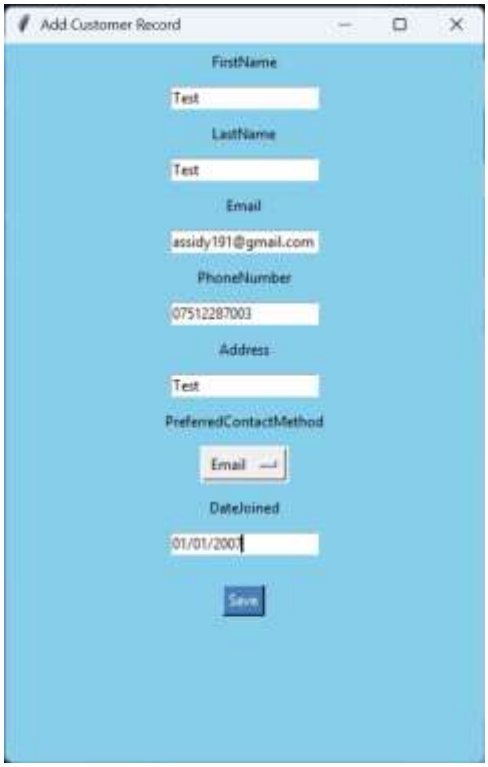

No Records Found
No record for VehicleID = 'Tesla'.
OK

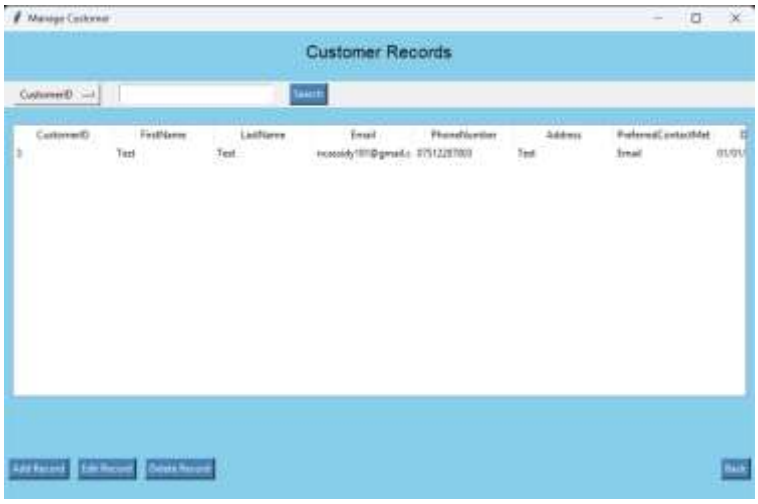
3. Customer Management (10 Tests)

Form: manageTable.py (Customer)

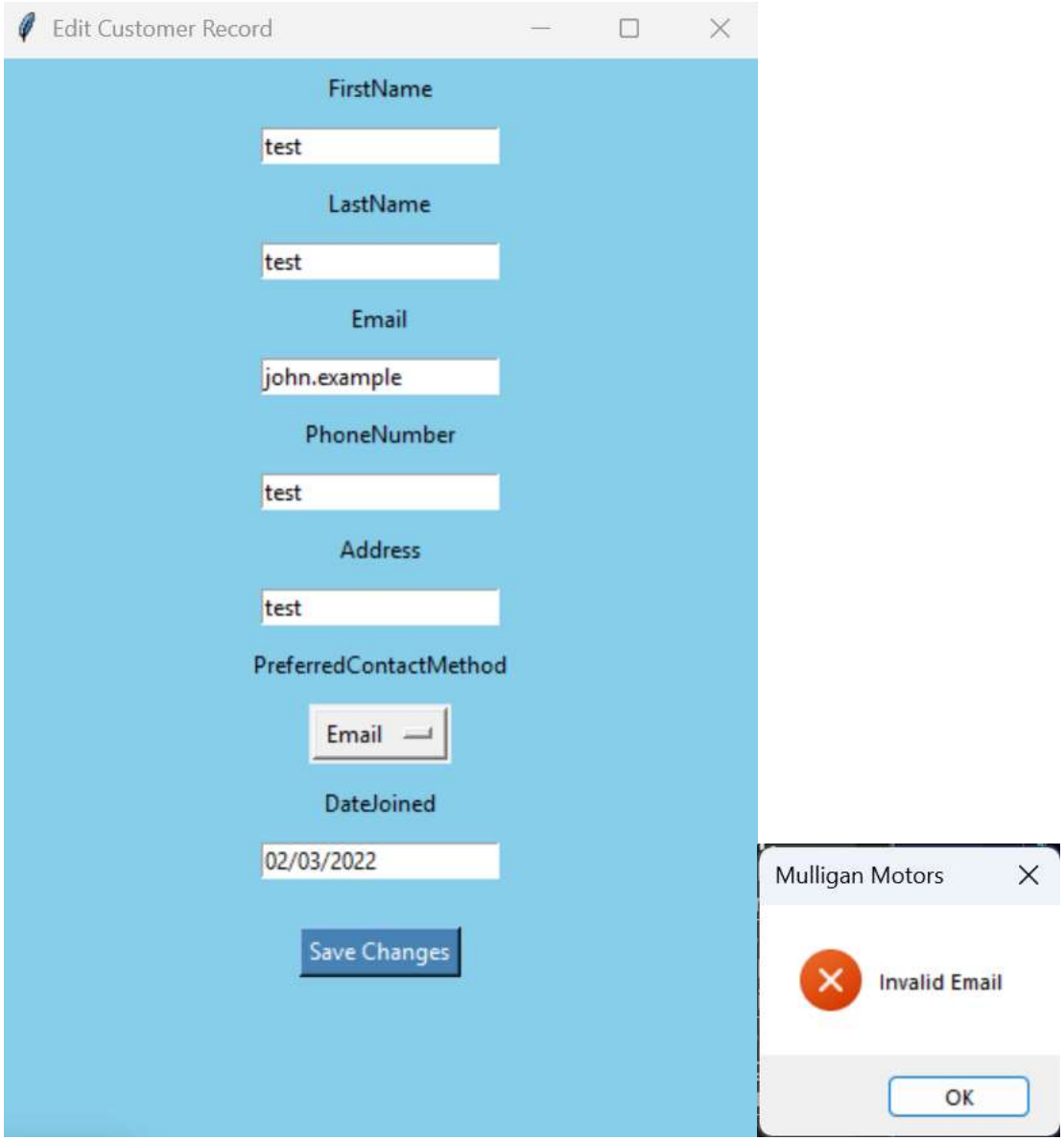
TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
21	Add Customer: Valid Email	Email=john@example.com	Functional	T	Record added.	SUCCESS
22	Add Customer: Invalid Email	Email=john.example	Negative	I	Error: "Invalid email."	SUCCESS
23	Add Customer: Drop down box preference	N/A	Functional	T	A dropdown box with 3 options appears	SUCCESS
24	Edit Customer: Change Contact Method	Contact Method=SMS	Functional	T	Record updated.	SUCCESS
25	Delete Customer with Orders	CustomerID linked to Orders	Functional	T	Deleted (FK cascade).	SUCCESS
26	Add Customer: Empty FirstName	FirstName=""	Negative	I	Error: "FirstName required."	SUCCESS
27	Edit Customer record to be blank	N/A	Functional	I	Error	SUCCESS
28	Search Customer: By Phone (1234567890)	Phone=1234567890	Functional	T	Matching record.	SUCCESS
29	Add Customer: Extreme Address (50 chars)	Address=A" * 80	Boundary	E	Added .	SUCCESS
30	Delete Customer: Non-Existent ID	CustomerID=9999	Negative	I	Error: No record.	SUCCESS

TEST 21


Add Customer Record
FirstName
Test
LastName
Test
Email
assidy191@gmail.com
PhoneNumber
07512287003
Address
Test
PreferredContactMethod
Email
DateJoined
01/01/2001
Save


Customer Records

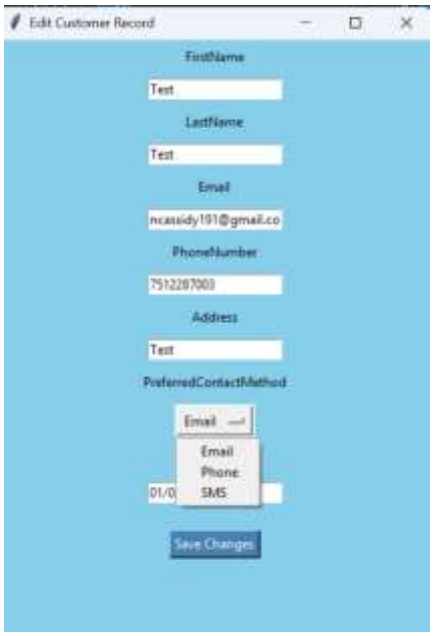
TEST 22

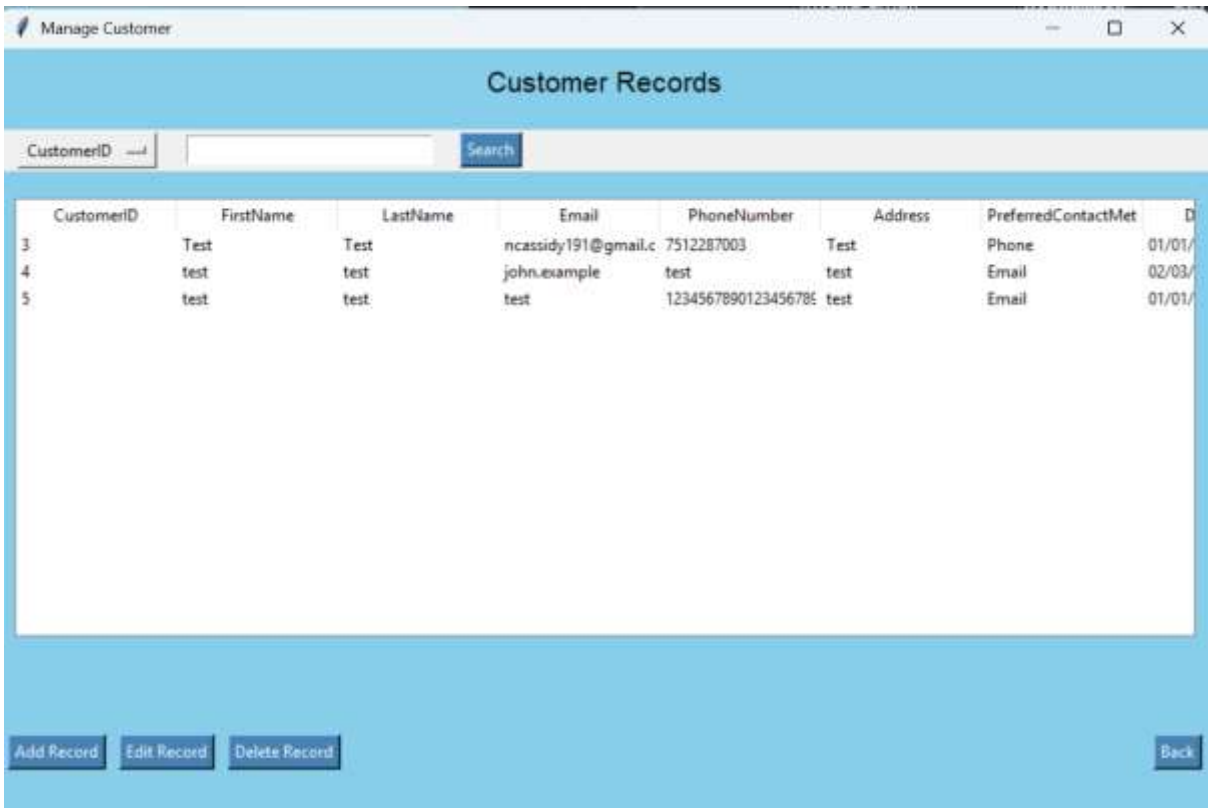

Edit Customer Record
FirstName
test
LastName
test
Email
john.example
PhoneNumber
test
Address
test
PreferredContactMethod
Email
DateJoined
02/03/2022
Save Changes
Mulligan Motors
Invalid Email
OK

TEST 23



TEST 24


Edit Customer Record
FirstName
Test
LastName
Test
Email
ncassidy191@gmail.co
PhoneNumber
7512287003
Address
Test
PreferredContactMethod
Email
Email
Phone
SMS
Save Changes


Manage Customer
Customer Records
CustomerID
Search
CustomerID FirstName LastName Email PhoneNumber Address PreferredContactMet
3 Test Test ncassidy191@gmail.c 7512287003 Test Phone
4 test test john.example test test Email
5 test test test 1234567890123456785 test Email
Add Record
Edit Record
Delete Record
Back

TEST 25

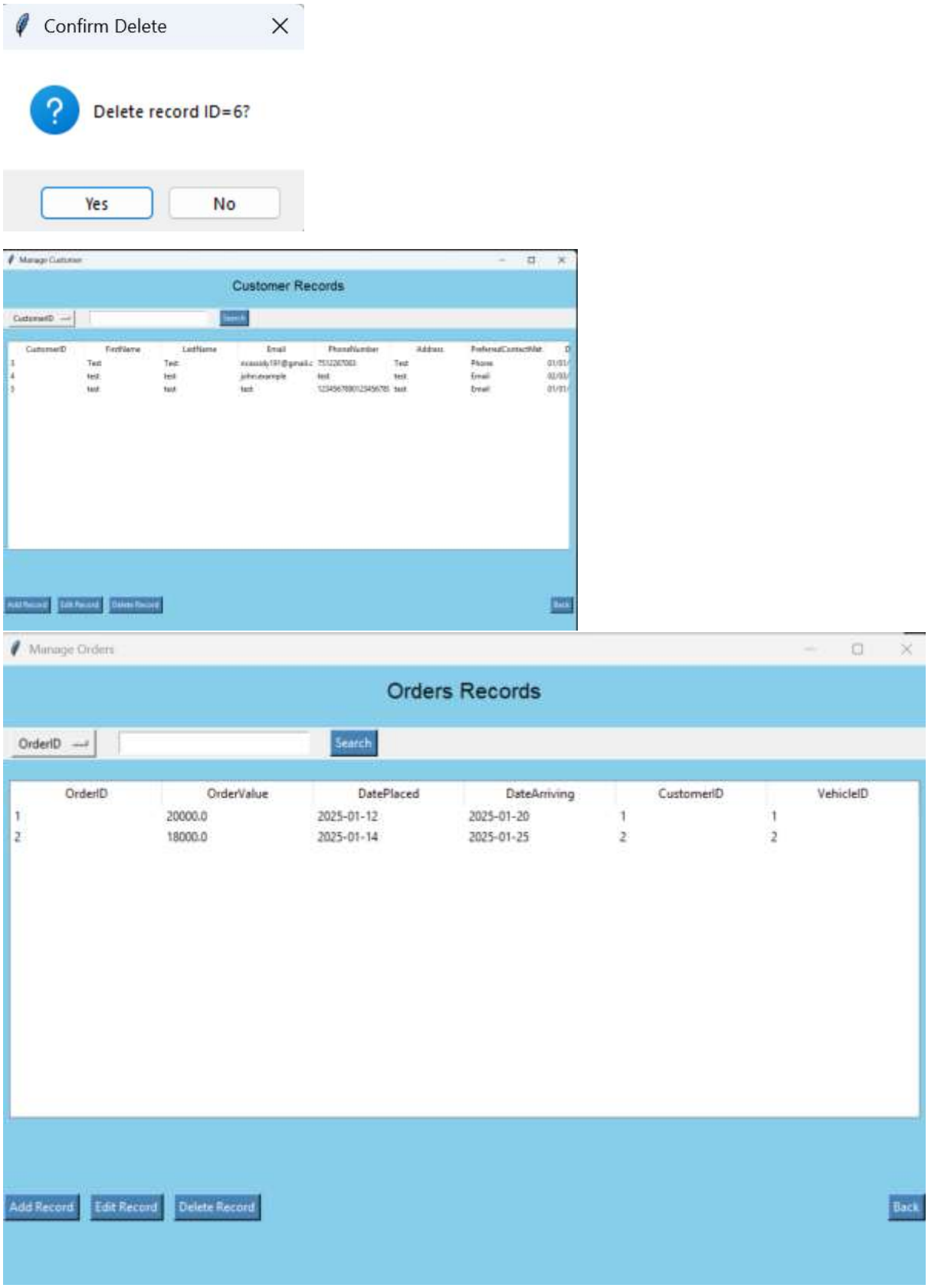
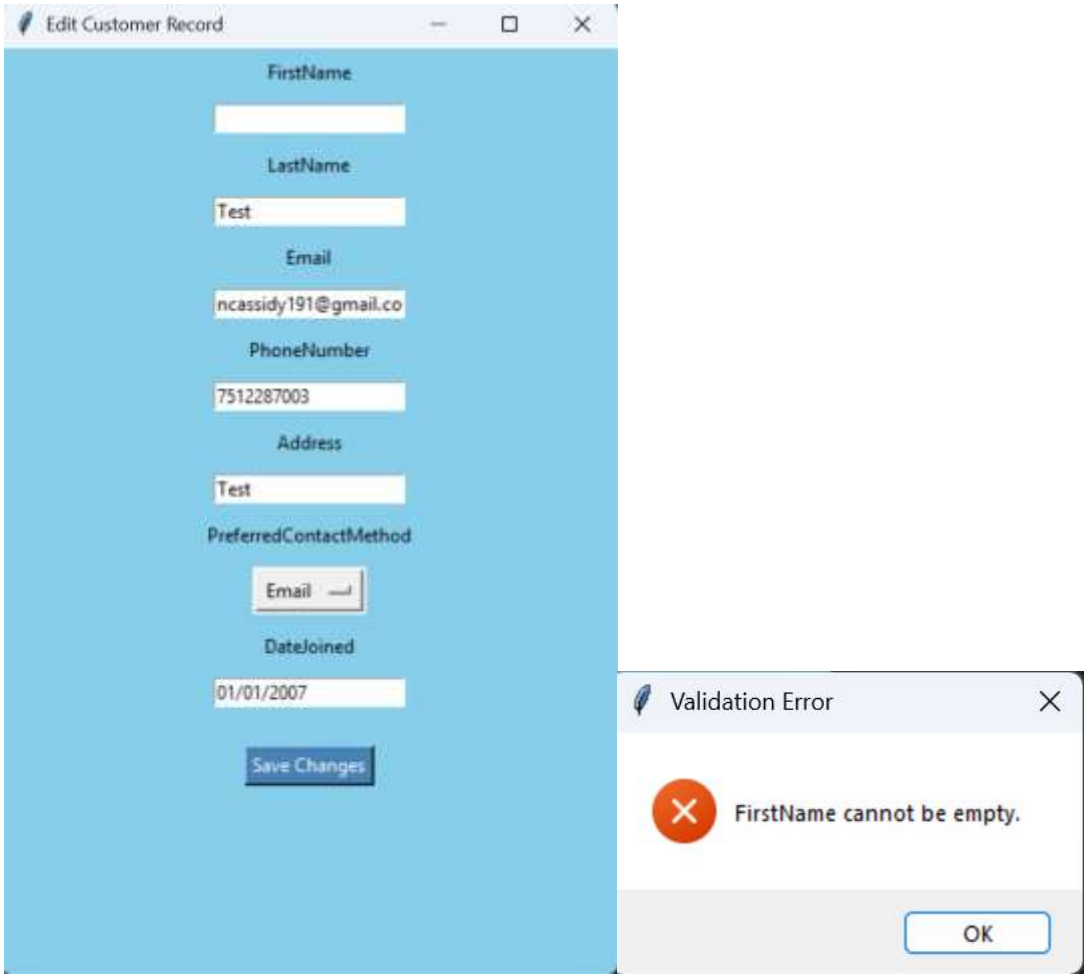
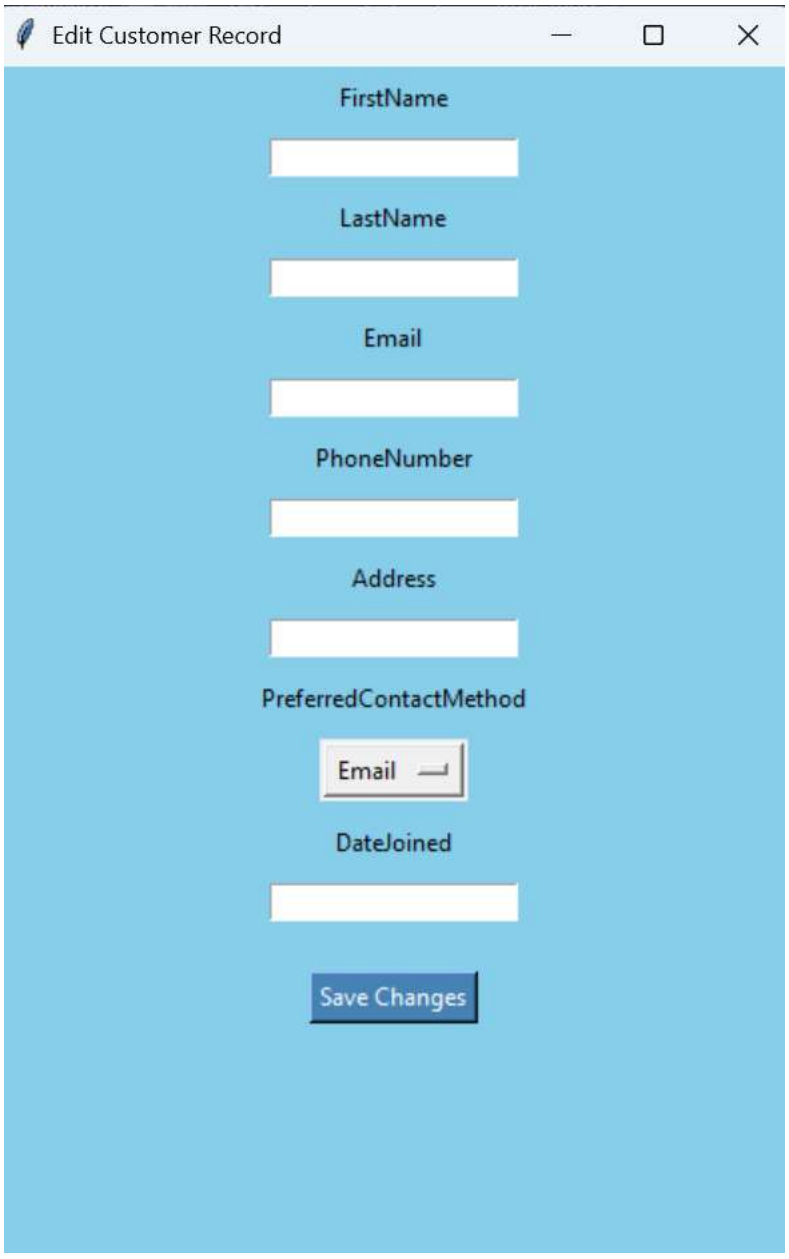


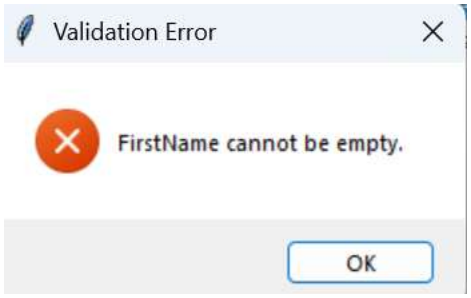
Figure 1: The related order to the customer has been deleted

TEST 26

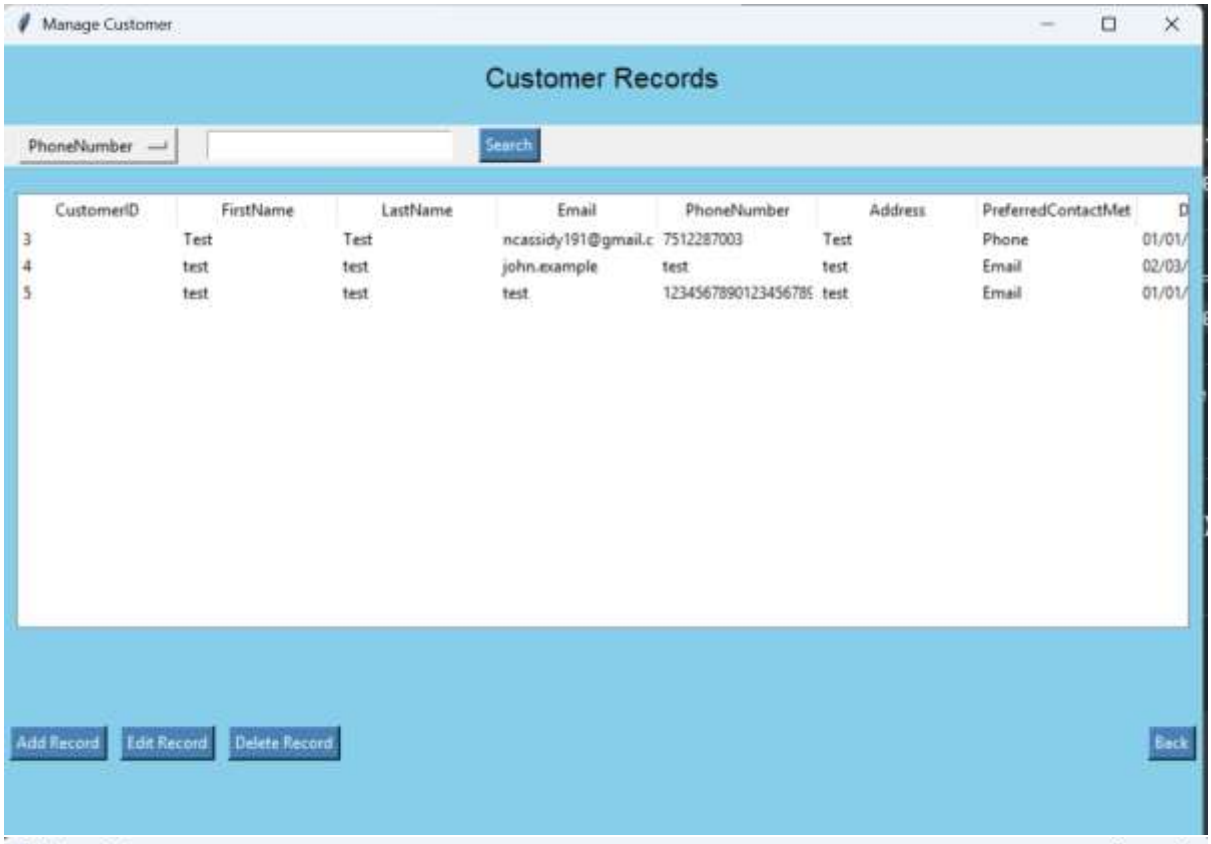

Edit Customer Record
FirstName
LastName
Test
Email
ncassidy191@gmail.co
PhoneNumber
7512287003
Address
Test
PreferredContactMethod
Email
DateJoined
01/01/2007
Save Changes
Validation Error
FirstName cannot be empty.
OK

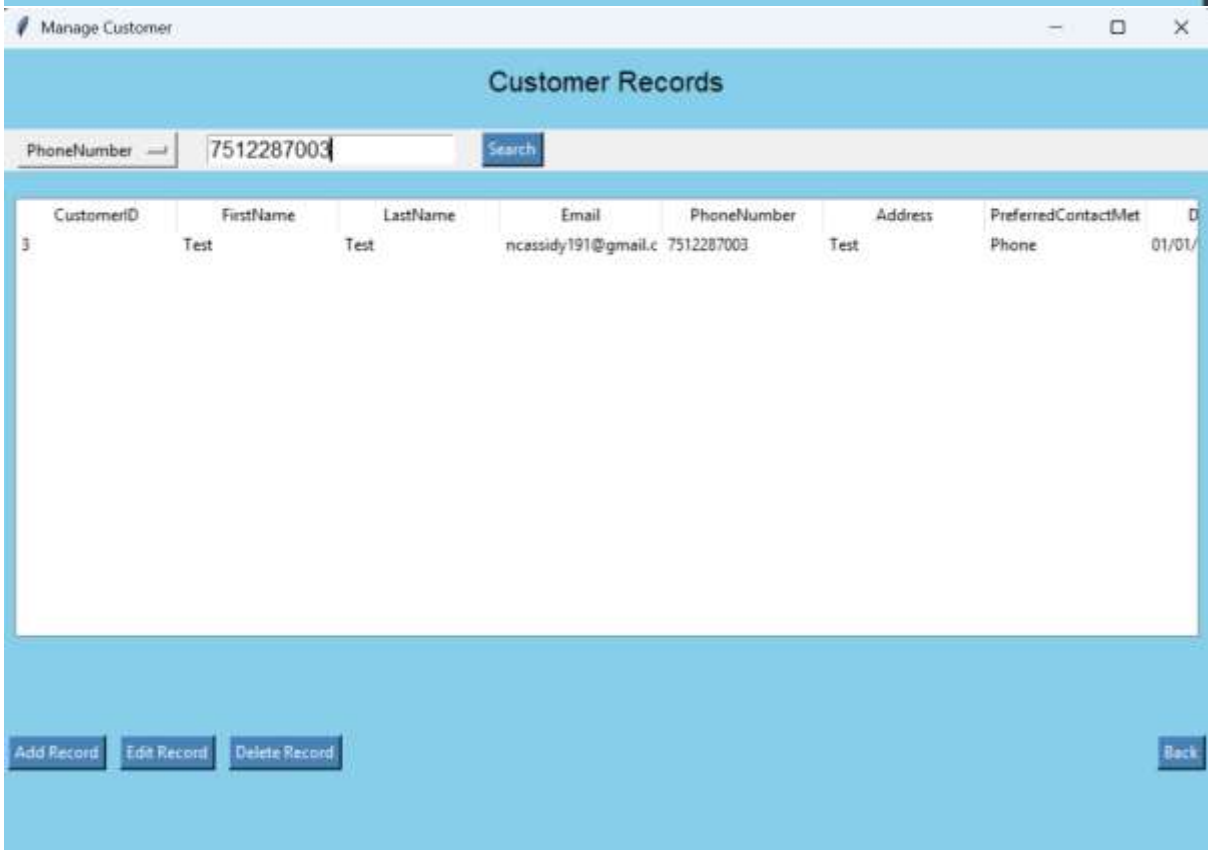
TEST 27


Edit Customer Record
FirstName
LastName
Email
PhoneNumber
Address
PreferredContactMethod
Email
DateJoined
Save Changes


Validation Error
FirstName cannot be empty.
OK

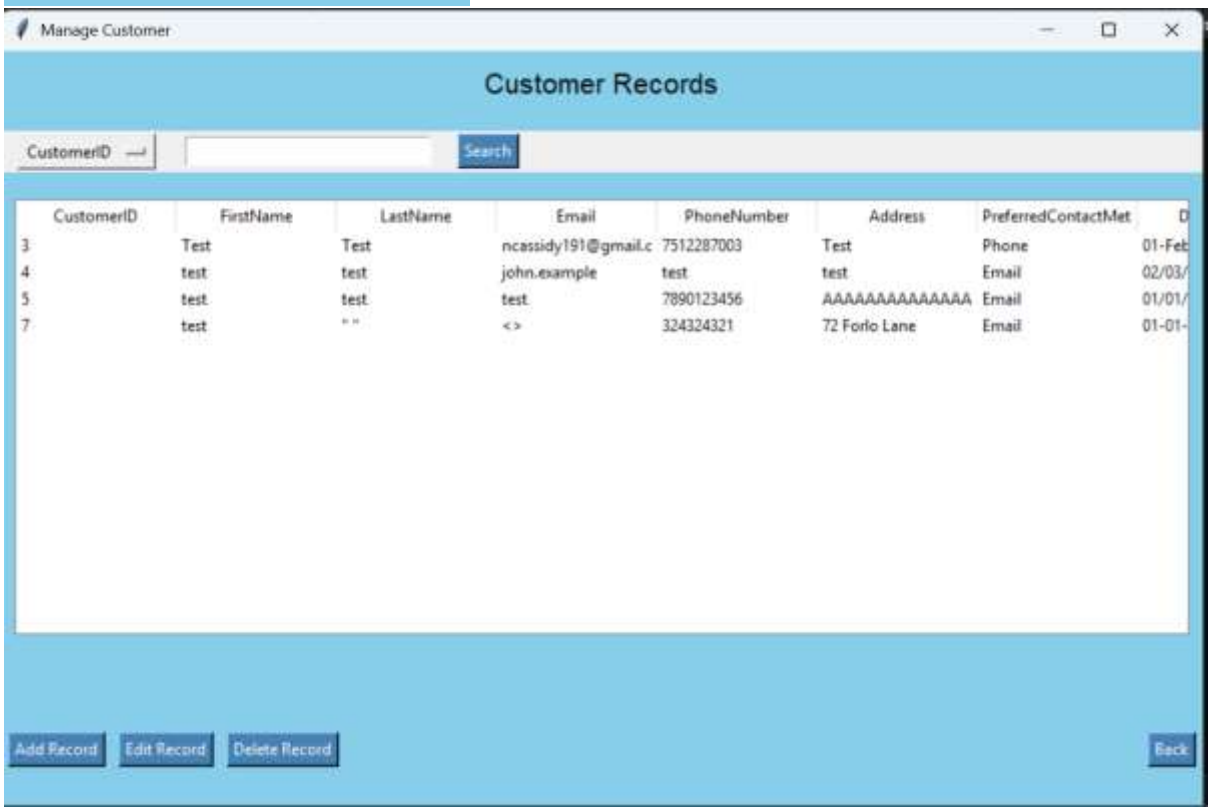
TEST 28


Manage Customer
Customer Records
PhoneNumber
Search
CustomerID
FirstName
LastName
Email
PhoneNumber
Address
PreferredContactMet
3
Test
Test
ncassidy191@gmail.c
7512287003
Test
Phone
4
test
test
john.example
test
test
Email
5
test
test
test
test
Email
Add Record
Edit Record
Delete Record
Back

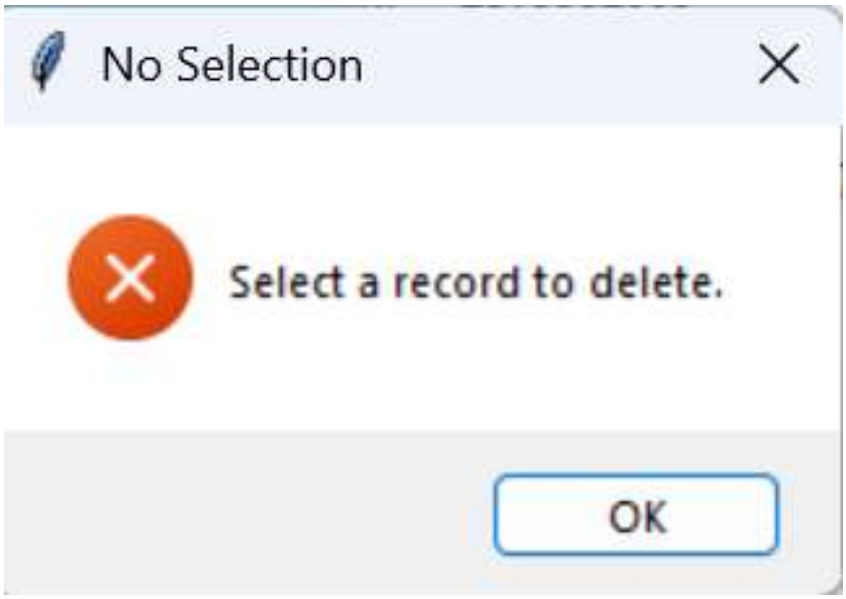

Manage Customer
Customer Records
PhoneNumber
7512287003
Search
CustomerID
FirstName
LastName
Email
PhoneNumber
Address
PreferredContactMet
3
Test
Test
ncassidy191@gmail.c
7512287003
Test
Phone
Add Record
Edit Record
Delete Record
Back

TEST 29


Edit Customer Record
FirstName
test
LastName
test
Email
test
PhoneNumber
7890123456
Address
AAAAAAAAAAAAAA
PreferredContactMethod
Email
DateJoined
01/01/2001
Save Changes


Manage Customer
Customer Records
CustomerID
Search
CustomerID FirstName LastName Email PhoneNumber Address PreferredContactMet D
3 Test Test ncassidy191@gmail.c 7512287003 Test Phone 01-Feb
4 test test john.example test test Email 02/03/
5 test test test 7890123456 AAAAAAAAAAAAAA Email 01/01/
7 test "" <> 324324321 72 Forlo Lane Email 01-01-
Add Record
Edit Record
Delete Record
Back

TEST 30

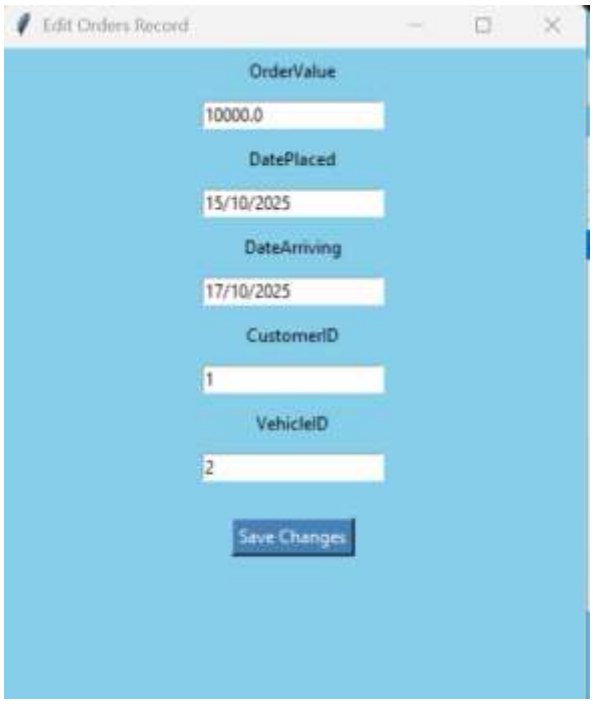


4. Orders Management (10 Tests)

Form: manageTable.py (Orders)

TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
31	Add Order: Valid CustomerID/VehicleID	CustomerID=1, VehicleID=1	Functional	T	Order added.	SUCCESS
32	Select a record.	Clicking on a record	Functional	T	Record is highlighted in blue	SUCCESS
33	Add Order: Extreme OrderValue (1e9)	OrderValue=1000000000	Boundary	E	Added .	SUCCESS
34	Edit Order: Update Arrival Date	DateArriving=2025-02-01	Functional	T	Record updated.	SUCCESS
35	Delete Order: Valid OrderID	OrderID=1	Functional	T	Record deleted.	SUCCESS
36	Try to add a blank record	N/A	Negative	I	Error: Please enter info.	SUCCESS
37	Search Order: By VehicleID=2	VehicleID=2	Functional	T	Matching records.	SUCCESS
38	Delete a record without selecting it first	N/A	Negative	I	Error: Select a record	SUCCESS
39	Add Order: Missing DatePlaced	DatePlaced=""	Negative	I	Error: "DatePlaced required."	SUCCESS
40	Search Order: Non-Existent CustomerID	CustomerID=999	Functional	T	No results.	SUCCESS

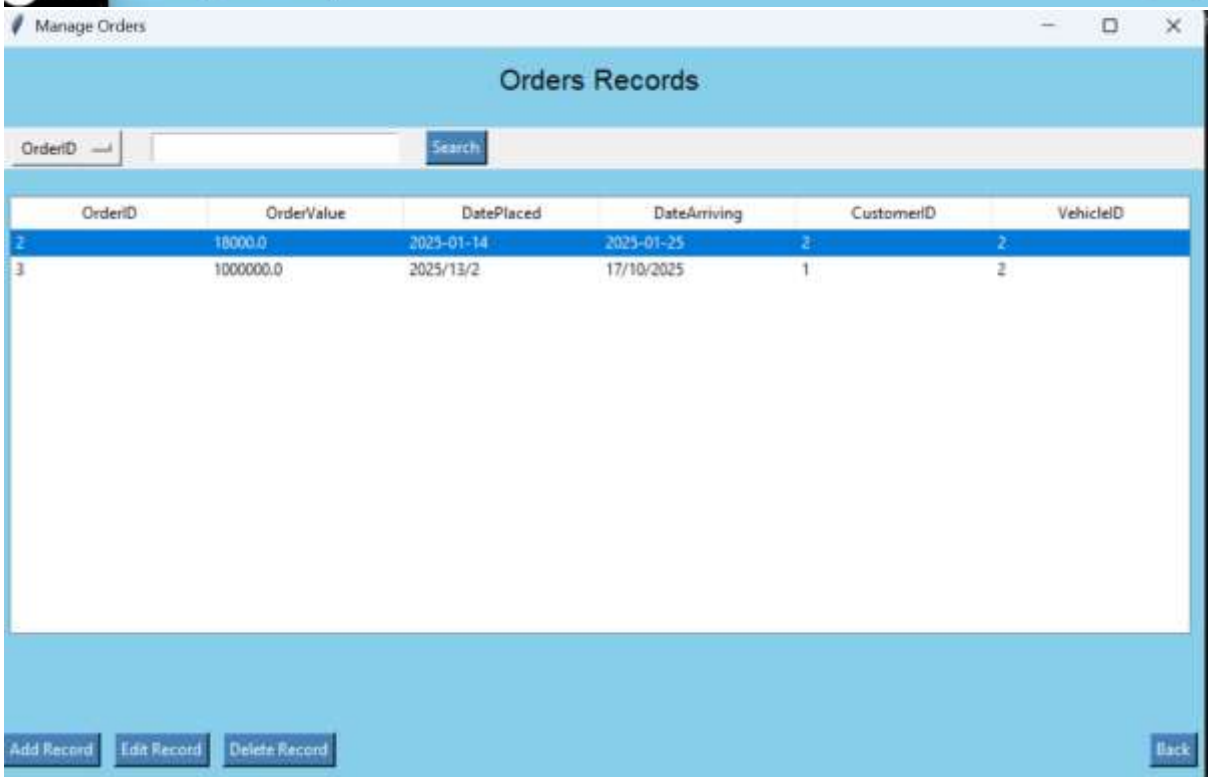
TEST 31


Edit Orders Record
OrderValue
10000.0
DatePlaced
15/10/2025
DateArriving
17/10/2025
CustomerID
1
VehicleID
2
Save Changes


Manage Orders
Orders Records
OrderID
Search
OrderID OrderValue DatePlaced DateArriving CustomerID VehicleID
1 20000.0 2025-01-12 2025-01-20 1 1
2 18000.0 2025-01-14 2025-01-25 2 2
3 10000.0 15/10/2025 17/10/2025 1 2
Add Record
Edit Record
Delete Record
Back

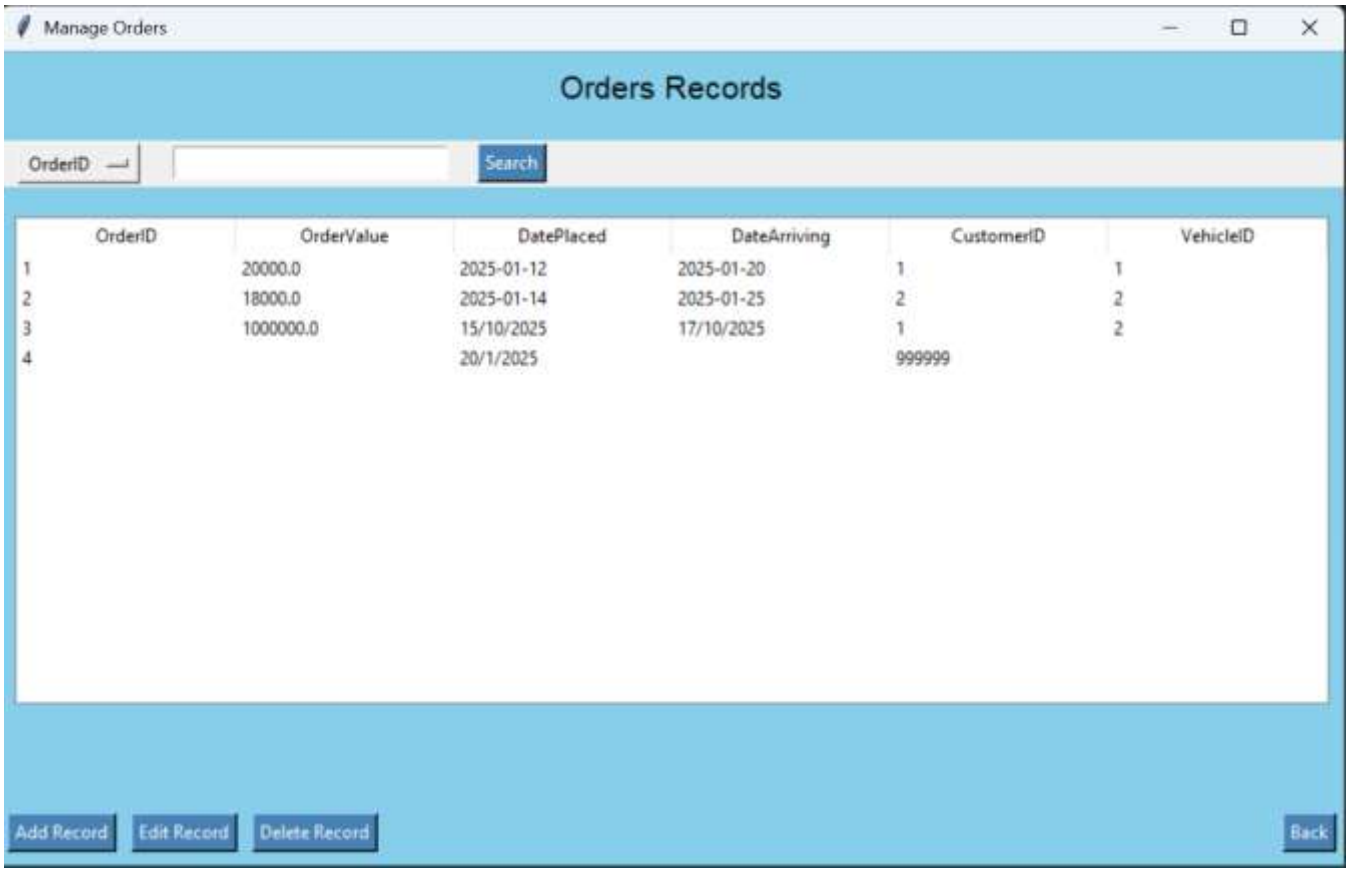
TEST 32


Manage Orders
Orders Records
OrderID
Search
OrderID
OrderValue
DatePlaced
DateArriving
CustomerID
VehicleID
2
18000.0
2025-01-14
2025-01-25
2
2
3
1000000.0
2025/13/2
17/10/2025
1
2
Edit Record
Delete Record
Back

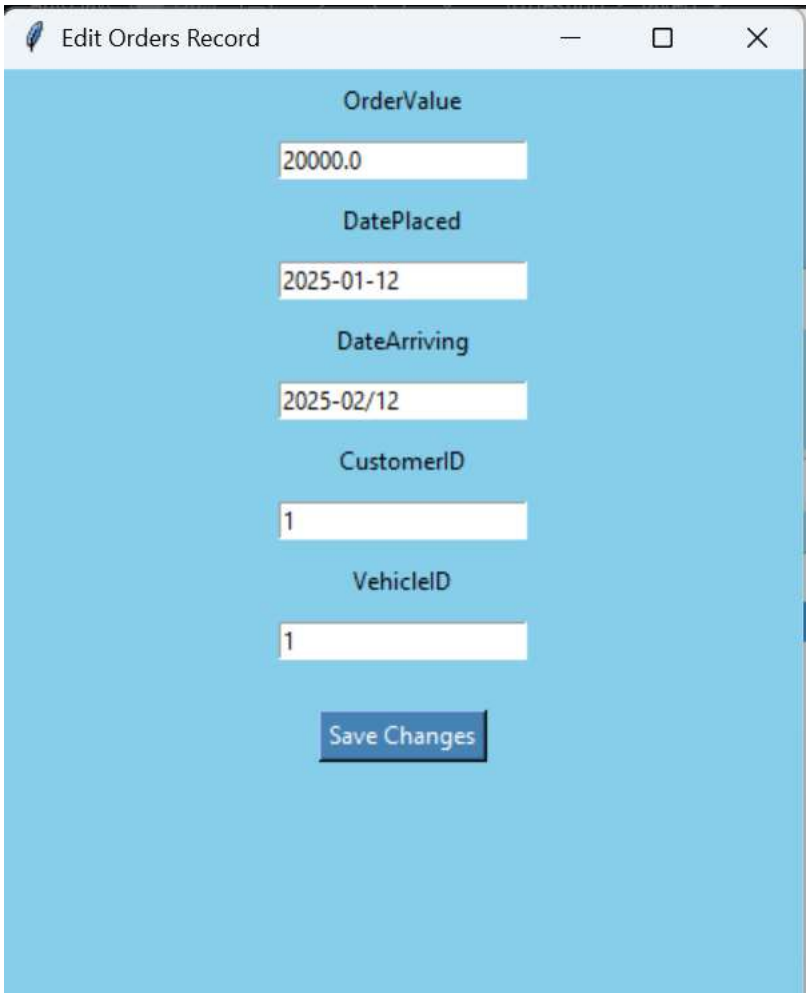

Manage Orders
Orders Records
OrderID
Search
OrderID
OrderValue
DatePlaced
DateArriving
CustomerID
VehicleID
2
18000.0
2025-01-14
2025-01-25
2
2
3
1000000.0
2025/13/2
17/10/2025
1
2
Add Record
Edit Record
Delete Record
Back

TEST 33


Edit Orders Record
OrderValue
1000000
DatePlaced
15/10/2025
DateArriving
17/10/2025
CustomerID
1
VehicleID
2
Save Changes

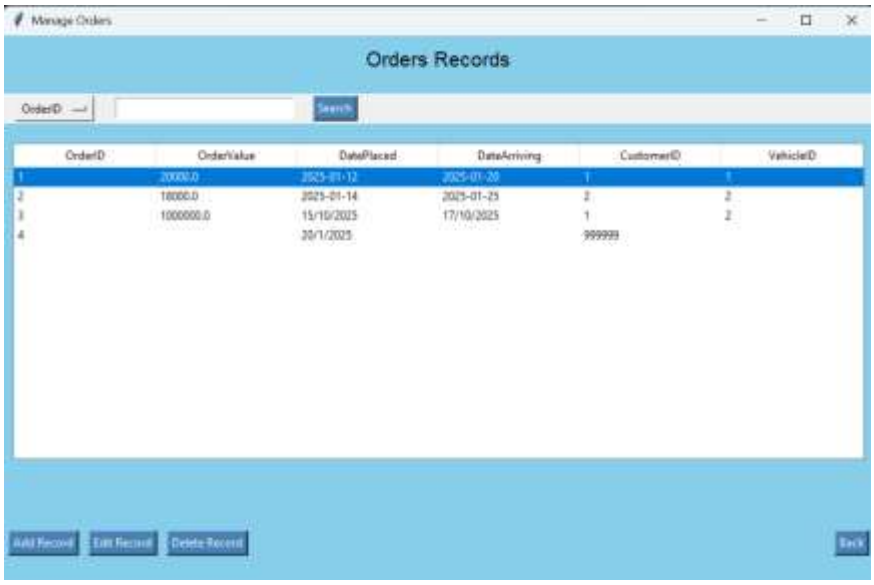

Manage Orders
Orders Records
OrderID
Search
OrderID OrderValue DatePlaced DateArriving CustomerID VehicleID
1 20000.0 2025-01-12 2025-01-20 1 1
2 18000.0 2025-01-14 2025-01-25 2 2
3 1000000.0 15/10/2025 17/10/2025 1 2
4 20/1/2025 999999
Add Record
Edit Record
Delete Record
Back

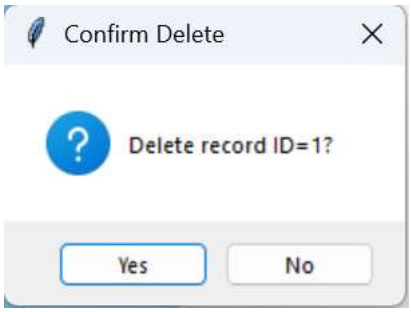
TEST 34


Edit Orders Record
OrderValue
20000.0
DatePlaced
2025-01-12
DateArriving
2025-02/12
CustomerID
1
VehicleID
1
Save Changes


Manage Orders
Orders Records
OrderID
Search
OrderID OrderValue DatePlaced DateArriving CustomerID VehicleID
1 20000.0 2025-01-12 2025-01-20 1 1
2 18000.0 2025-01-14 2025-01-25 2 2
3 1000000.0 15/10/2025 17/10/2025 1 2
4 20/1/2025 999999

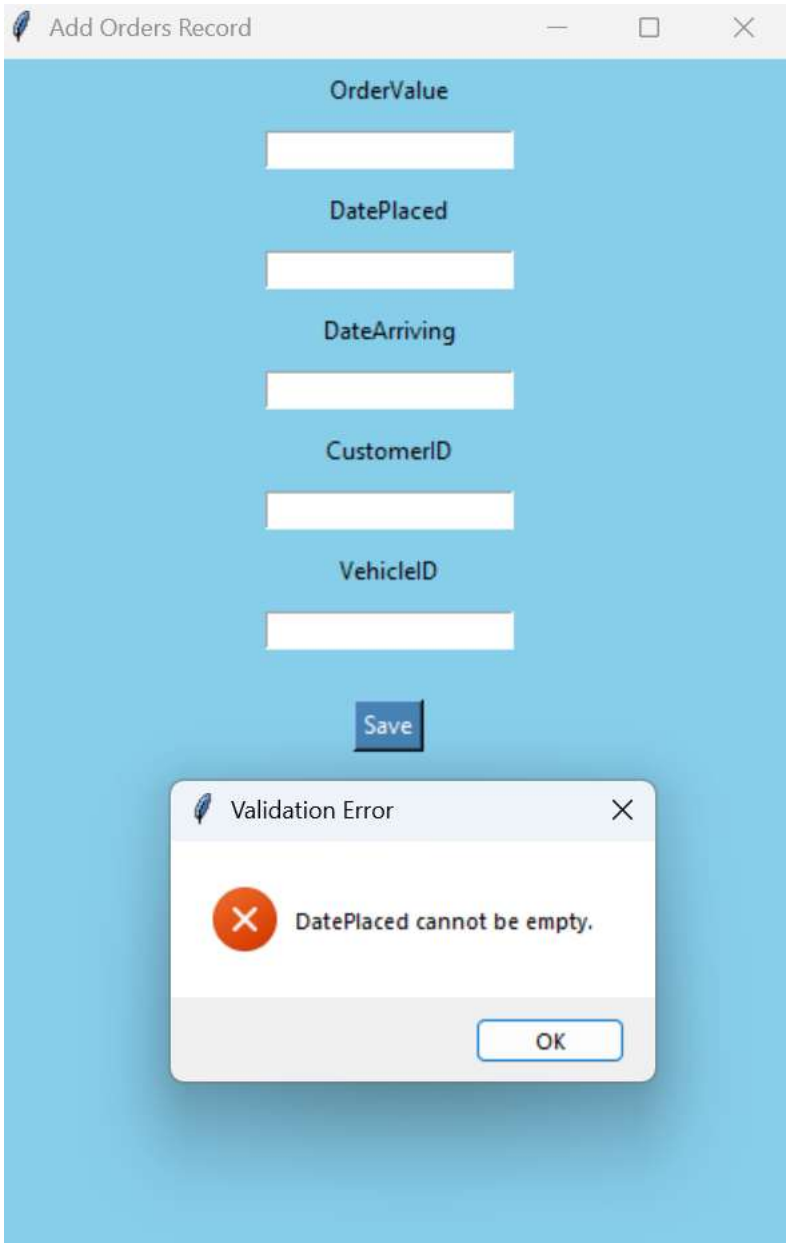
TEST 35


Manage Orders
Orders Records
OrderID
Search
OrderID OrderValue DatePlaced DateArriving CustomerID VehicleID
1 20000.0 2025-01-12 2025-01-20 1 1
2 18000.0 2025-01-14 2025-01-25 2 2
3 1000000.0 15/10/2025 17/10/2025 1 2
4 20/1/2025 999999
Add Record
Edit Record
Delete Record
Back


Confirm Delete
Delete record ID=1?
Yes
No

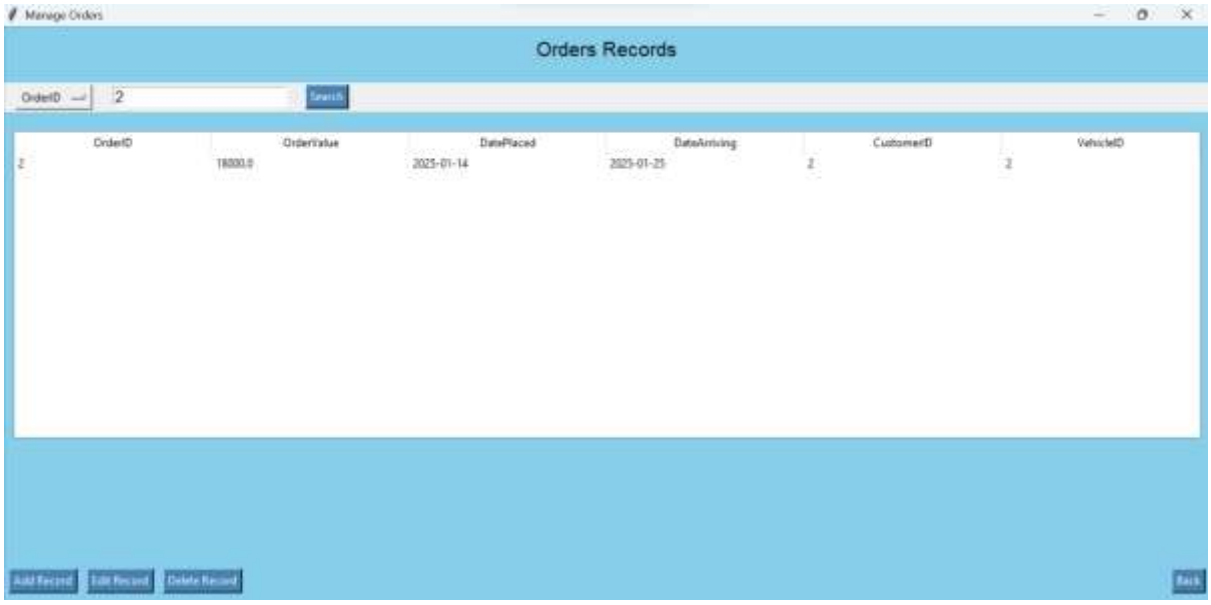

Manage Orders
Orders Records
OrderID
Search
OrderID OrderValue DatePlaced DateArriving CustomerID VehicleID
2 18000.0 2025-01-14 2025-01-25 2 2
3 1000000.0 15/10/2025 17/10/2025 1 2
4 20/1/2025 999999
5 412412.0 tommorow 124 124 1241
Add Record
Edit Record
Delete Record
Back

TEST 36

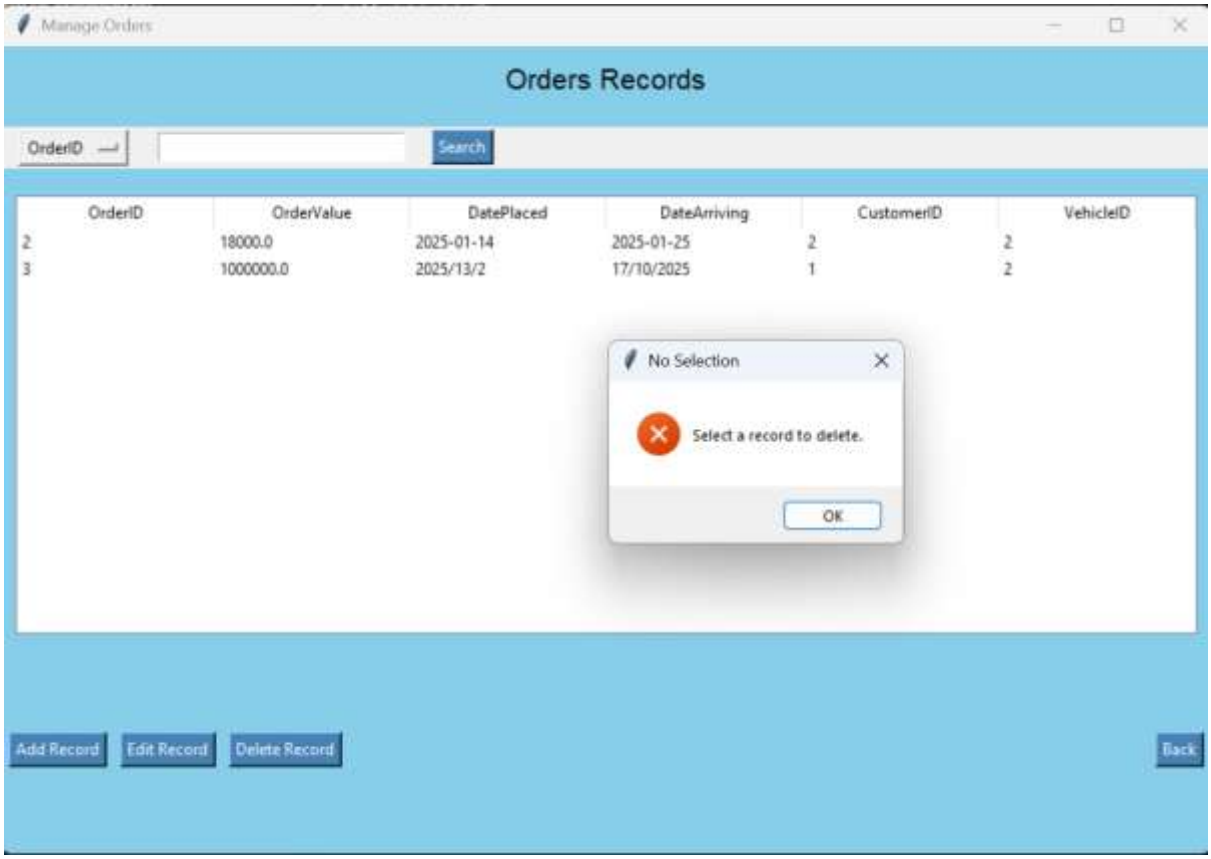

Add Orders Record
OrderValue
DatePlaced
DateArriving
CustomerID
VehicleID
Save
Validation Error
DatePlaced cannot be empty.
OK

TEST 37


Manage Orders
Orders Records
OrderID
Search
OrderID
OrderValue
DatePlaced
DateArriving
CustomerID
VehicleID
2
18000.0
2025-01-14
2025-01-25
2
2
3
1000000.0
15/10/2025
17/10/2025
1
2
Add Record
Edit Record
Delete Record
Back

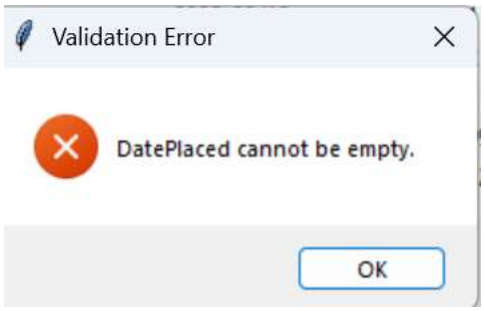

Manage Orders
Orders Records
OrderID
2
Search
OrderID
OrderValue
DatePlaced
DateArriving
CustomerID
VehicleID
2
18000.0
2025-01-14
2025-01-25
2
2
Add Record
Edit Record
Delete Record
Back

TEST 38

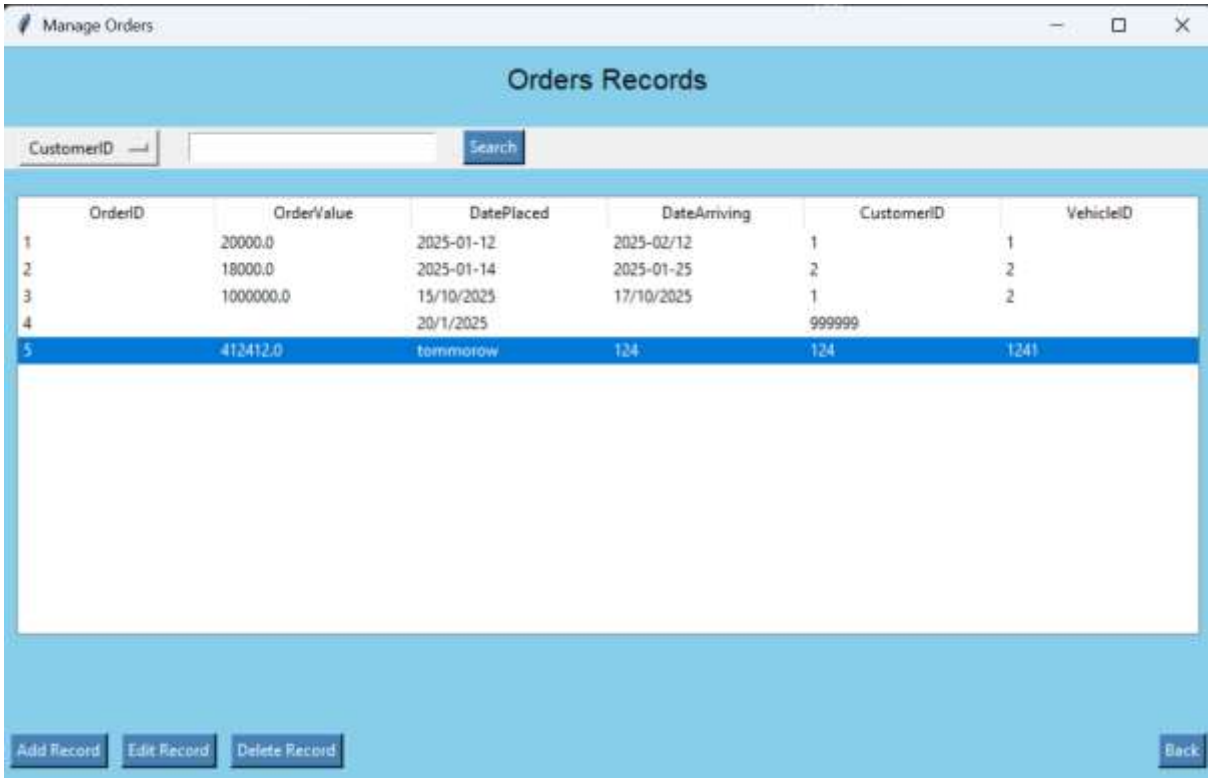

Manage Orders
Orders Records
OrderID
Search
OrderID
OrderValue
DatePlaced
DateArriving
CustomerID
VehicleID
2
18000.0
2025-01-14
2025-01-25
2
2
3
1000000.0
2025/13/2
17/10/2025
1
2
No Selection
Select a record to delete.
OK
Add Record
Edit Record
Delete Record
Back

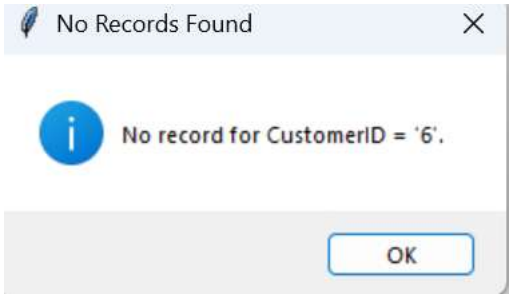
TEST 39


Add Orders Record
OrderValue
1
DatePlaced
DateArriving
10/10/2022
CustomerID
1
VehicleID
1
Save


Validation Error
DatePlaced cannot be empty.
OK

TEST 40


Manage Orders
Orders Records
CustomerID
Search
OrderID
OrderValue
DatePlaced
DateArriving
CustomerID
VehicleID
1 20000.0 2025-01-12 2025-02/12 1 1
2 18000.0 2025-01-14 2025-01-25 2 2
3 1000000.0 15/10/2025 17/10/2025 1 2
4 20/1/2025 999999
5 412412.0 tommorow 124 124 1241
Add Record
Edit Record
Delete Record
Back

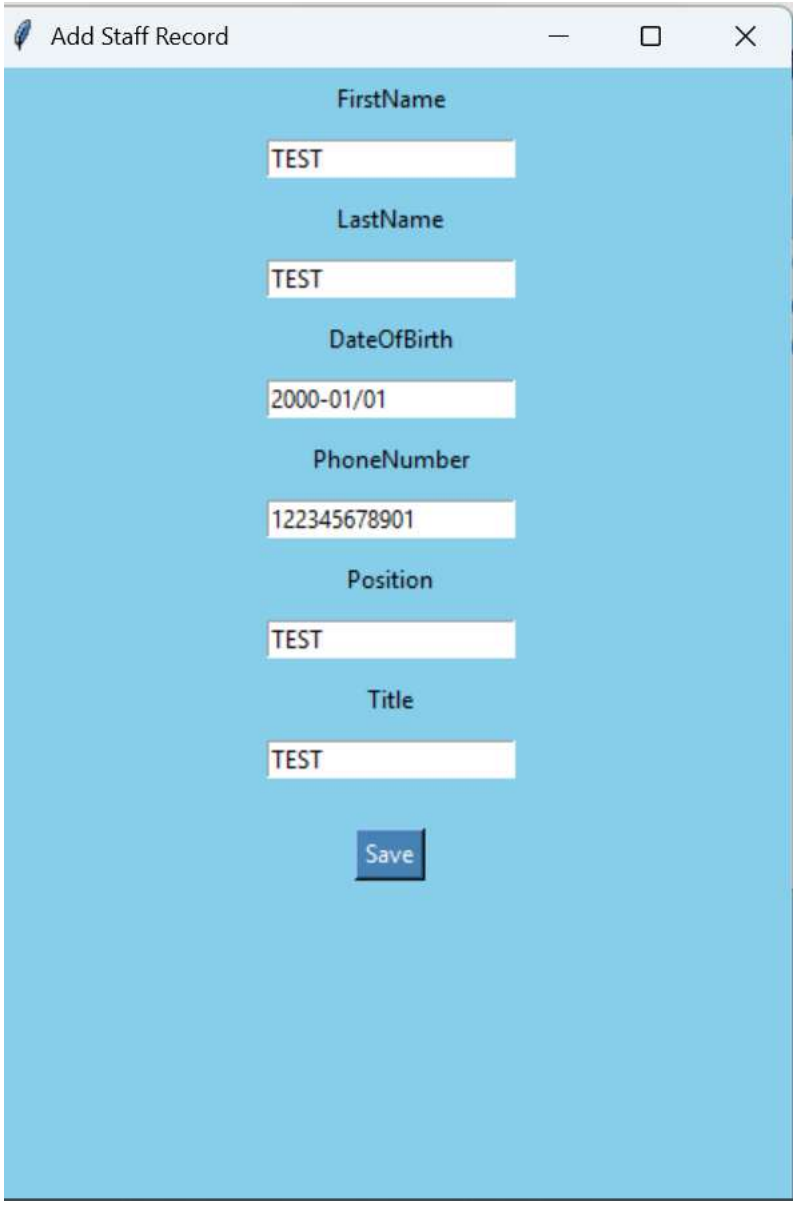

No Records Found
No record for CustomerID = '6'.
OK

5. Staff Management (10 Tests)

Form: manageTable.py (Staff)

TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
41	Add Staff: Valid DOB (1990-01-01)	DOB=1990-01-01	Functional	T	Record added.	SUCCESS
42	Close the form using the back button	N/A	Functional	T	Form is closed to main menu.	SUCCESS
43	Edit Staff: Change Position	Position=Senior Sales	Functional	T	Record updated.	SUCCESS
44	Delete Staff: Valid ID	StaffID=1	Functional	T	Record deleted	SUCCESS
45	Add Staff: Empty FirstName	FirstName=""	Negative	I	Error: "FirstName required."	SUCCESS
46	Edit Staff: Invalid Phone (Letters)	Phone=abc	Negative	I	Error (if validated).	SUCCESS
47	Search Staff: By Position=Manager	Position=Manager	Functional	T	Matching records.	SUCCESS
48	Add Staff: long last name	Last name = “A” *30	Boundary	E	Added.	SUCCESS
49	Delete Staff: Non-Existent ID	StaffID=999	Negative	I	Error: No record.	SUCCESS
50	Search Staff: Empty Query	N/A	Negative	E	Error, Select a query	SUCCESS

TEST 41


Add Staff Record
FirstName
TEST
LastName
TEST
DateOfBirth
2000-01/01
PhoneNumber
122345678901
Position
TEST
Title
TEST
Save

TEST 42


Manage Staff
Staff Records
StaffID
Search
StaffID FirstName LastName DateOfBirth PhoneNumber Position Title
2 Bob Williams 1990-03-02 1231243242 Manager Mr.
3 TEST TEST 2000-01/01 122345678901 TEST TEST
4 Dylan Convery 01/01/01 1432432 Sales MR
Add Record
Edit Record
Delete Record
Back

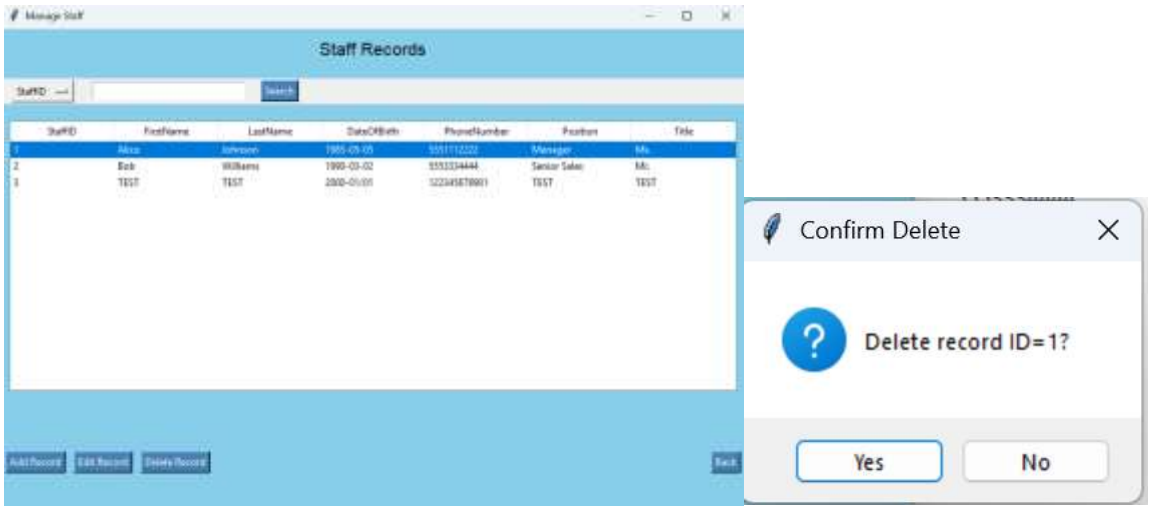

Mulligan Motors - Manager
M
Mulligan Motors Management (ADMIN)
Manage Vehicle Stock
Manage Customers
Manage Orders
Manage Staff
Manage Test Drives
Manage Access
Manage Payslips
Generate Report
Backup Database
Help
Log Off

TEST 43


Edit Staff Record
FirstName
Bob
LastName
Williams
DateOfBirth
1990-03-02
PhoneNumber
5553334444
Position
Senior Sales
Title
Mr.
Save Changes

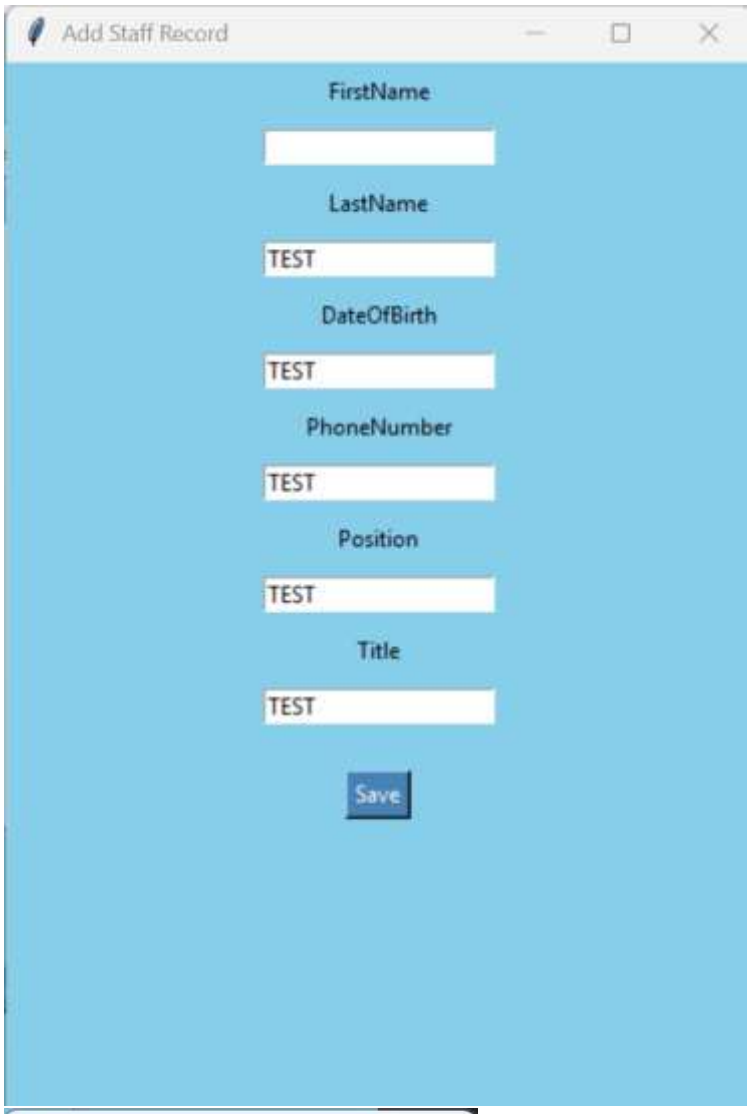

Manage Staff
Staff Records
StaffID
Search
StaffID FirstName LastName DateOfBirth PhoneNumber Position Title
1 Alice Johnson 1985-05-05 5551112222 Manager Ms.
2 Bob Williams 1990-03-02 5553334444 Senior Sales Mr.
3 TEST TEST 2000-01/01 122345678901 TEST TEST
Add Record
Edit Record
Delete Record
Back

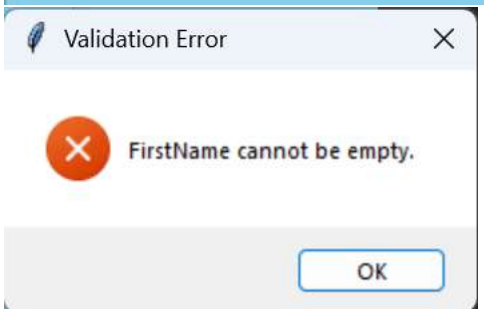
TEST 44


Staff Records
Search
Confirm Delete
Delete record ID=1?
Yes
No

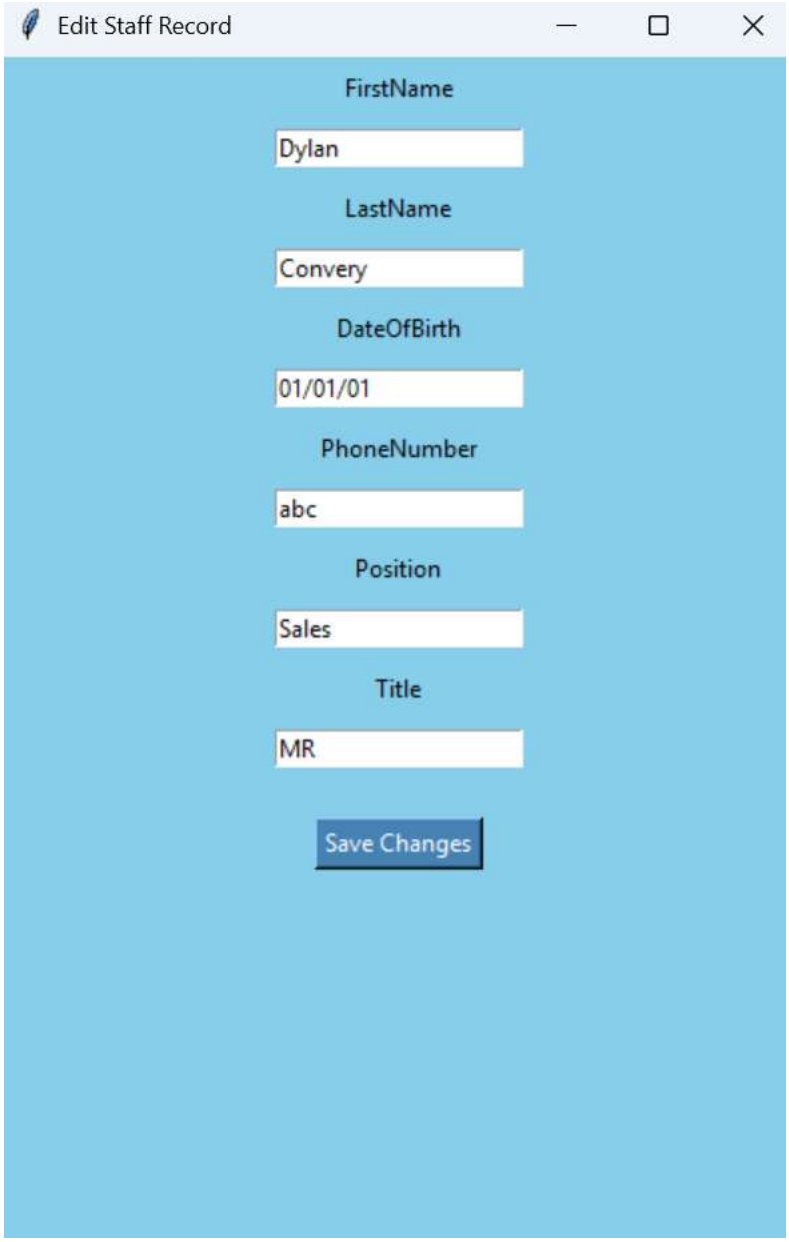

Manage Staff
Staff Records
StaffID
Search
StaffID FirstName LastName DateOfBirth PhoneNumber Position Title
2 Bob Williams 1990-03-02 5553334444 Senior Sales Mr.
3 TEST TEST 2000-01/01 122345678901 TEST TEST
Add Record
Edit Record
Delete Record
Back

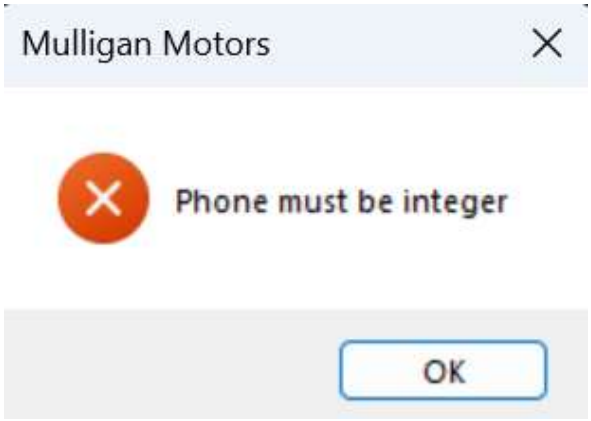
TEST 45


Add Staff Record
FirstName
LastName
TEST
DateOfBirth
TEST
PhoneNumber
TEST
Position
TEST
Title
TEST
Save

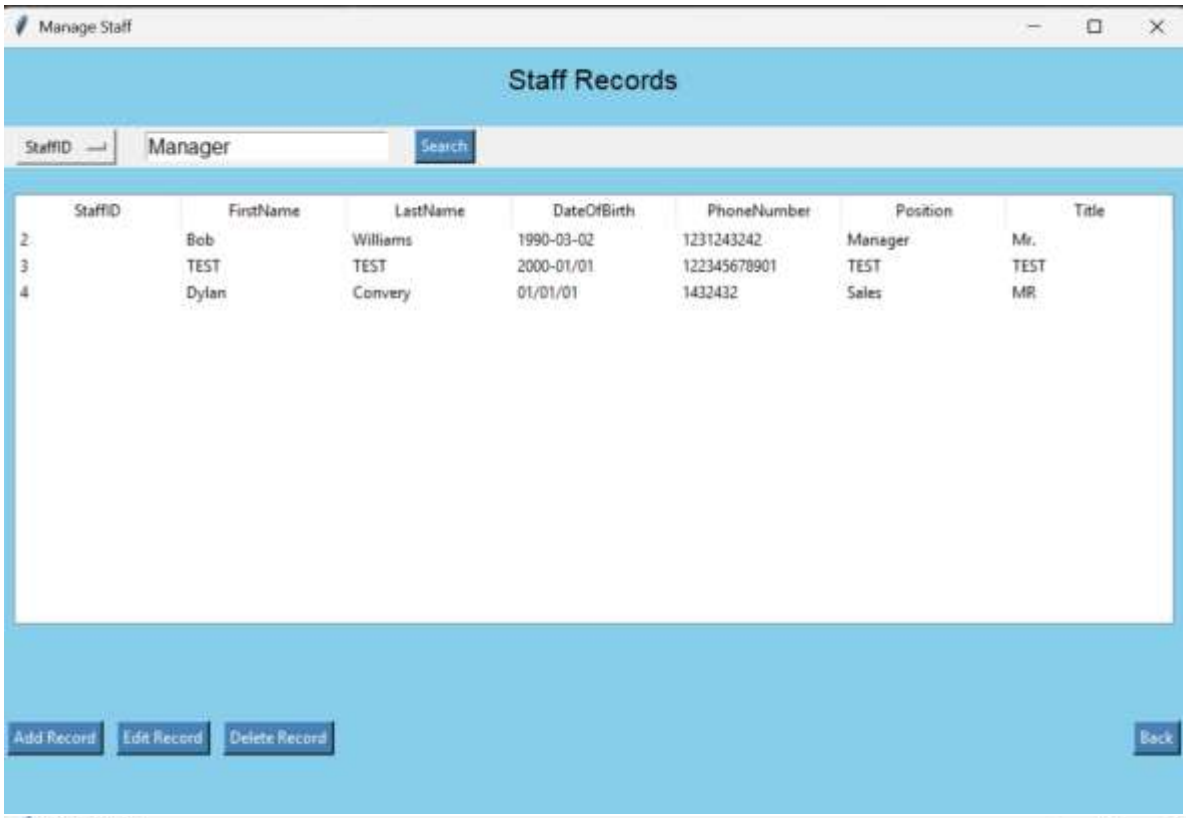

Validation Error
FirstName cannot be empty.
OK

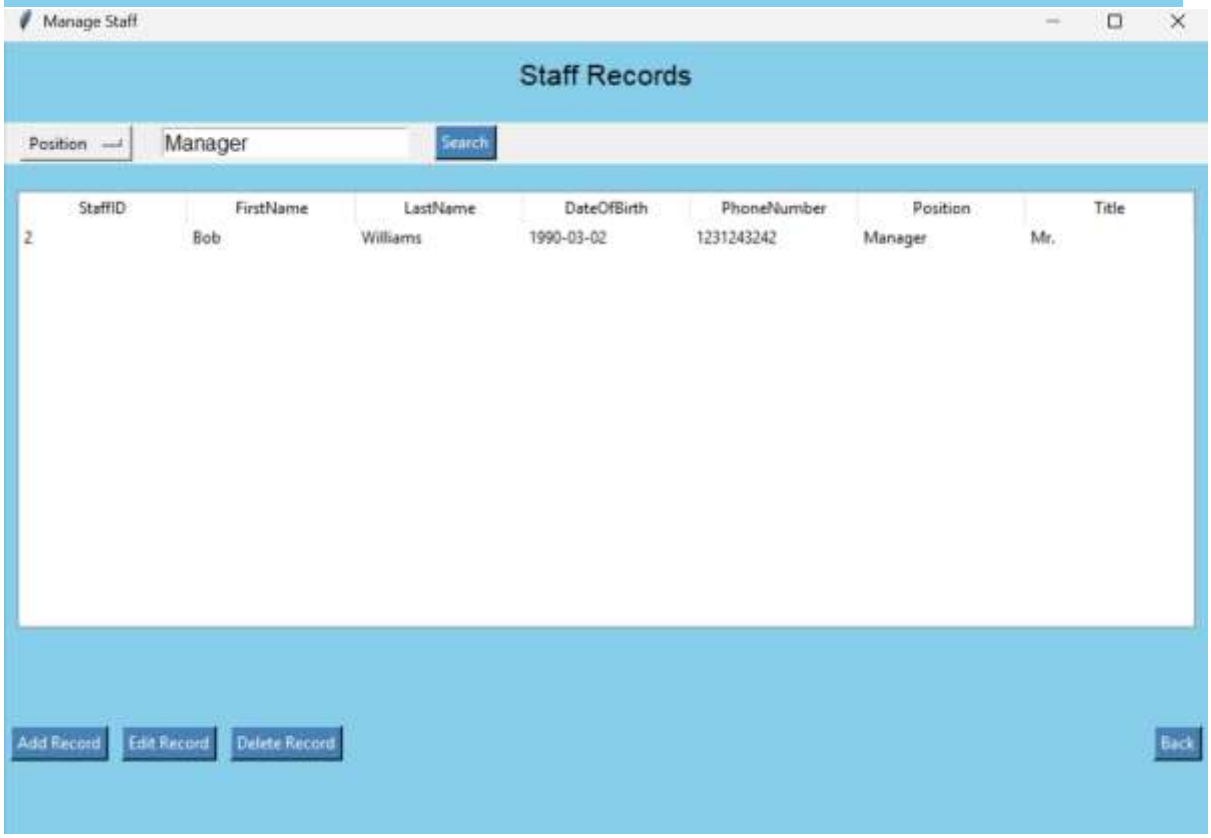
TEST 46


Edit Staff Record
FirstName
Dylan
LastName
Convery
DateOfBirth
01/01/01
PhoneNumber
abc
Position
Sales
Title
MR
Save Changes


Mulligan Motors
Phone must be integer
OK

TEST 47


Manage Staff
Staff Records
StaffID
Manager
Search
StaffID FirstName LastName DateOfBirth PhoneNumber Position Title
2 Bob Williams 1990-03-02 1231243242 Manager Mr.
3 TEST TEST 2000-01/01 122345678901 TEST TEST
4 Dylan Convery 01/01/01 1432432 Sales MR
Add Record
Edit Record
Delete Record
Back


Manage Staff
Staff Records
Position
Manager
Search
StaffID FirstName LastName DateOfBirth PhoneNumber Position Title
2 Bob Williams 1990-03-02 1231243242 Manager Mr.
Add Record
Edit Record
Delete Record
Back

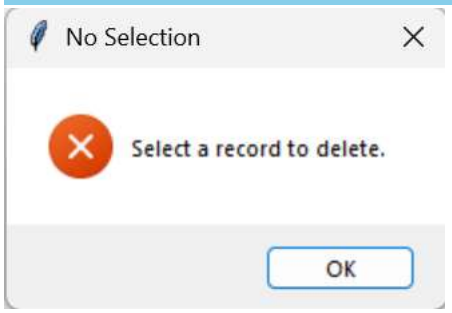
TEST 48


Edit Staff Record
FirstName
Dylan
LastName
aaaaaaaaaaaaaaaaaaa
DateOfBirth
01/01/01
PhoneNumber
1432432
Position
Sales
Title
MR
Save Changes

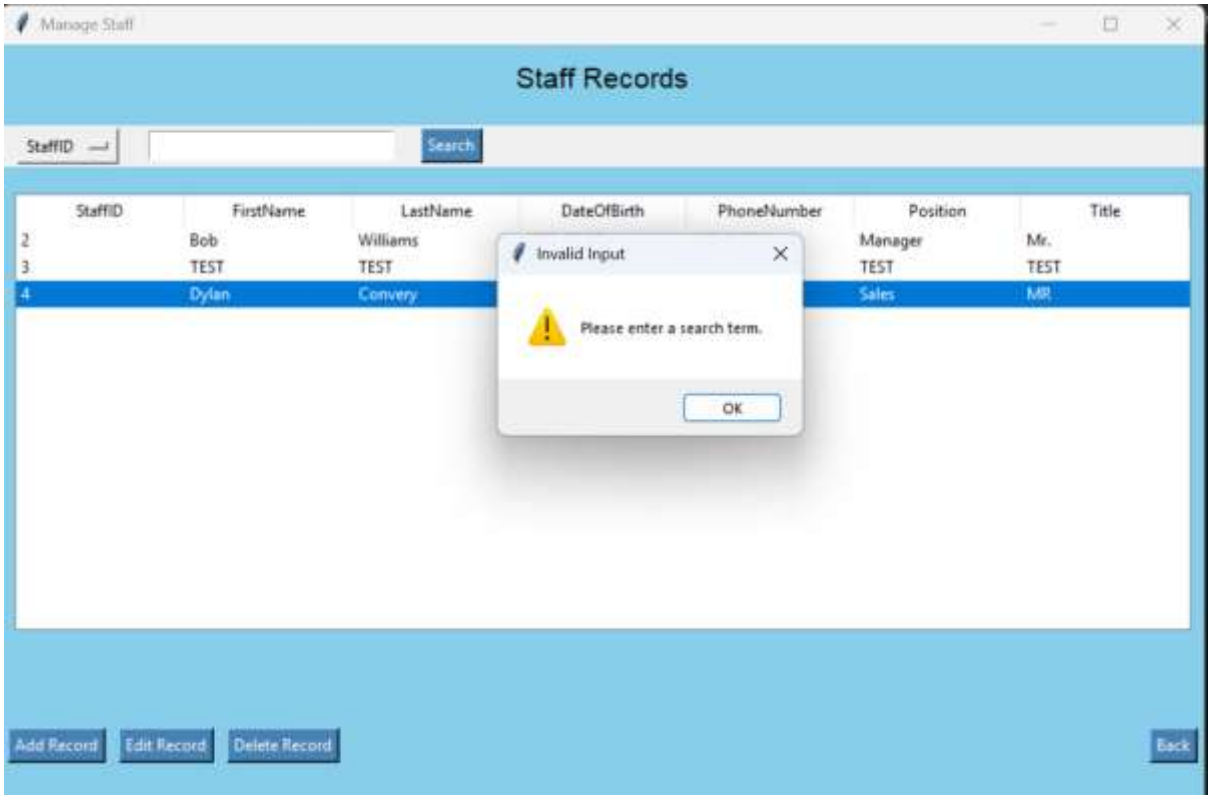

Manage Staff
Staff Records
StaffID
Search
StaffID FirstName LastName DateOfBirth PhoneNumber Position Title
2 Bob Williams 1990-03-02 1231243242 Manager Mr.
3 TEST TEST 2000-01/01 122345678901 TEST TEST
4 Dylan aaaaaaaaaaaaaaaaaa 01/01/01 1432432 Sales MR
Add Record
Edit Record
Delete Record
Back

TEST 49


Staff Records
StaffID
Search
StaffID
FirstName
LastName
DateOfBirth
PhoneNumber
Position
Title
2
Bob
Williams
1990-03-02
1231243242
Manager
Ms.
3
TEST
TEST
2000-01-01
12345678901
TEST
TEST
4
Dylan
Convery
01/01/01
1432432
Sales
MR
Add Record
Edit Record
Delete Record
Back


No Selection
Select a record to delete.
OK

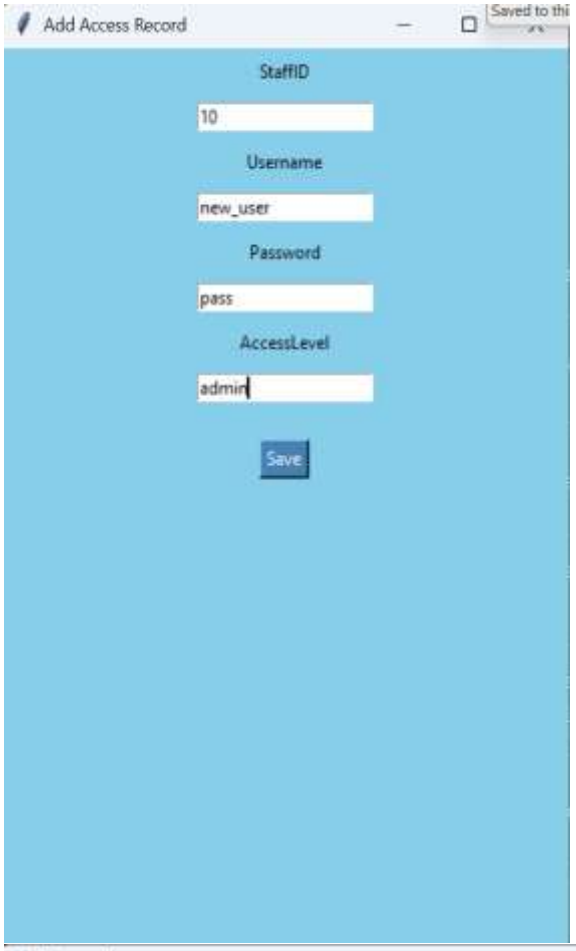
TEST 50

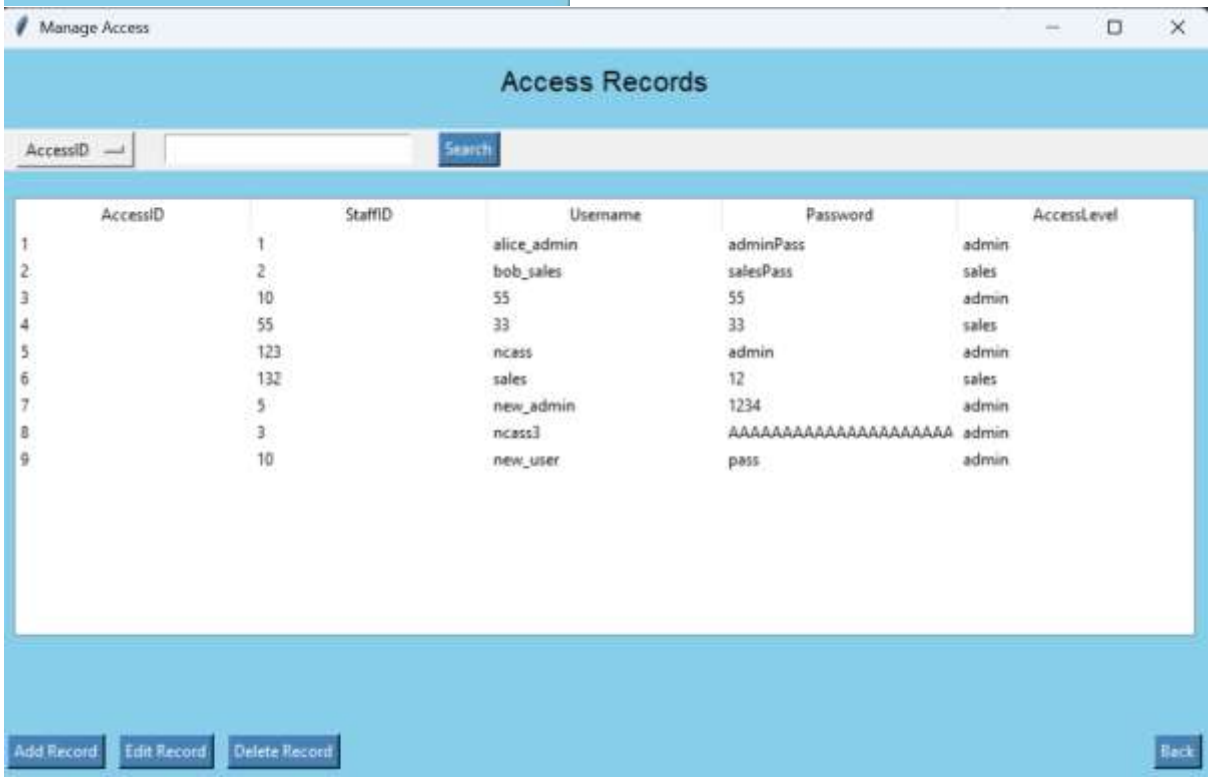

Manage Staff
Staff Records
StaffID
Search
StaffID
FirstName
LastName
DateOfBirth
PhoneNumber
Position
Title
2
Bob
Williams
Manager
Mr.
3
TEST
TEST
TEST
TEST
4
Dylan
Convery
Sales
MR
Invalid Input
Please enter a search term.
OK
Add Record
Edit Record
Delete Record
Back

Form: manageTable.py (Access)

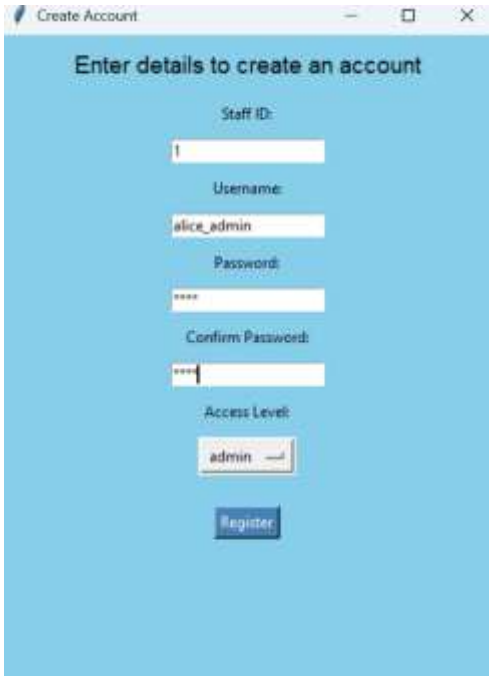
TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
51	Add Access: Valid Data	Username=new_user, Password=pass	Functional	T	Record added.	SUCCESS
52	Add Access: Duplicate Username	Username=alice_admin	Negative	I	Error: "Username taken."	SUCCESS
53	Edit Access: Change Level to admin	AccessLevel=admin	Functional	T	Record updated.	SUCCESS
54	Delete Access: Valid ID	AccessID=1	Functional	T	Record deleted.	SUCCESS
55	Add Access: Empty Username	Username=""	Negative	I	Error: "Username required."	SUCCESS
56	Attempt to edit without selecting a record	N/A	Negative	I	Error: Select Record	SUCCESS
57	Search Access: By AccessLevel=sales	AccessLevel=sales	Functional	T	Matching records.	SUCCESS
58	Add Access: Extreme Password (30 chars)	Password=A" * 30	Boundary	E	Added	SUCCESS
59	Delete Access: Non-Existent ID	AccessID=999	Negative	I	Error: No record.	SUCCESS
60	Edit Access: Empty Password	Password=""	Negative	I	Error: "Password required."	SUCCESS

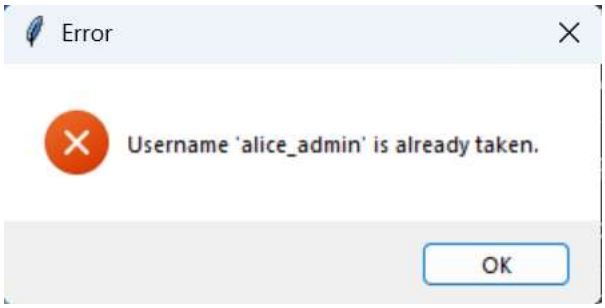
TEST 51


Add Access Record
Saved to thi
StaffID
10
Username
new_user
Password
pass
AccessLevel
admin
Save


Manage Access
Access Records
AccessID
Search
AccessID StaffID Username Password AccessLevel
1 1 alice_admin adminPass admin
2 2 bob_sales salesPass sales
3 10 55 55 admin
4 55 33 33 sales
5 123 ncass admin admin
6 132 sales 12 sales
7 5 new_admin 1234 admin
8 3 ncass3 AAAAAAAAAAAAAAAAAAAA admin
9 10 new_user pass admin
Add Record
Edit Record
Delete Record
Back

TEST 52


Create Account
Enter details to create an account
Staff ID:
1
Username:
alice_admin
Password:
Confirm Password:
Access Level:
admin
Register


Error
Username 'alice_admin' is already taken.
OK

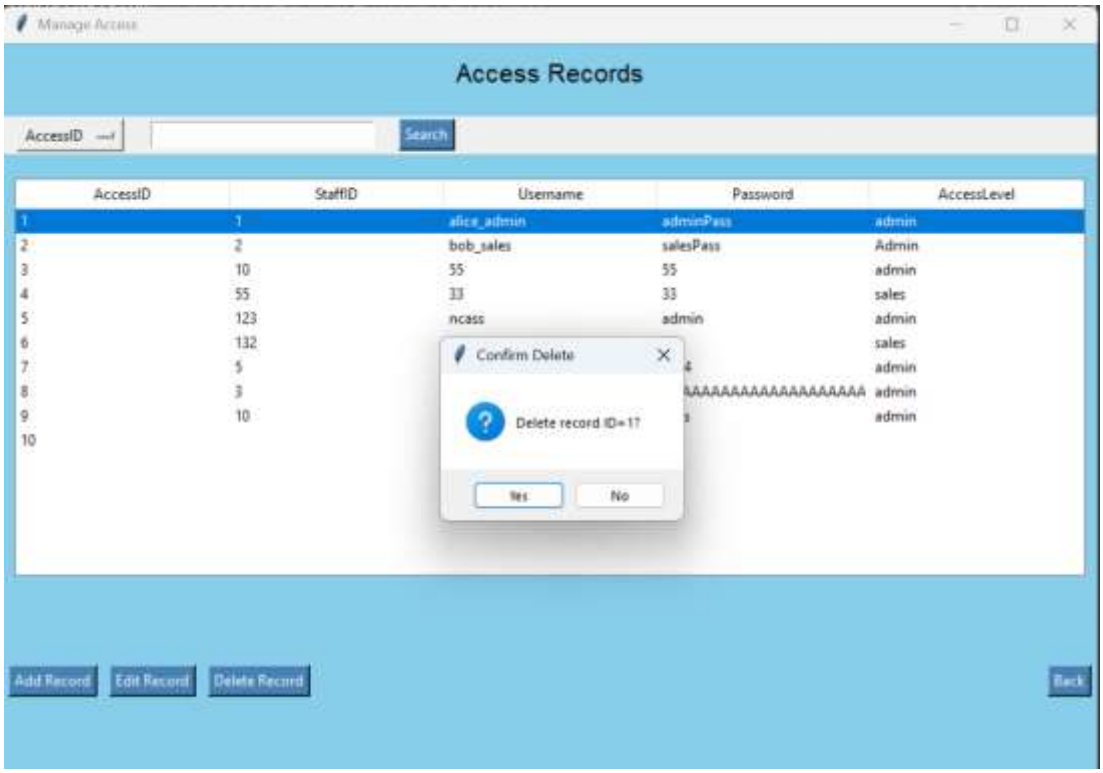
TEST 53





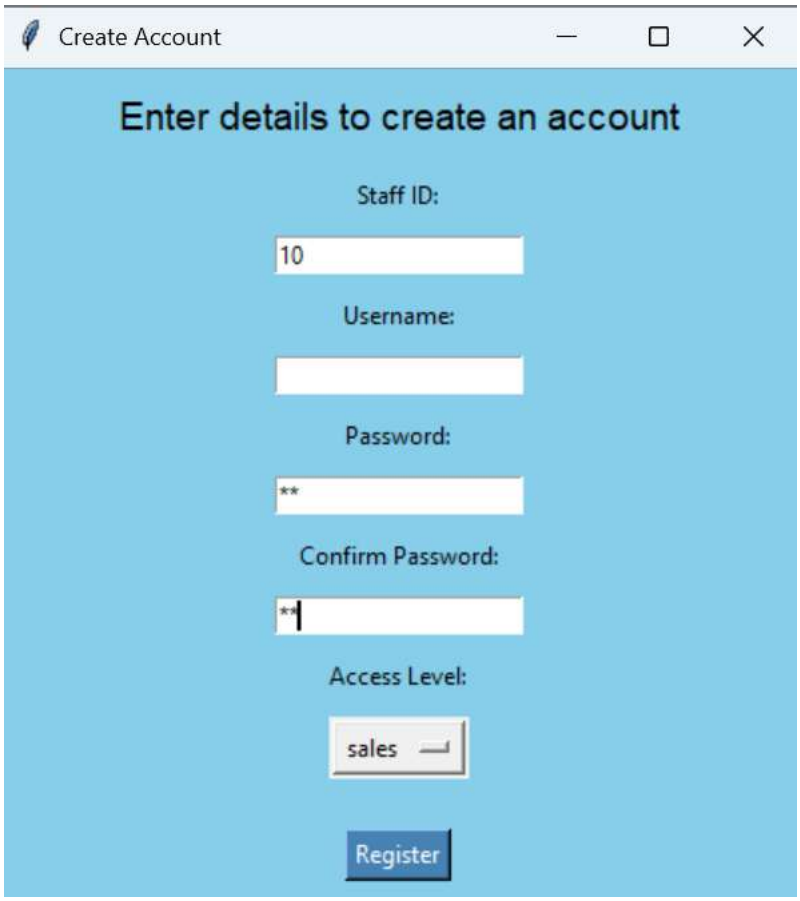
2	2	bob_sales	salesPass	Admin

TEST 54


Manage Access
Access Records
AccessID
Search
AccessID StaffID Username Password AccessLevel
1 1 alice_admin adminPass admin
2 2 bob_sales salesPass Admin
3 10 55 55 admin
4 55 33 33 sales
5 123 ncass admin admin
6 132 sales
7 5 admin
8 3 admin
9 10 admin
10
Confirm Delete
Delete record ID=1?
Yes
No
Add Record
Edit Record
Delete Record
Back


Manage Access
Access Records
AccessID
Search
AccessID StaffID Username Password AccessLevel
2 2 bob_sales salesPass Admin
3 10 55 55 admin
4 55 33 33 sales
5 123 ncass admin admin
6 132 sales 12 sales
7 5 new_admin 1234 admin
8 3 ncass3 AAAAAAAAAAAAAAAAAAAA admin
9 10 new_user pass admin
10 alice_admin
Add Record
Edit Record
Delete Record
Back

TEST 55


Create Account
Enter details to create an account
Staff ID:
10
Username:
Password:
**
Confirm Password:
**
Access Level:
sales
Register


Validation Error
All fields are required.
OK

TEST 56

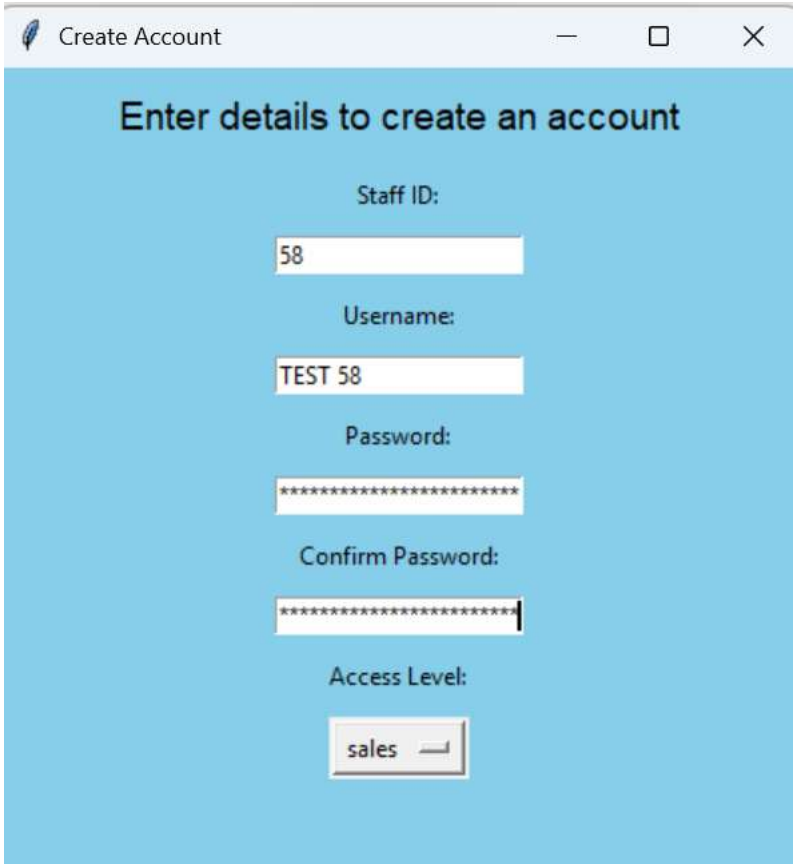

Manage Access
Access Records
AccessID
Search
AccessID StaffID Username Password AccessLevel
2 2 bob_sales salesPass Admin
3 10 55 55 admin
4 55 33 33 sales
5 123 ncass admin admin
6 132 sales sales
7 5 new_admin admin
8 3 ncass3 admin
9 10 new_user admin
10 alice_admin
11 55 test guest
No Selection
Select a record to edit.
OK
Add Record
Edit Record
Delete Record
Back

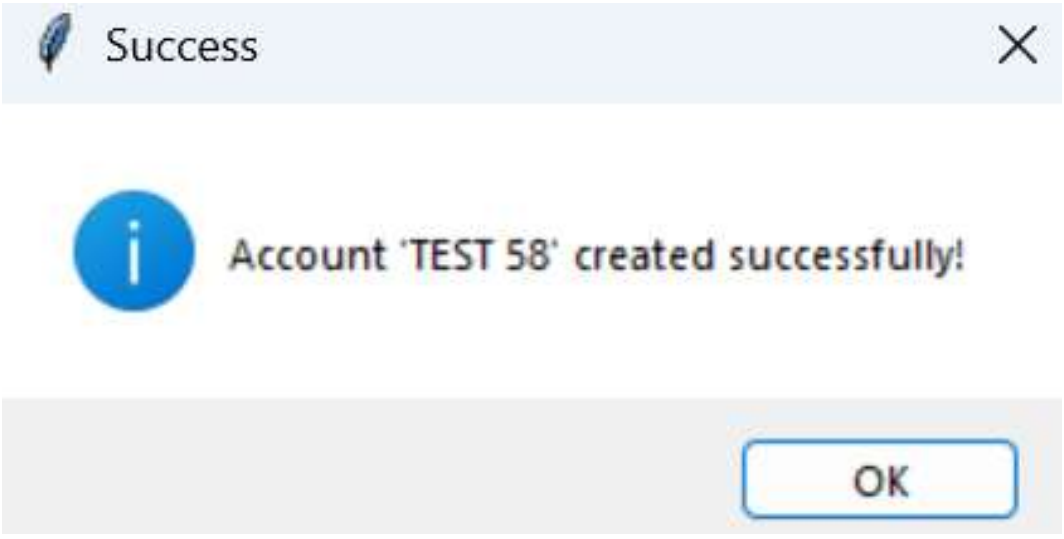
TEST 57


Manage Access
Access Records
AccessLevel
sales
Search
AccessID
StaffID
Username
Password
AccessLevel
2 2 bob_sales salesPass Admin
3 10 55 55 admin
4 55 33 33 sales
5 123 ncass admin admin
6 132 sales 12 sales
7 5 new_admin 1234 admin
8 3 ncass3 AAAAAAAAAAAAAAAAAAAA admin
9 10 new_user pass admin
10 alice_admin
Add Record
Edit Record
Delete Record
Back


Manage Access
Access Records
AccessLevel
sales
Search
AccessID
StaffID
Username
Password
AccessLevel
4 55 33 33 sales
6 132 sales 12 sales
Add Record
Edit Record
Delete Record
Back

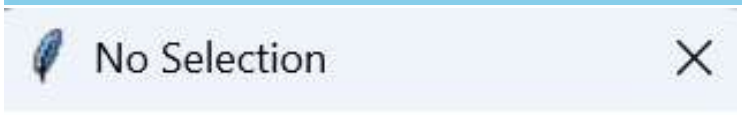
TEST 58

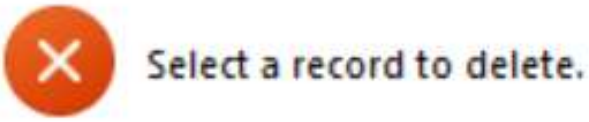

Create Account
Enter details to create an account
Staff ID:
58
Username:
TEST 58
Password:
Confirm Password:
Access Level:
sales


Success
Account 'TEST 58' created successfully!
OK

TEST 59

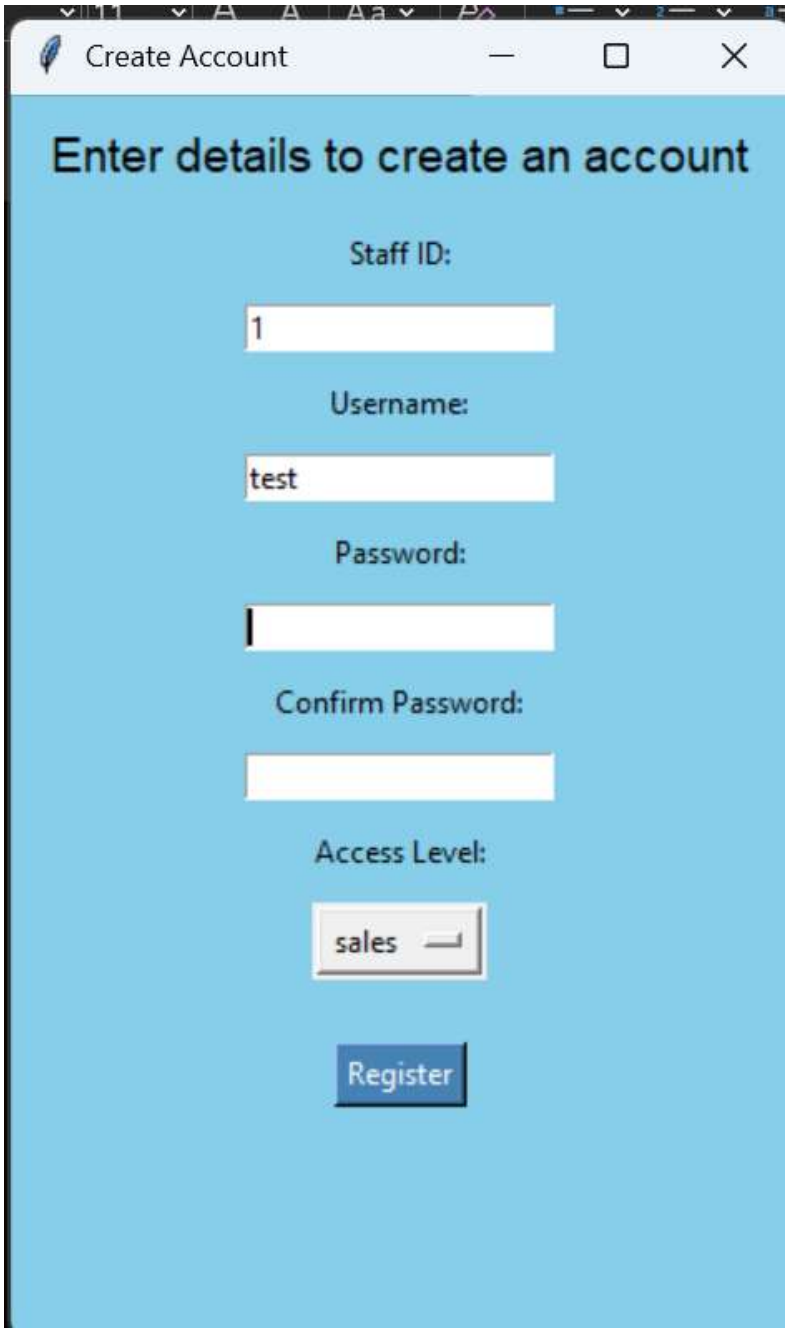

Manage Access
Access Records
AccessID
Search
AccessID StaffID Username Password AccessLevel
2 2 bob_sales salesPass Admin
3 10 55 55 admin
4 55 33 33 sales
5 123 ncass admin admin
6 132 sales 12 sales
7 5 new_admin 1234 admin
8 3 ncass3 AAAAAAAAAAAAAAAAAAAA admin
9 10 new_user pass admin
10 alice_admin
Add Record
Edit Record
Delete Record
Back

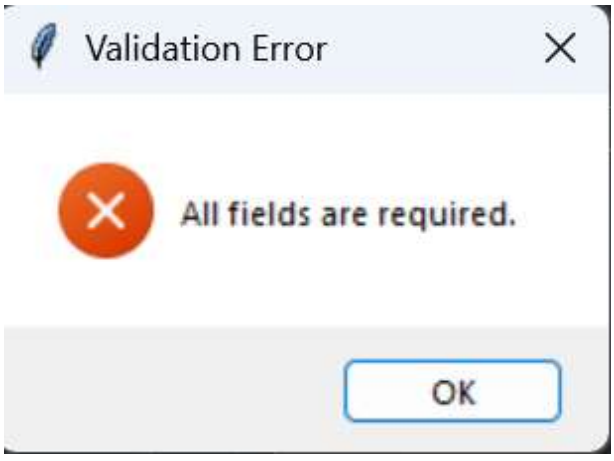

No Selection


Select a record to delete.


OK

TEST 60


Create Account
Enter details to create an account
Staff ID:
1
Username:
test
Password:
Confirm Password:
Access Level:
sales
Register

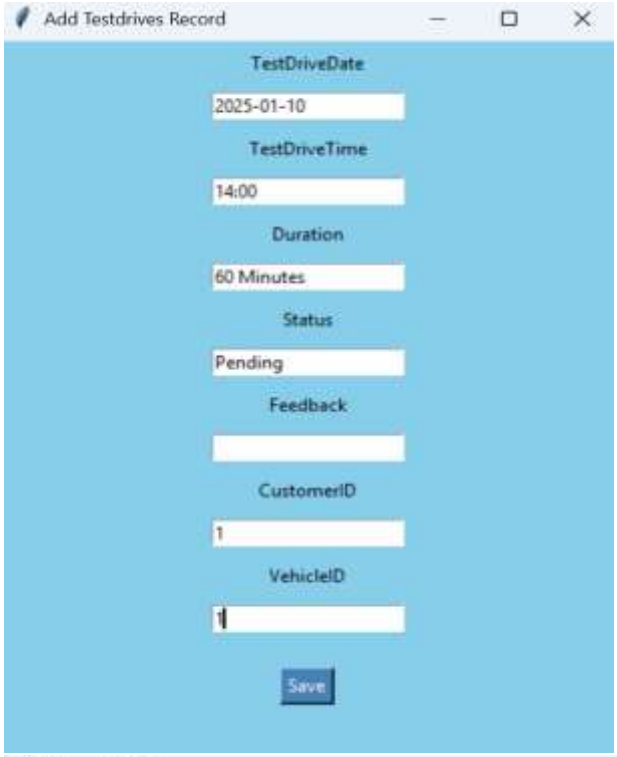

Validation Error
All fields are required.
OK

7. TestDrives Management (10 Tests)

Form: manageTable.py (TestDrives)

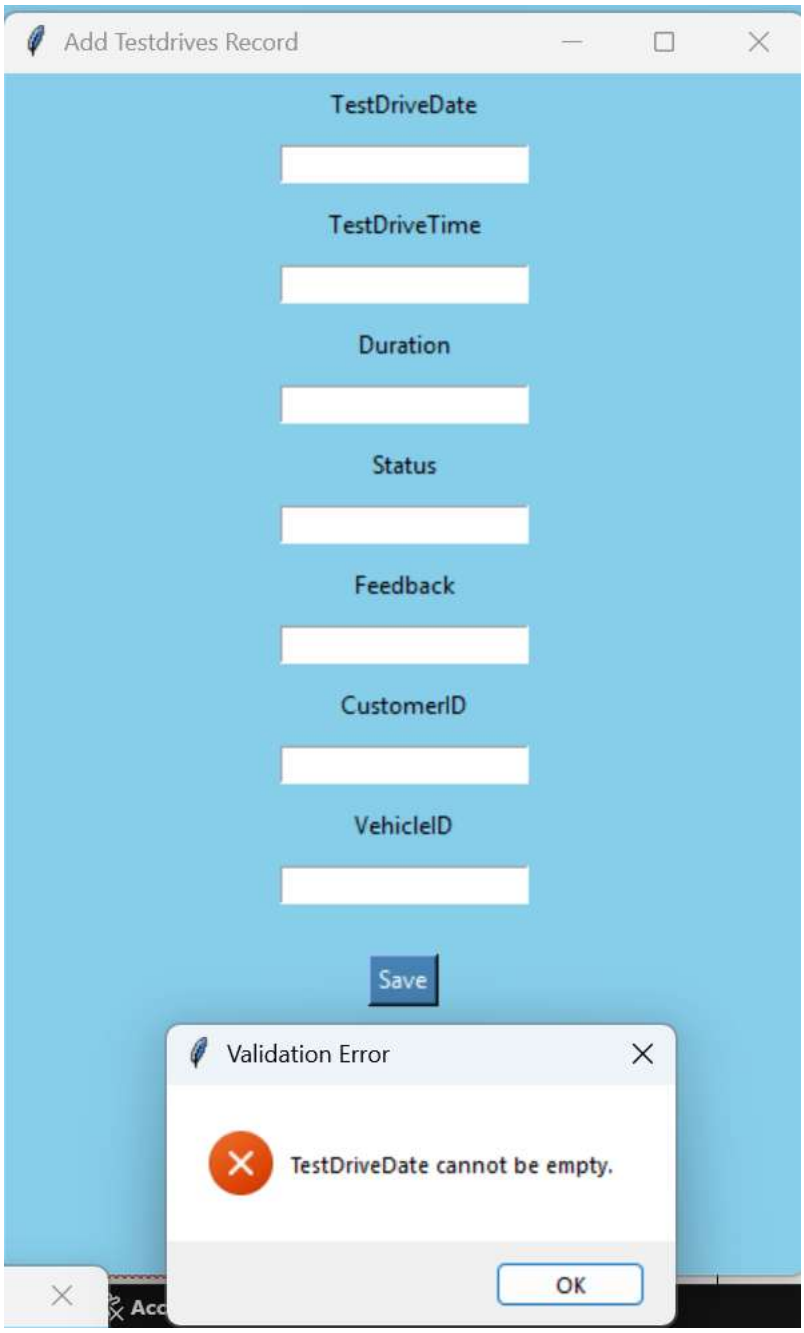
TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
61	Add TestDrive: Valid Data	Date=2025-03-01, Time=14:00	Functional	T	Record added.	SUCCESS
62	Add TestDrive: Blank form	N/A	Negative	I	Error: Please input info	SUCCESS
63	Edit TestDrive: Update Status	Status=Canceled	Functional	T	Record updated.	SUCCESS
64	Delete TestDrive: Valid ID	TestDriveID=1	Functional	T	Record deleted.	SUCCESS
65	Edit Feedback:	“ “ -> Fantastic	Negative	I		SUCCESS
66	Search TestDrive: By Status=Pending	Status=Pending	Functional	T	Matching records.	SUCCESS
67	Attempt to delete a record without selecting a record	N/A	Negative	I	Error (Select a record).	SUCCESS
68	Edit TestDrive: Empty Feedback	Feedback=""	Functional	T	Record updated.	SUCCESS
69	Search for non existent attribute.	e.g LastServiced	Negative	I	Drop down box should limit selection	SUCCESS
70	Add TestDrive: Extreme Duration (24h)	Duration=24 hours	Boundary	E	Added	SUCCESS

TEST 61

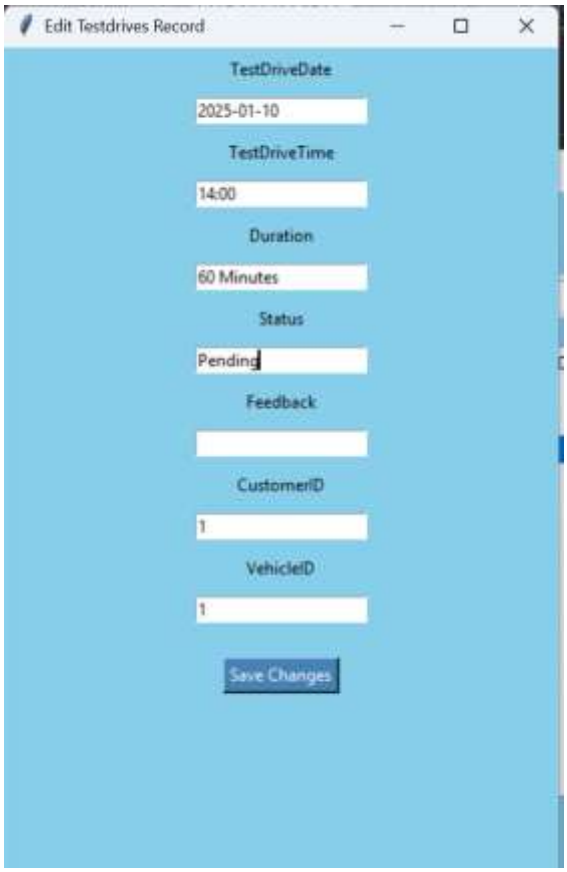

Add Testdrives Record
TestDriveDate
2025-01-10
TestDriveTime
14:00
Duration
60 Minutes
Status
Pending
Feedback
CustomerID
1
VehicleID
1
Save


Manage Testdrives
Testdrives Records
TestDriveID
Search
TestDriveID TestDriveDate TestDriveTime Duration Status Feedback CustomerID
1 2025-02-01 10:00 30 minutes Completed Loved it! 1 2
2 2025-02-05 14:00 45 minutes Pending 2 3
3 2025-01-10 14:00 60 Minutes Pending 1 1
Add Record
Edit Record
Delete Record
Back

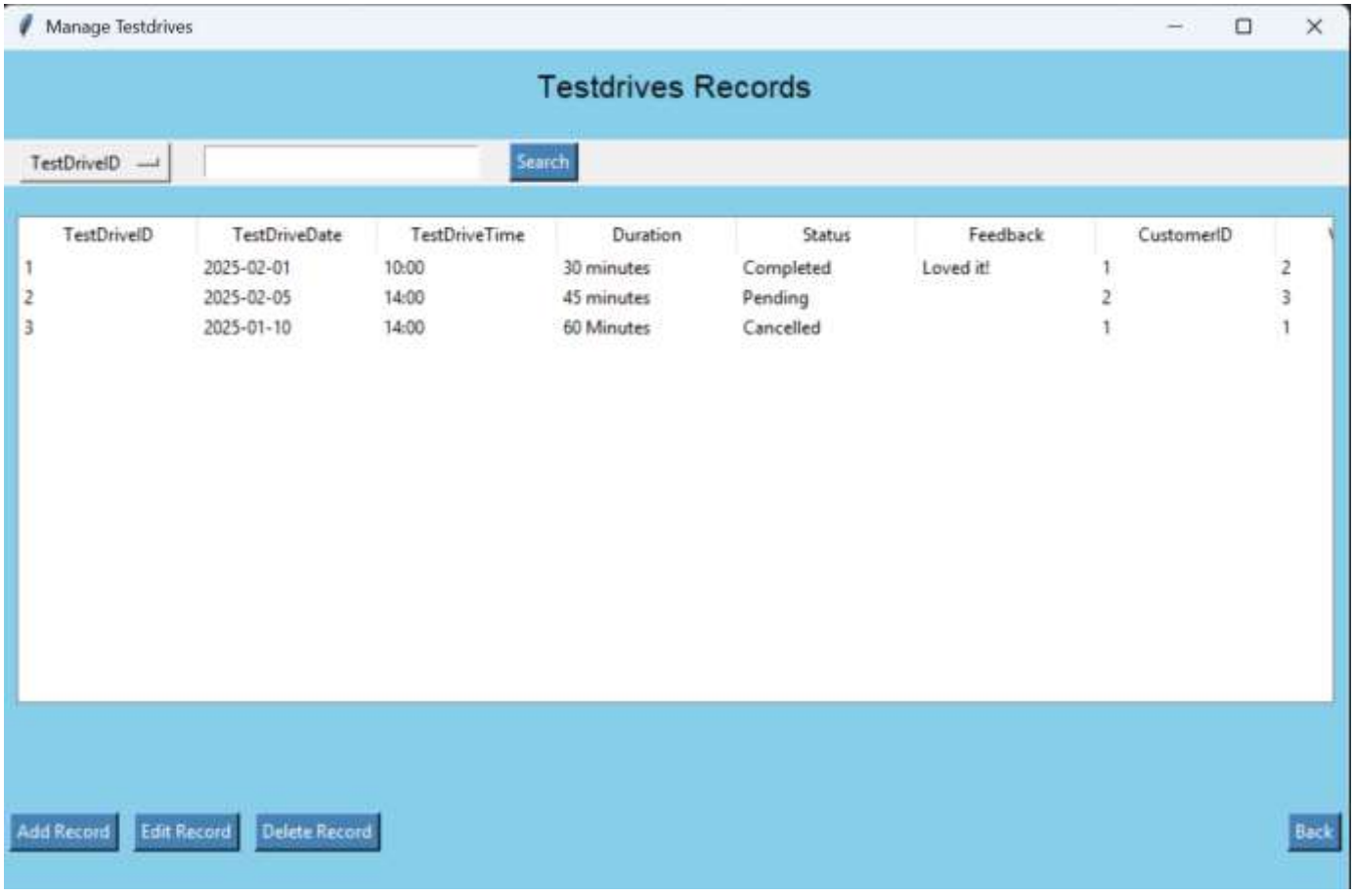
TEST 62


Add Testdrives Record
TestDriveDate
TestDriveTime
Duration
Status
Feedback
CustomerID
VehicleID
Save
Validation Error
TestDriveDate cannot be empty.
OK

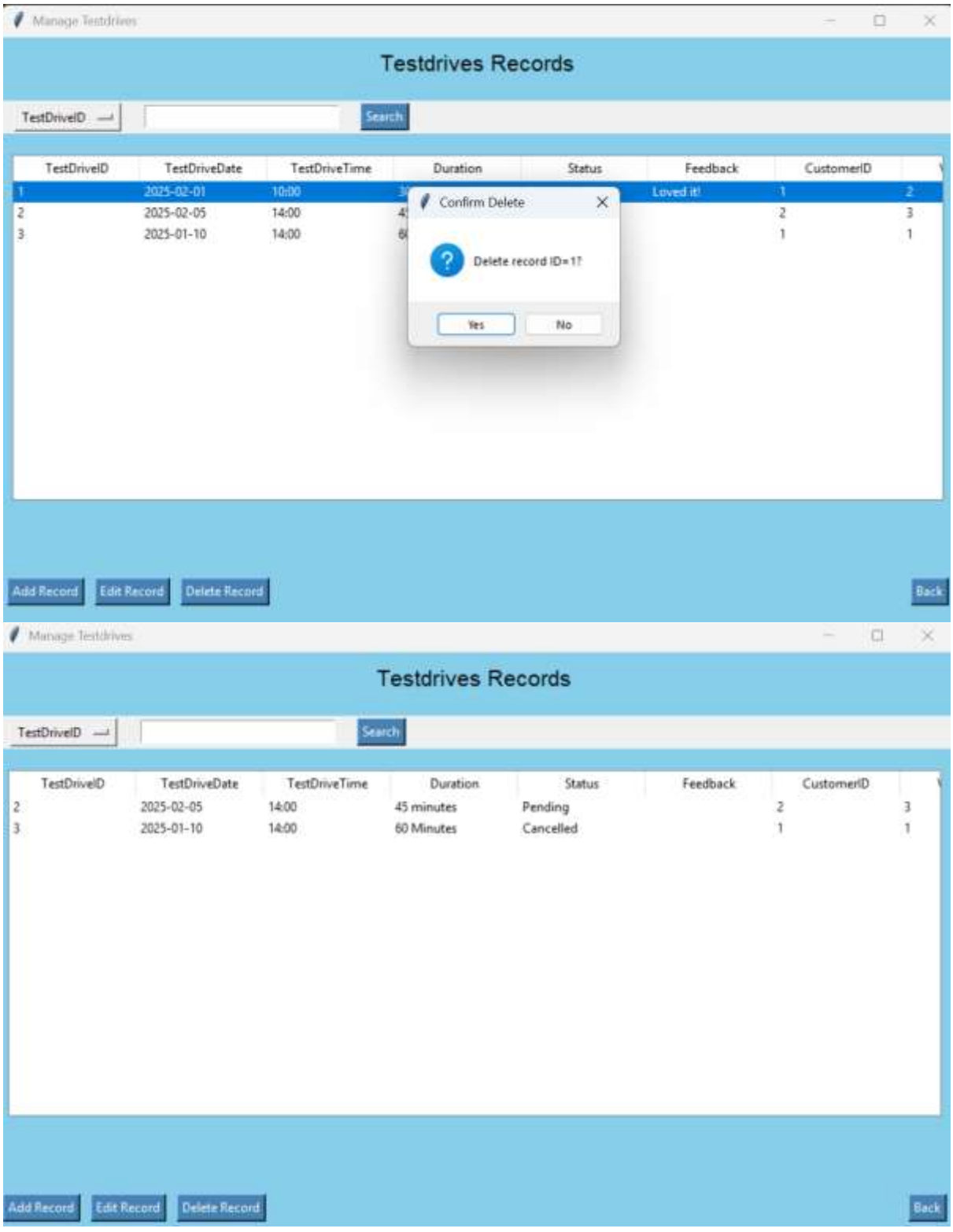
TEST 63


Edit Testdrives Record
TestDriveDate
2025-01-10
TestDriveTime
14:00
Duration
60 Minutes
Status
Pending
Feedback
CustomerID
1
VehicleID
1
Save Changes

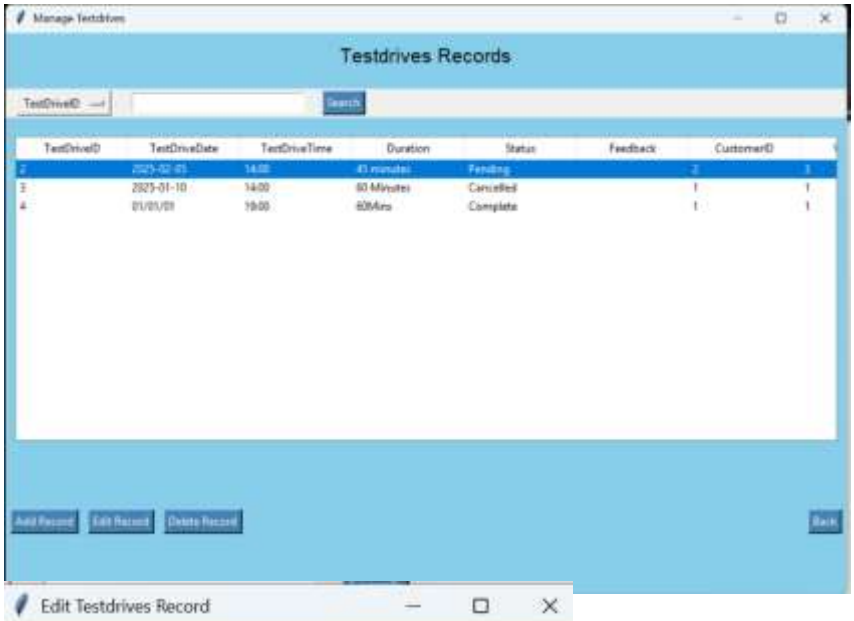

Edit Testdrives Record
TestDriveDate
2025-01-10
TestDriveTime
14:00
Duration
60 Minutes
Status
Cancelled
Feedback
CustomerID
1
VehicleID
1
Save Changes

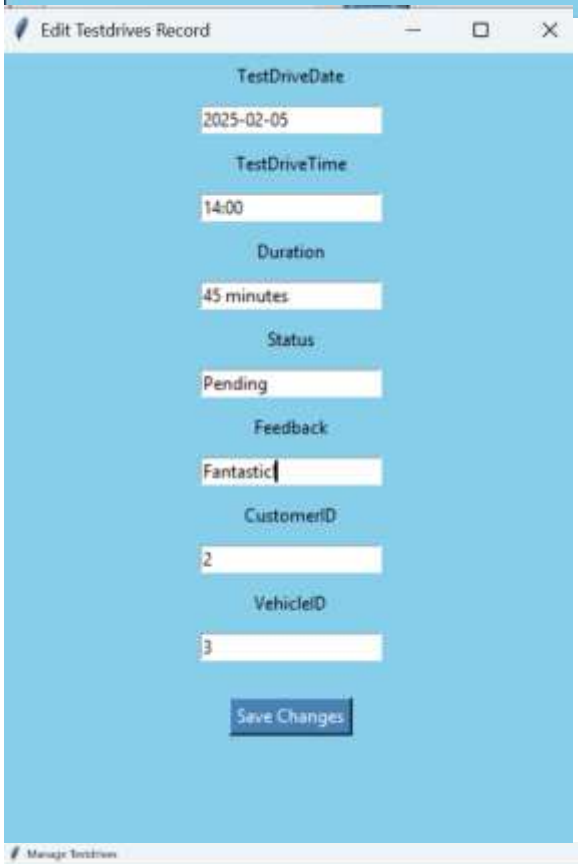

Manage Testdrives
Testdrives Records
TestDriveID
Search
TestDriveID TestDriveDate TestDriveTime Duration Status Feedback CustomerID
1 2025-02-01 10:00 30 minutes Completed Loved it! 1 2
2 2025-02-05 14:00 45 minutes Pending 2 3
3 2025-01-10 14:00 60 Minutes Cancelled 1 1
Add Record
Edit Record
Delete Record
Back

TEST 64


Manage Testdrives
Testdrives Records
TestDriveID
Search
TestDriveID TestDriveDate TestDriveTime Duration Status Feedback CustomerID
1 2025-02-01 10:00 Loved it! 1 2
2 2025-02-05 14:00 2 3
3 2025-01-10 14:00 1 1
Confirm Delete
Delete record ID=1?
Yes
No
Add Record
Edit Record
Delete Record
Back
Manage Testdrives
Testdrives Records
TestDriveID
Search
TestDriveID TestDriveDate TestDriveTime Duration Status Feedback CustomerID
2 2025-02-05 14:00 45 minutes Pending 2 3
3 2025-01-10 14:00 60 Minutes Cancelled 1 1
Add Record
Edit Record
Delete Record
Back

TEST 65


Testdrives Records
TestDriveID
Search
TestDriveID
TestDriveDate
TestDriveTime
Duration
Status
Feedback
CustomerID
2
2025-02-05
14:00
45 minutes
Pending
2
3
3
2025-01-10
14:00
60 Minutes
Cancelled
1
1
4
01/01/01
19:00
60Mins
Complete
1
1
Add Record
Edit Record
Delete Record
Back


Edit Testdrives Record
TestDriveDate
2025-02-05
TestDriveTime
14:00
Duration
45 minutes
Status
Pending
Feedback
Fantastic
CustomerID
2
VehicleID
3
Save Changes

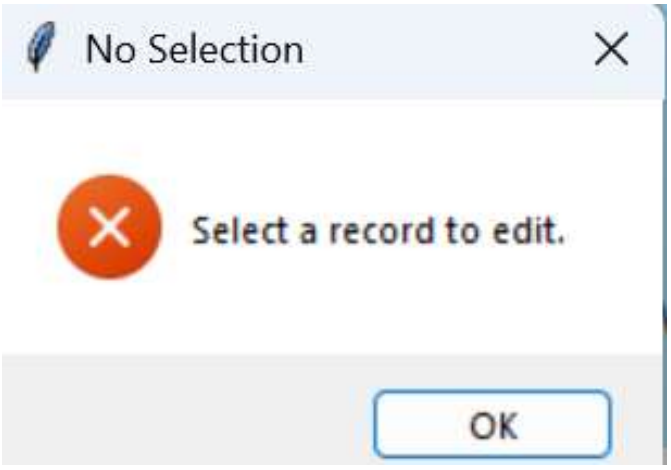

Testdrives Records
TestDriveID
Search
TestDriveID
TestDriveDate
TestDriveTime
Duration
Status
Feedback
CustomerID
2
2025-02-05
14:00
45 minutes
Pending
Fantastic!
2
3
3
2025-01-10
14:00
60 Minutes
Cancelled
1
1
4
01/01/01
19:00
60Mins
Complete
1
1
Add Record
Edit Record
Delete Record
Back

TEST 66

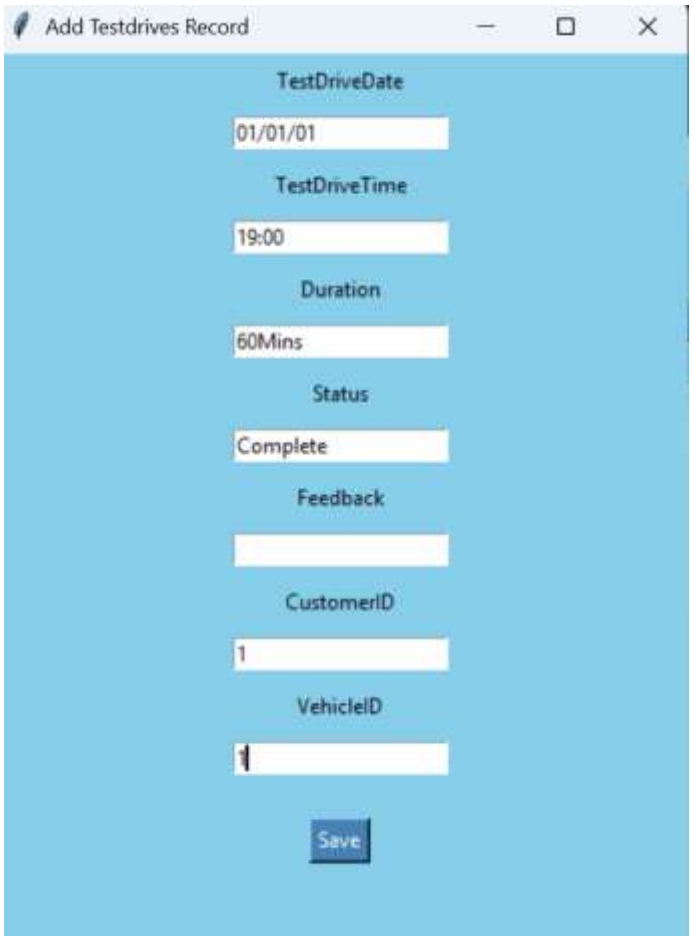

Manage Testdrives
Testdrives Records
TestDriveID
pending
Search
TestDriveID TestDriveDate TestDriveTime Duration Status Feedback CustomerID
2 2025-02-05 14:00 45 minutes Pending 2 3
3 2025-01-10 14:00 60 Minutes Cancelled 1 1
Add Record
Edit Record
Delete Record
Back


Manage Testdrives
Testdrives Records
Status
pending
Search
TestDriveID TestDriveDate TestDriveTime Duration Status Feedback CustomerID
2 2025-02-05 14:00 45 minutes Pending 2 3
Add Record
Edit Record
Delete Record
Back

TEST 67

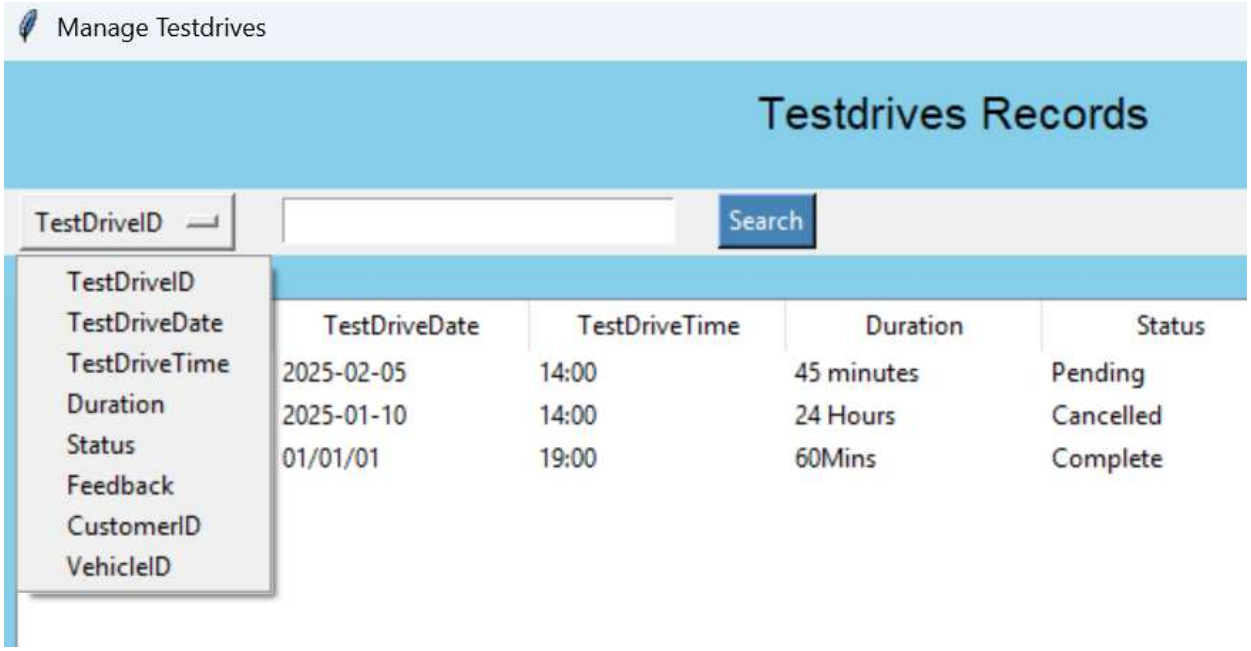


TEST 68


Add Testdrives Record
TestDriveDate
01/01/01
TestDriveTime
19:00
Duration
60Mins
Status
Complete
Feedback
CustomerID
1
VehicleID
1
Save


Manage Testdrives
Testdrives Records
Status
pending
Search
TestDriveID
TestDriveDate
TestDriveTime
Duration
Status
Feedback
CustomerID
2
2025-02-05
14:00
45 minutes
Pending
2
3
3
2025-01-10
14:00
60 Minutes
Cancelled
1
1
4
01/01/01
19:00
60Mins
Complete
1
1
Add Record
Edit Record
Delete Record
Back

TEST 69


Manage Testdrives
Testdrives Records
TestDriveID
Search
TestDriveID
TestDriveDate
TestDriveTime
Duration
Status
Feedback
CustomerID
VehicleID
TestDriveDate
TestDriveTime
Duration
Status
2025-02-05
14:00
45 minutes
Pending
2025-01-10
14:00
24 Hours
Cancelled
01/01/01
19:00
60Mins
Complete

TEST 70


Edit Testdrives Record
TestDriveDate
2025-01-10
TestDriveTime
14:00
Duration
24 Hours
Status
Cancelled
Feedback
CustomerID
1
VehicleID
1
Save Changes

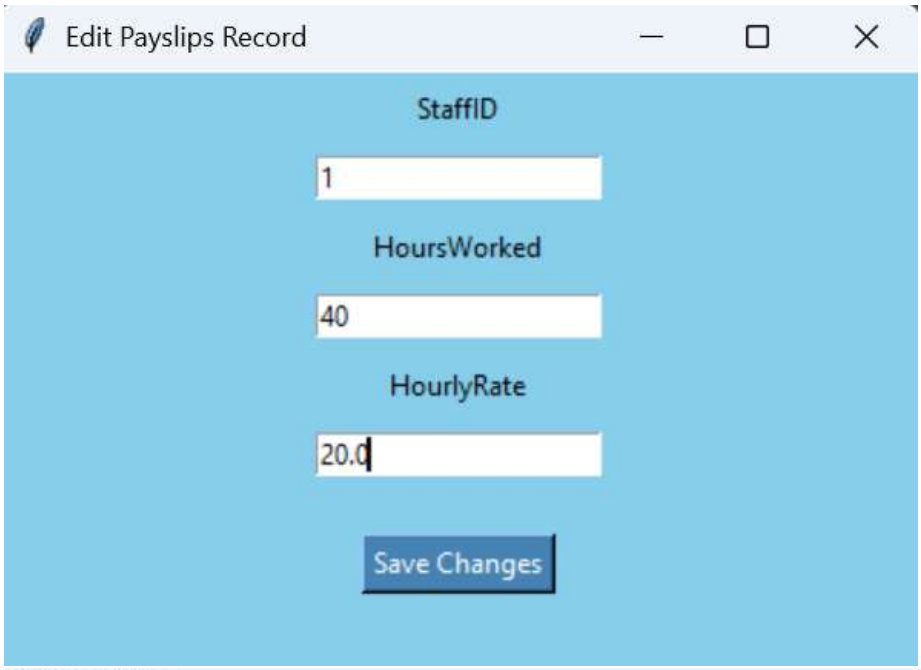

Manage Testdrives
Testdrives Records
TestDriveID
Search
TestDriveID TestDriveDate TestDriveTime Duration Status Feedback CustomerID
2 2025-02-05 14:00 45 minutes Pending Fantastic! 2 3
3 2025-01-10 14:00 24 Hours Cancelled 1 1
4 01/01/01 19:00 60Mins Complete 1 1
Add Record
Edit Record
Delete Record
Back

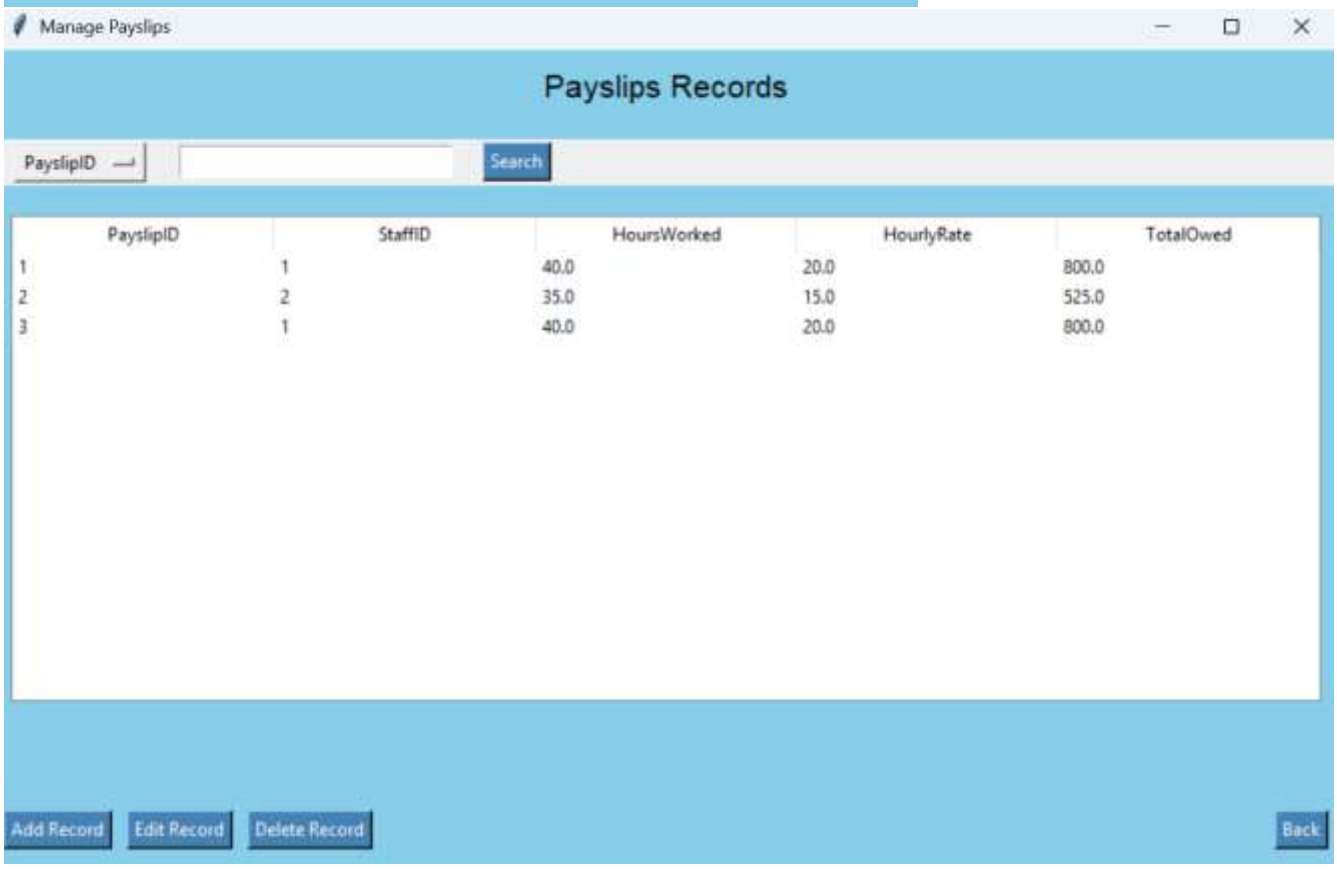
8. Payslips Management (10 Tests)

Form: manageTable.py (Payslips)

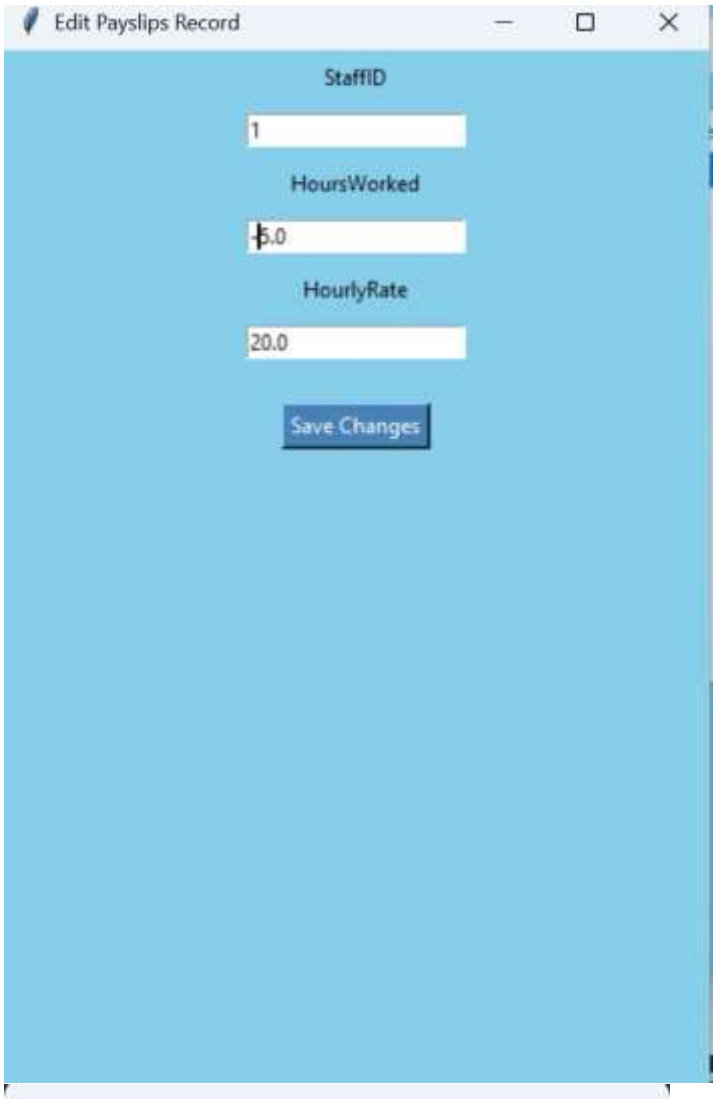
TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
71	Add Payslip: Auto-Calculate TotalOwed	Hours=40, Rate=20	Functional	T	TotalOwed=800.0.	SUCCESS
72	Add Payslip: Negative Hours	Hours=-10	Negative	I	Error: Invalid hours.	SUCCESS
73	Edit Payslip: Update HourlyRate	Rate=25	Functional	T	TotalOwed recalculated.	SUCCESS
74	Delete Payslip: Valid ID	PayslipID=1	Functional	T	Record deleted.	SUCCESS
75	Attempt to use back button in form	StaffID=9999	Functional	I	Form is closed back to main menu	SUCCESS
76	Search Payslip: By StaffID=2	StaffID=2	Functional	T	Matching records.	SUCCESS
77	Add Payslip: Extreme Rate (50)	Rate=50	Boundary	E	Added	SUCCESS
78	Edit Payslip: Hours=0	Hours=0	Functional	E	TotalOwed=0.0.	SUCCESS
79	Delete Payslip: Non-Existent ID	PayslipID=999	Negative	I	Error: No record.	SUCCESS
80	Add Payslip: Non-Numeric Rate	Rate=twenty	Negative	I	Error: Numeric required.	SUCCESS

TEST 71


Edit Payslips Record
StaffID
1
HoursWorked
40
HourlyRate
20.0
Save Changes


Manage Payslips
Payslips Records
PayslipID
Search
PayslipID StaffID HoursWorked HourlyRate TotalOwed
1 1 40.0 20.0 800.0
2 2 35.0 15.0 525.0
3 1 40.0 20.0 800.0
Add Record
Edit Record
Delete Record
Back

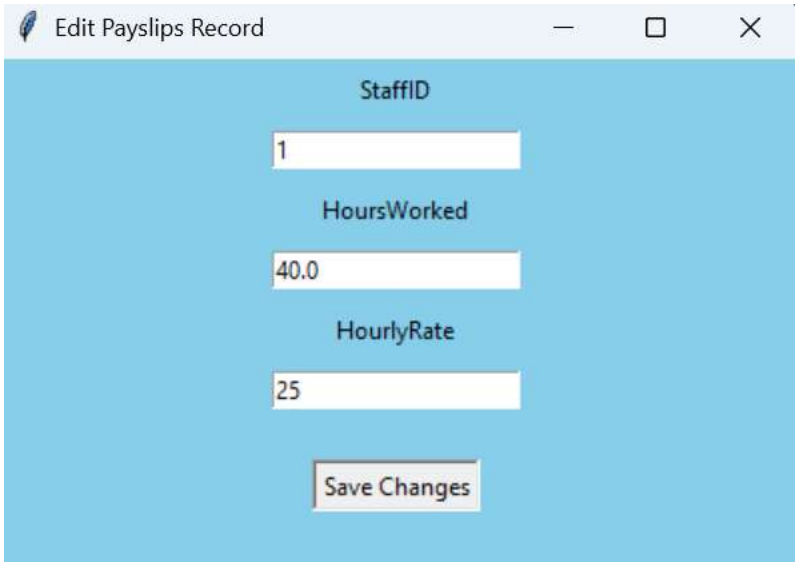
TEST 72

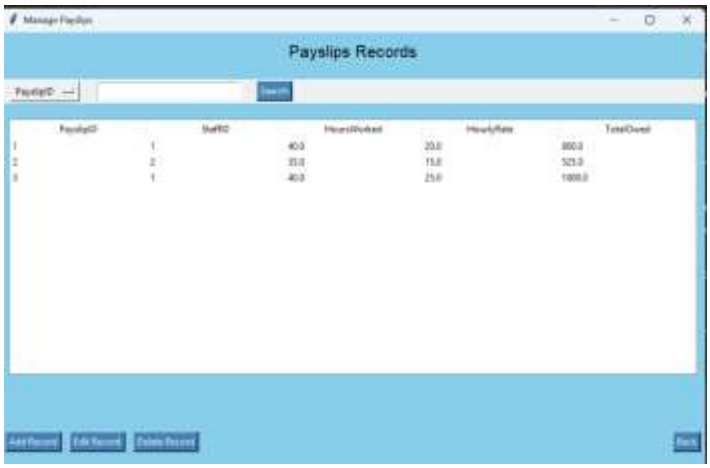

Edit Payslips Record
StaffID
1
HoursWorked
-5.0
HourlyRate
20.0
Save Changes


Mulligan Motors
Input must not be negative
OK

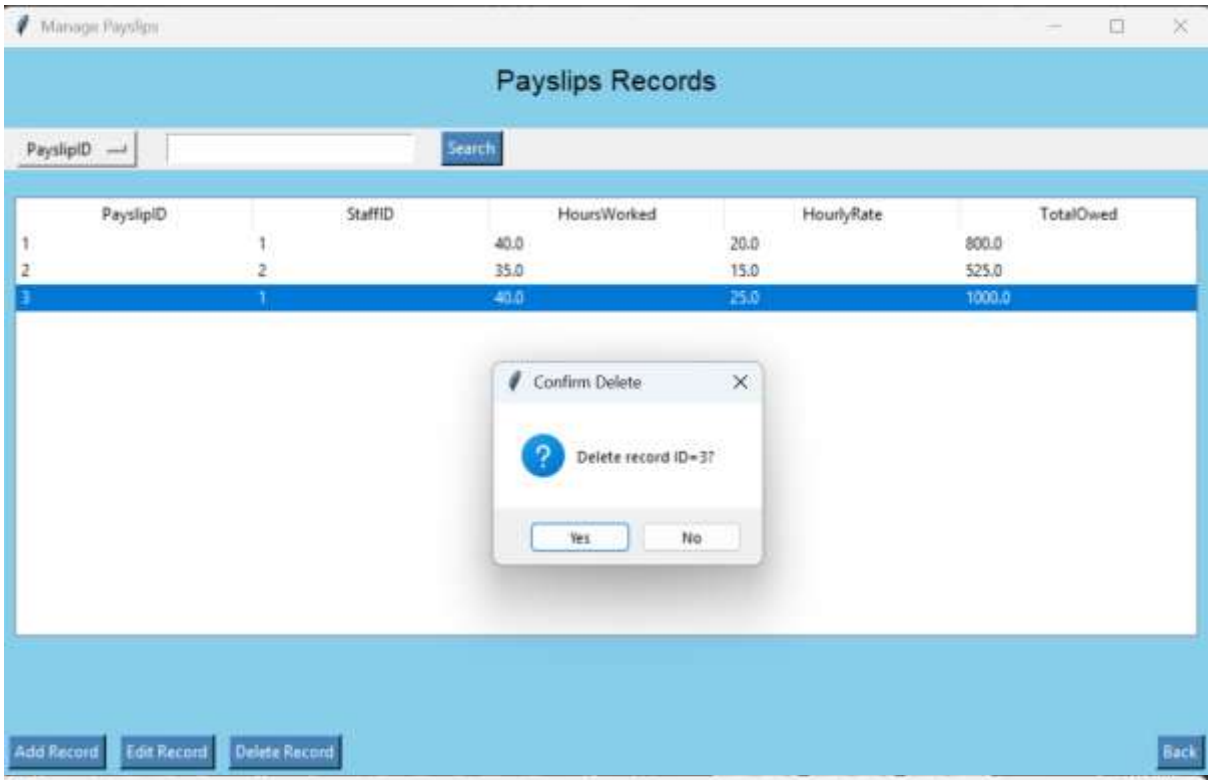
TEST 73


Payslips Records
Search
PayslipID
StaffID
HoursWorked
HourlyRate
TotalOwed
Add Record
Edit Record
Delete Record
Back


Edit Payslips Record
StaffID
1
HoursWorked
40.0
HourlyRate
25
Save Changes


Payslips Records
Search
PayslipID
StaffID
HoursWorked
HourlyRate
TotalOwed

TEST 74


Manage Payslips
Payslips Records
PayslipID
Search
PayslipID StaffID HoursWorked HourlyRate TotalOwed
1 1 40.0 20.0 800.0
2 2 35.0 15.0 525.0
3 1 40.0 25.0 1000.0
Confirm Delete
Delete record ID=3?
Yes
No
Add Record
Edit Record
Delete Record
Back

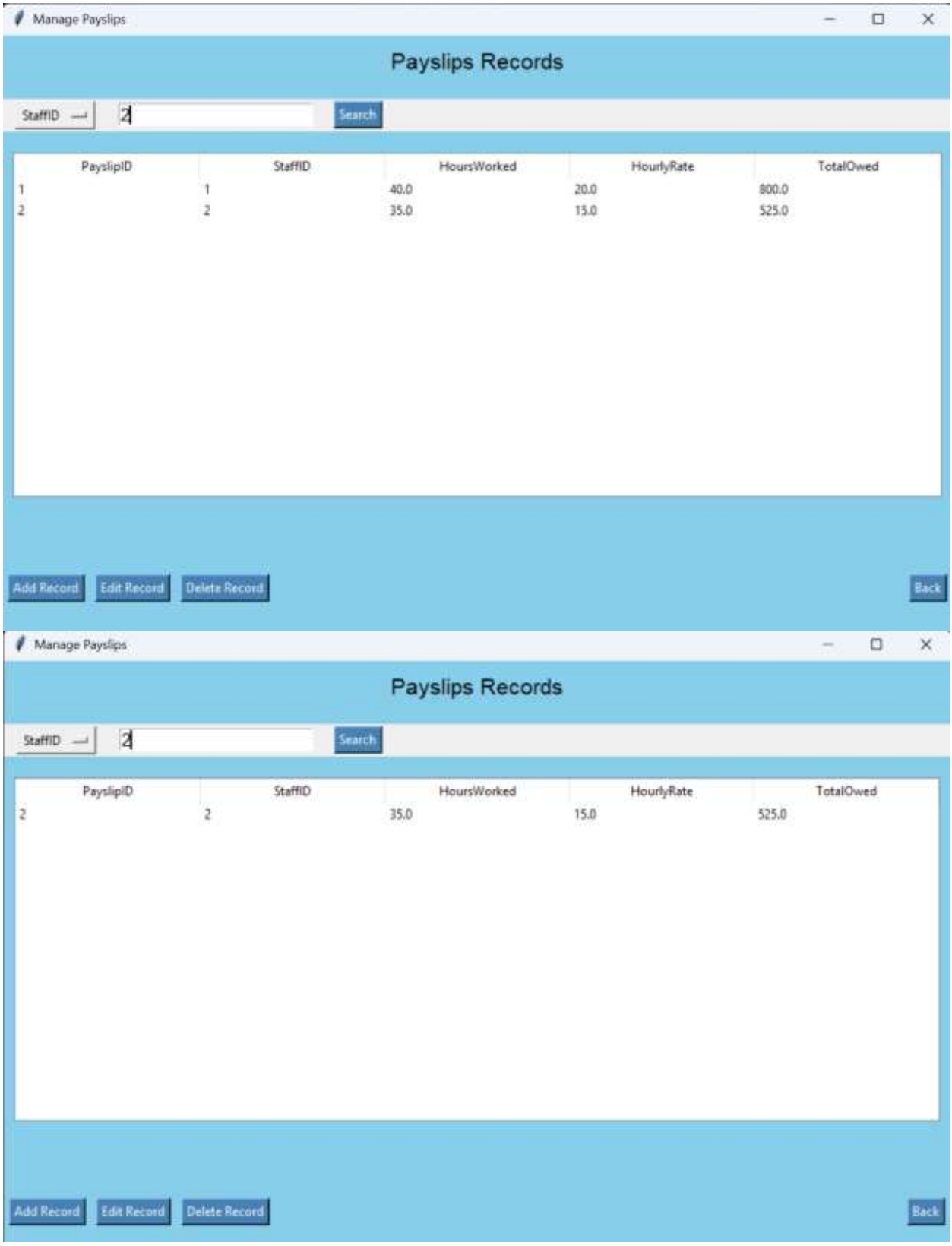

Manage Payslips
Payslips Records
PayslipID
Search
PayslipID StaffID HoursWorked HourlyRate TotalOwed
1 1 40.0 20.0 800.0
2 2 35.0 15.0 525.0
Add Record
Edit Record
Delete Record
Back

TEST 75

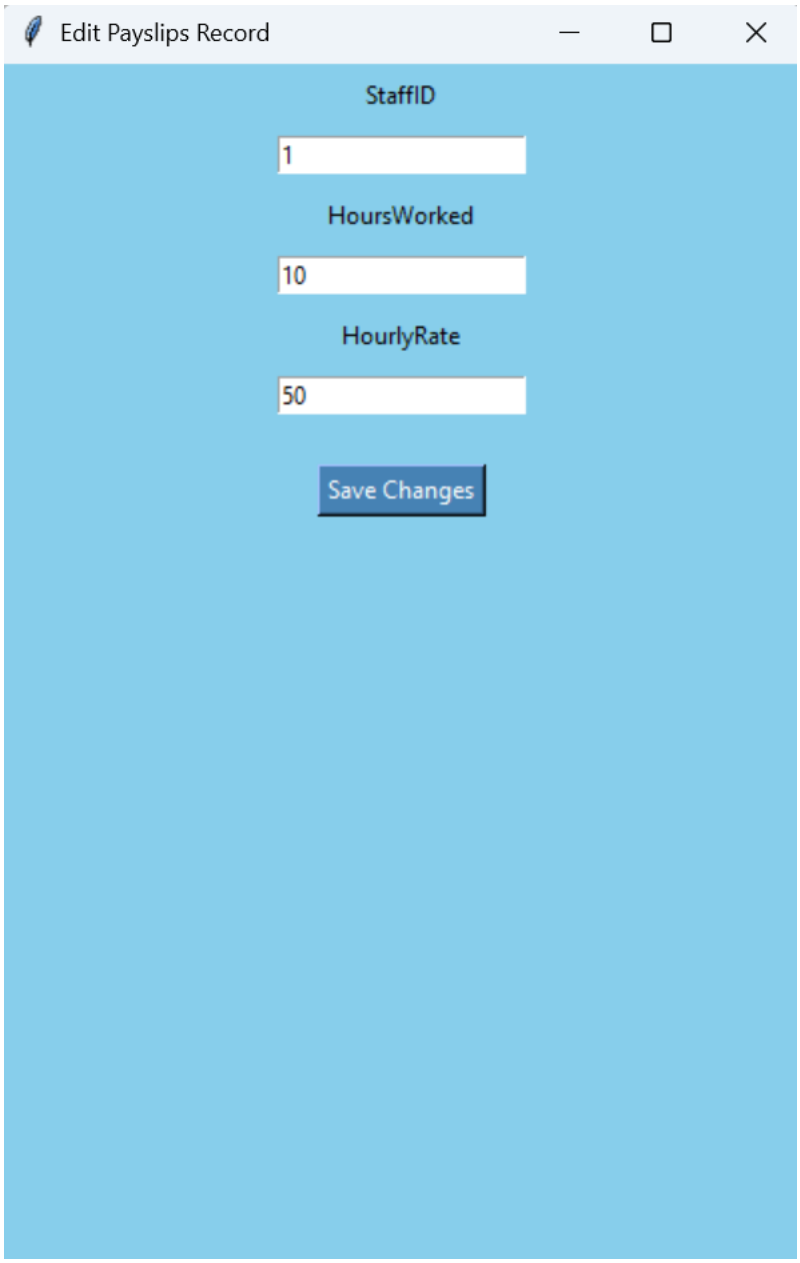

Payslips Records
Search
PayslipID
StaffID
HoursWorked
HourlyRate
TotalOwed
Back


Mulligan Motors - Manager
M
Mulligan Motors Management (ADMIN)
Manage Vehicle Stock
Manage Customers
Manage Orders
Manage Staff
Manage Test Drives
Manage Access
Manage Payslips
Generate Report
Backup Database
Help
Log Off

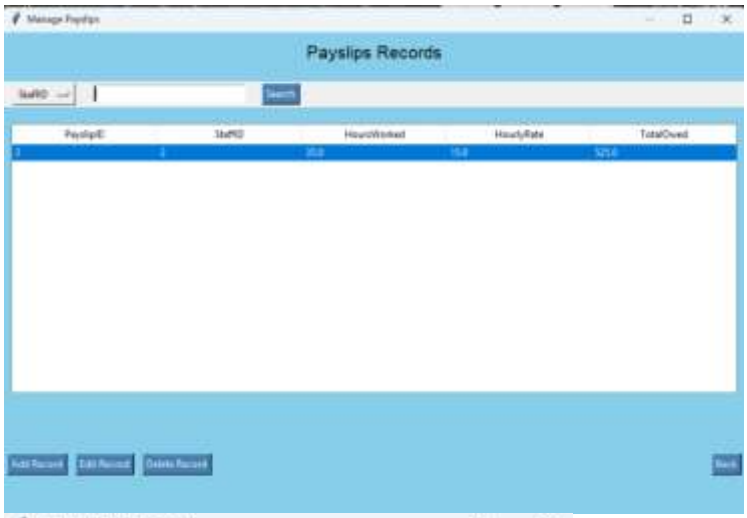
TEST 76

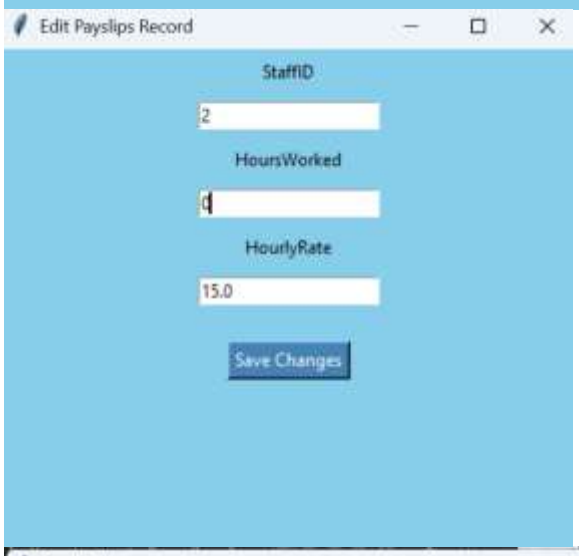

Manage Payslips
Payslips Records
StaffID
2
Search
PayslipID
StaffID
HoursWorked
HourlyRate
TotalOwed
1
1
40.0
20.0
800.0
2
2
35.0
15.0
525.0
Add Record
Edit Record
Delete Record
Back
Manage Payslips
Payslips Records
StaffID
2
Search
PayslipID
StaffID
HoursWorked
HourlyRate
TotalOwed
2
2
35.0
15.0
525.0
Add Record
Edit Record
Delete Record
Back

TEST 77


Edit Payslips Record
StaffID
1
HoursWorked
10
HourlyRate
50
Save Changes

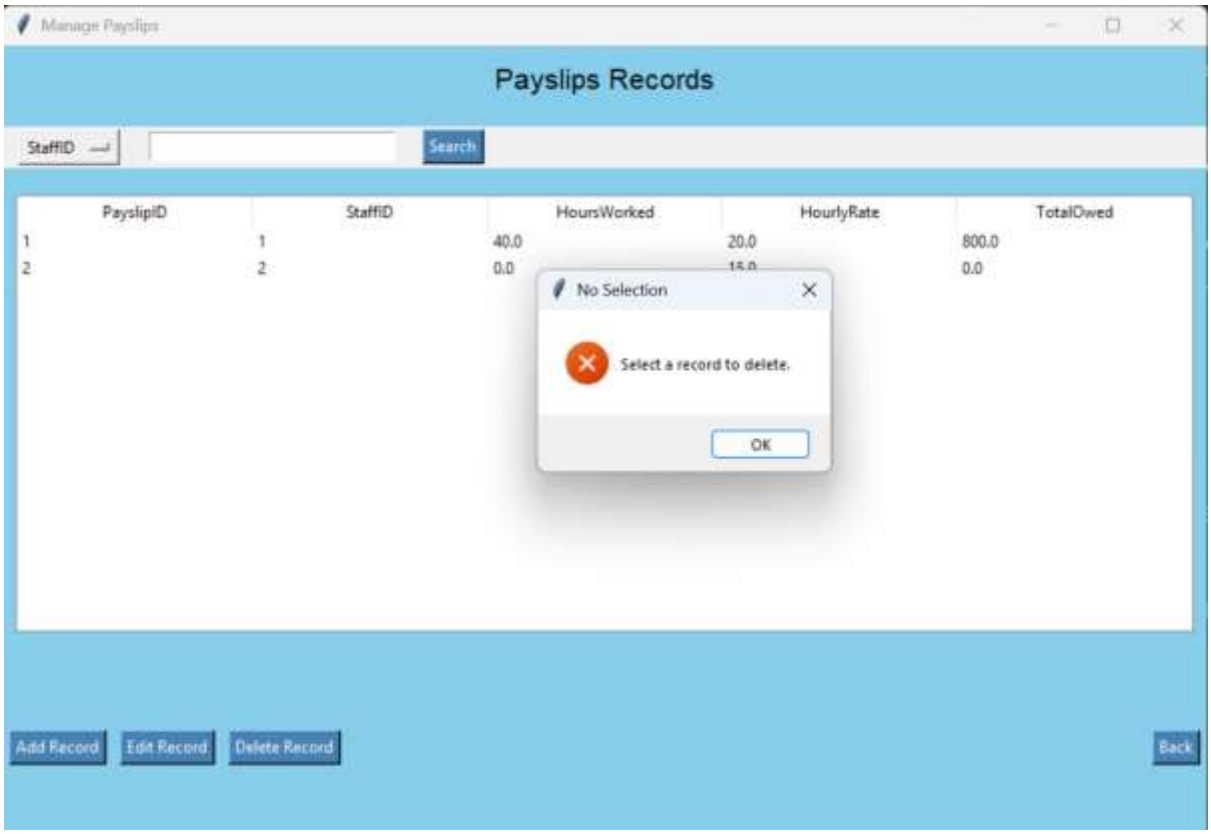
TEST 78


Payslips Records
Search
Add Record
Edit Record
Delete Record
Back


Edit Payslips Record
StaffID
2
HoursWorked
0
HourlyRate
15.0
Save Changes

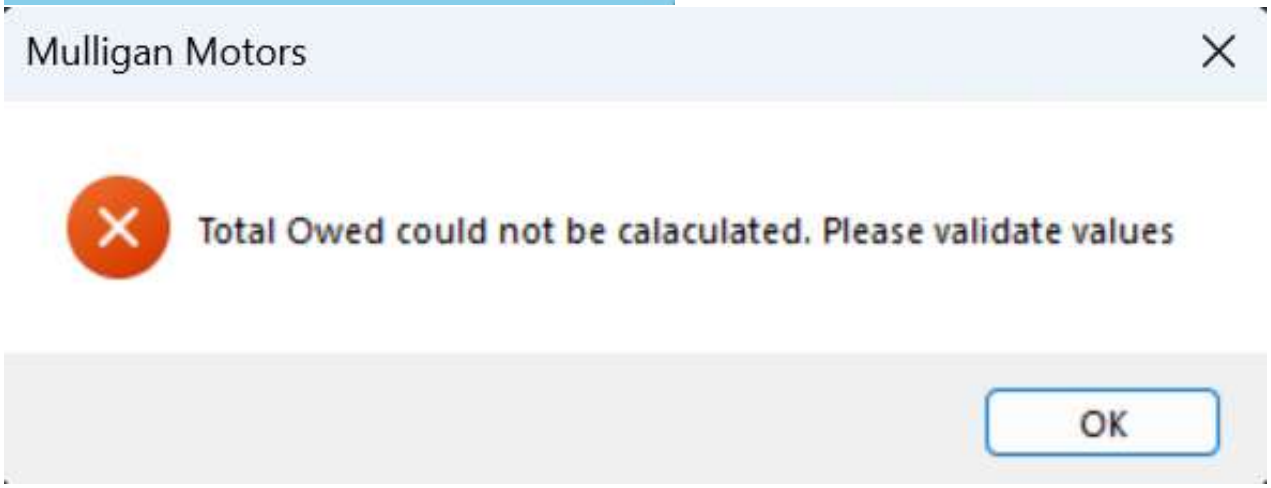

Payslips Records
Search
PayslipID
StaffID
HoursWorked
HourlyRate
TotalOwed
1
1
40.0
20.0
800.0
2
2
0.0
15.0
0.0
Add Record
Edit Record
Delete Record
Back

TEST 79


Manage Payslips
Payslips Records
StaffID
Search
PayslipID
StaffID
HoursWorked
HourlyRate
TotalOwed
1
1
40.0
20.0
800.0
2
2
0.0
0.0
No Selection
Select a record to delete.
OK
Add Record
Edit Record
Delete Record
Back

TEST 80


Add Payslips Record
StaffID
1
HoursWorked
15
HourlyRate
twenty
Save

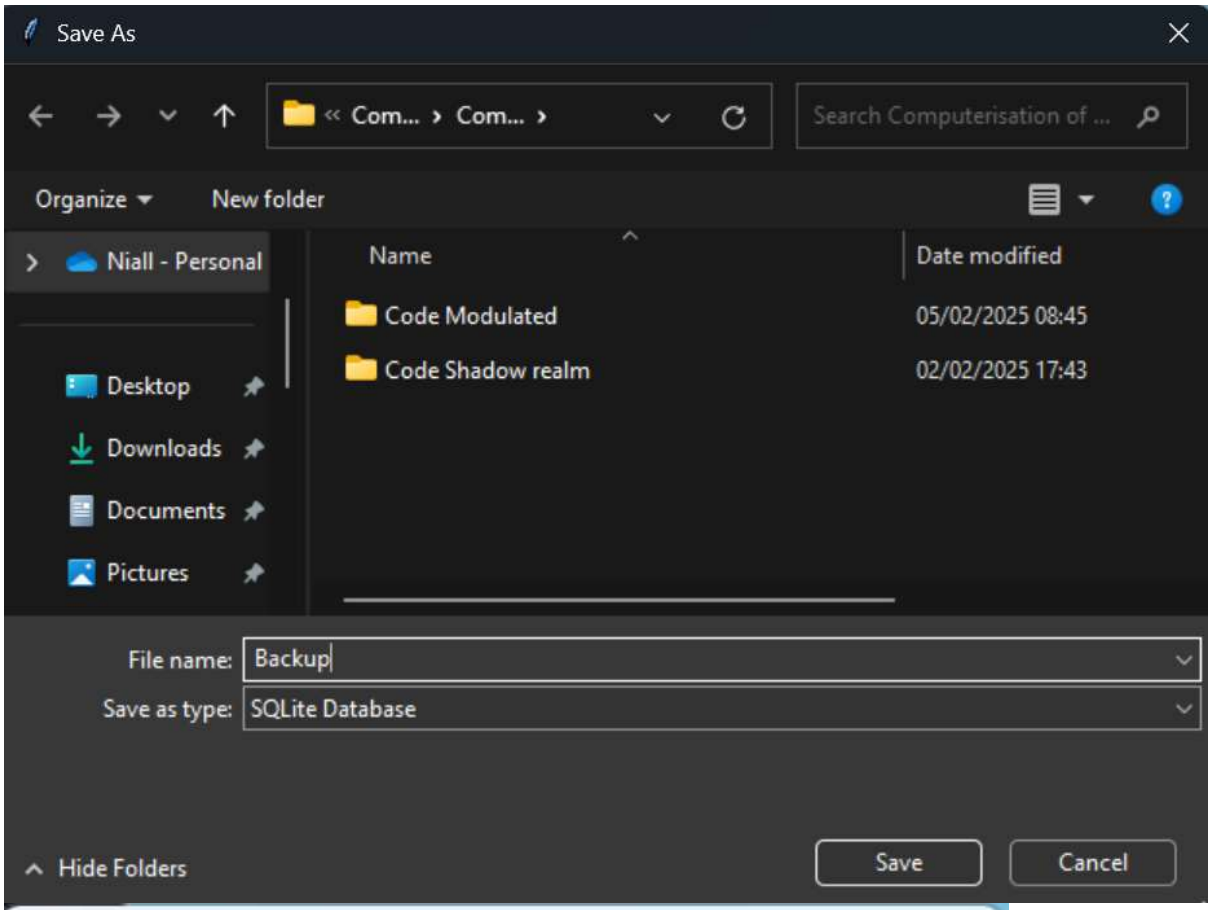

Mulligan Motors
Total Owed could not be calaculated. Please validate values
OK

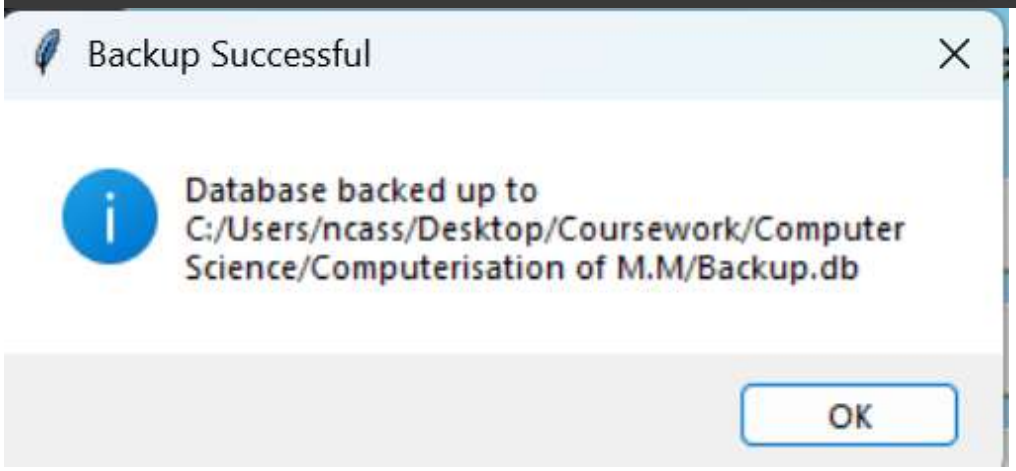
9. Backup & Reporting/Help page(10 Tests)

Form: HelpMenu.py

TEST#	DESCRIPTION	TESTING INPUTS	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
81	Backup: Valid Path	Path=C:/backup.db	Functional	T	"Backup Successful."	SUCCESS
82	Open the help page	Clicking the help button	Functional	I	Help page opens	SUCCESS
83	Generate Report: No selection	Table=Not selected	Negative	I	Error: Select a table	SUCCESS
84	Generate Report: Invalid Table	Table=NonExistent	Negative	I	Drop down box should prevent selection of invalid table.	SUCCESS
85	Generate Report: Valid Table (VehicleStock)	Table=VehicleStock	Functional	T	TXT file with data.	SUCCESS
86	Backup: Cancel File Dialog	Cancel during path selection	Functional	T	No backup created.	SUCCESS
87	Generate Report: Special Characters in Data	Feedback=Test & < > " '	Boundary	E	Escaped in TXT file.	SUCCESS
88	Generate report: Staff table with no data	Generate staff table	Boundary	E	Success	SUCCESS
89	Close the help page	Click ‘close’	Functional	T	Help page closes	SUCCESS
90	Backup: Overwrite Existing File	Path=existing_backup.db	Functional	T	Overwrite confirmed.	SUCCESS

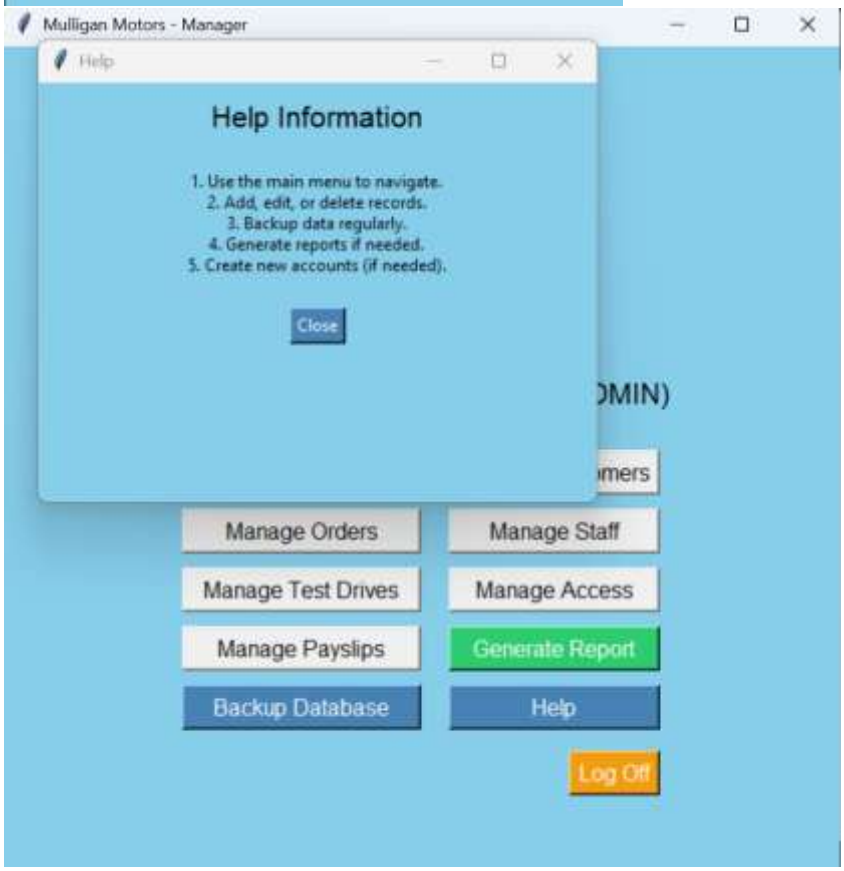
TEST 81


Save As
Com... › Com... ›
Search Computerisation of ...
Organize
New folder
Niall - Personal
Name
Date modified
Code Modulated
05/02/2025 08:45
Code Shadow realm
02/02/2025 17:43
Desktop
Downloads
Documents
Pictures
File name:
Backup
Save as type:
SQLite Database
Hide Folders
Save
Cancel

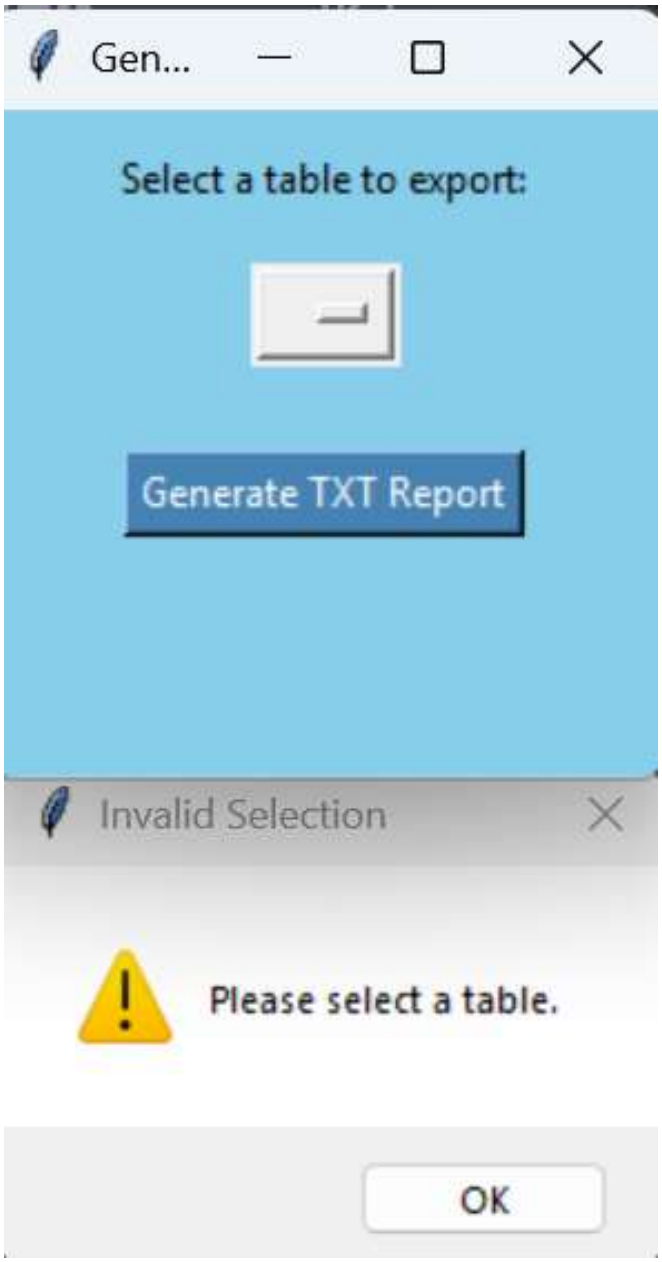

Backup Successful
Database backed up to
C:/Users/ncass/Desktop/Coursework/Computer
Science/Computerisation of M.M/Backup.db
OK

TEST 82

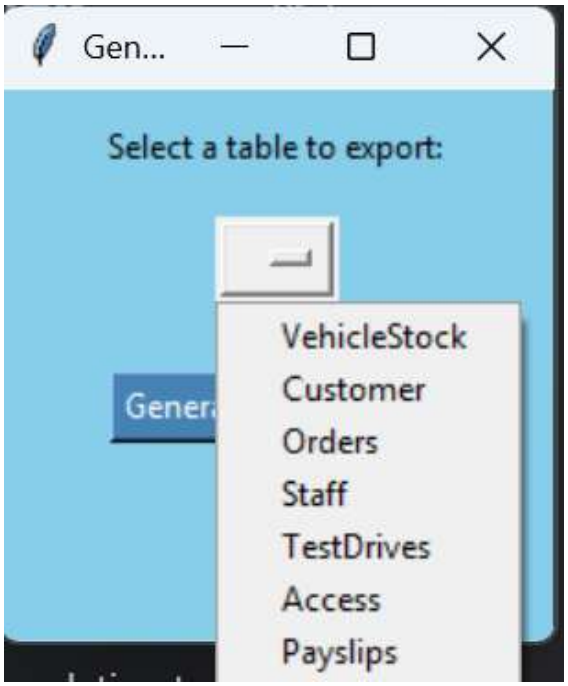

Mulligan Motors Management (ADMIN)
Manage Vehicle Stock
Manage Customers
Manage Orders
Manage Staff
Manage Test Drives
Manage Access
Manage Payslips
Generate Report
Backup Database
Help
Log Off


Mulligan Motors - Manager
Help
Help Information
1. Use the main menu to navigate.
2. Add, edit, or delete records.
3. Backup data regularly.
4. Generate reports if needed.
5. Create new accounts (if needed).
Close
Manage Orders
Manage Staff
Manage Test Drives
Manage Access
Manage Payslips
Generate Report
Backup Database
Help
Log Off

TEST 83

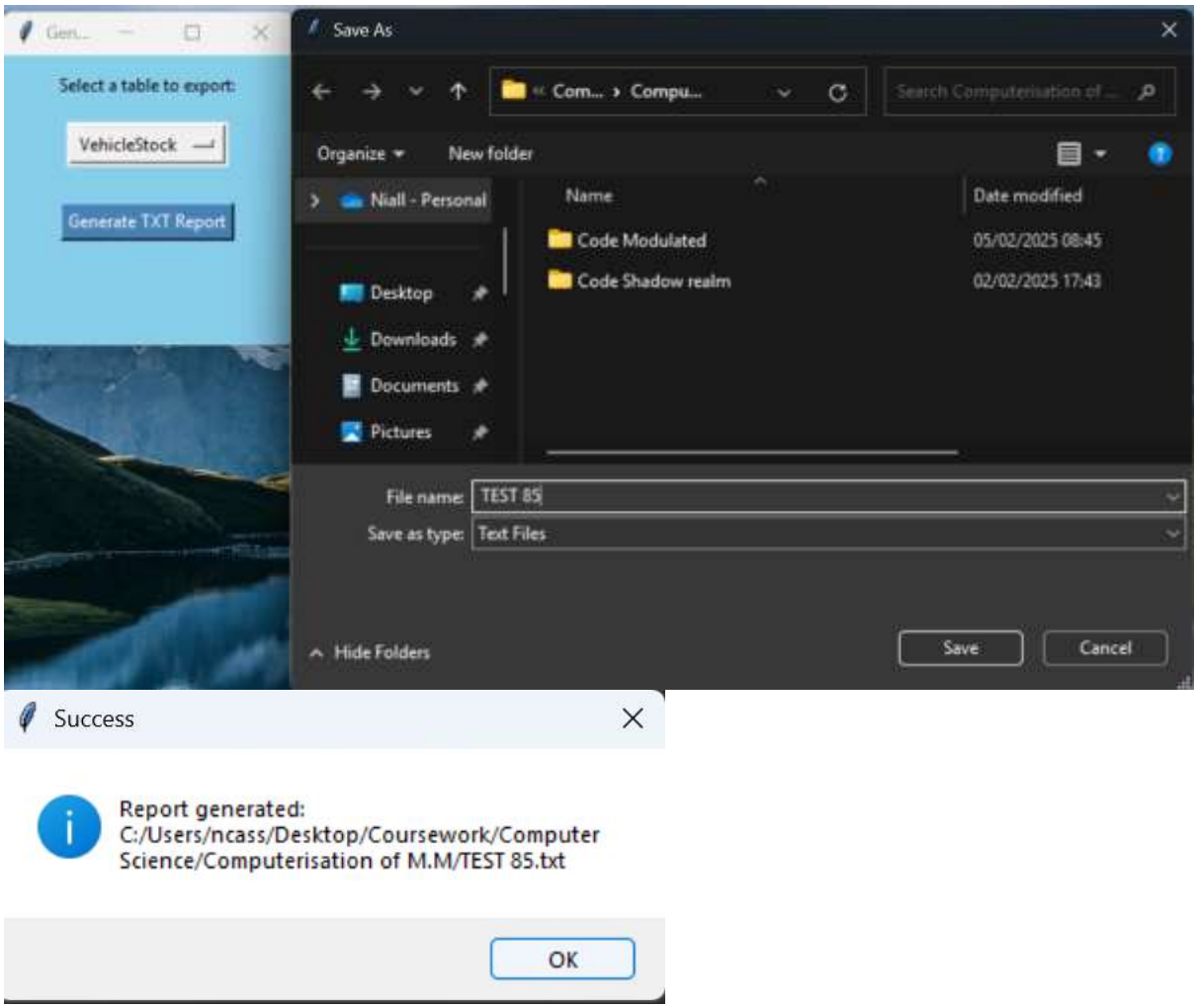

Gen...
Select a table to export:
Generate TXT Report
Invalid Selection
Please select a table.
OK

TEST 84



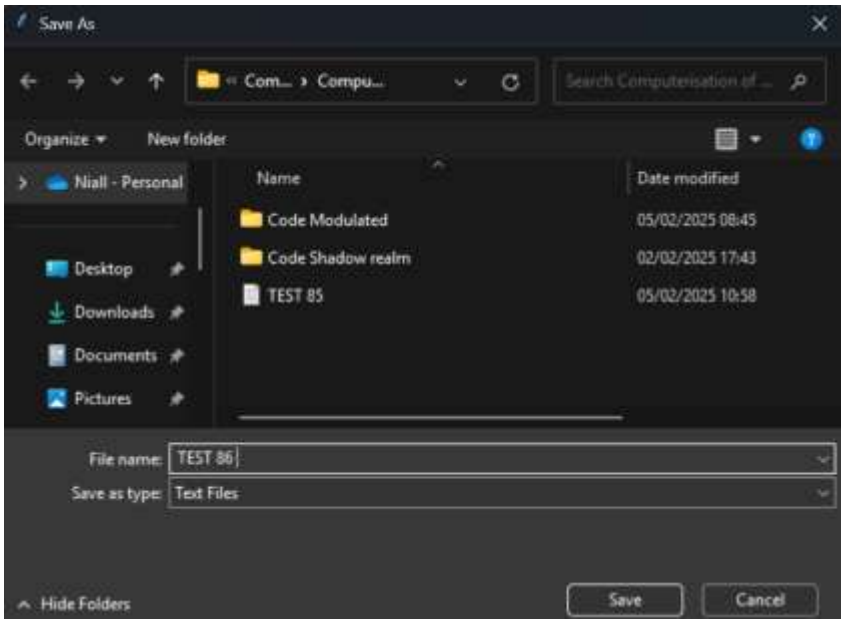
The drop down selection method ensures a non-existent table cannot be selected.

TEST 85

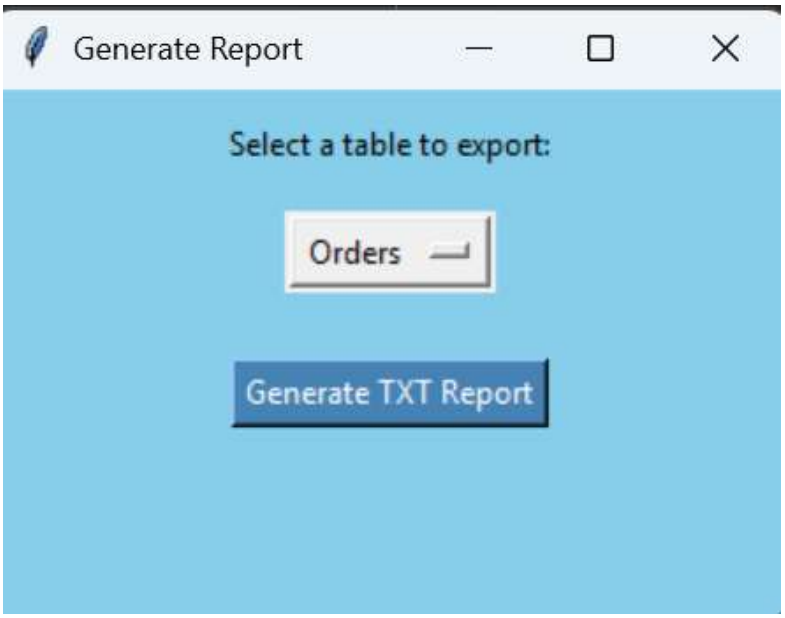

Gen...
Select a table to export:
VehicleStock
Generate TXT Report
Save As
Com... › Compu...
Search Computerisation of ...
Organize
New folder
Niall - Personal
Desktop
Downloads
Documents
Pictures
Name
Date modified
Code Modulated
05/02/2025 08:45
Code Shadow realm
02/02/2025 17:43
File name: TEST 85
Save as type: Text Files
Hide Folders
Save
Cancel

Success
Report generated:
C:/Users/ncass/Desktop/Coursework/Computer Science/Computerisation of M.M/TEST 85.txt
OK

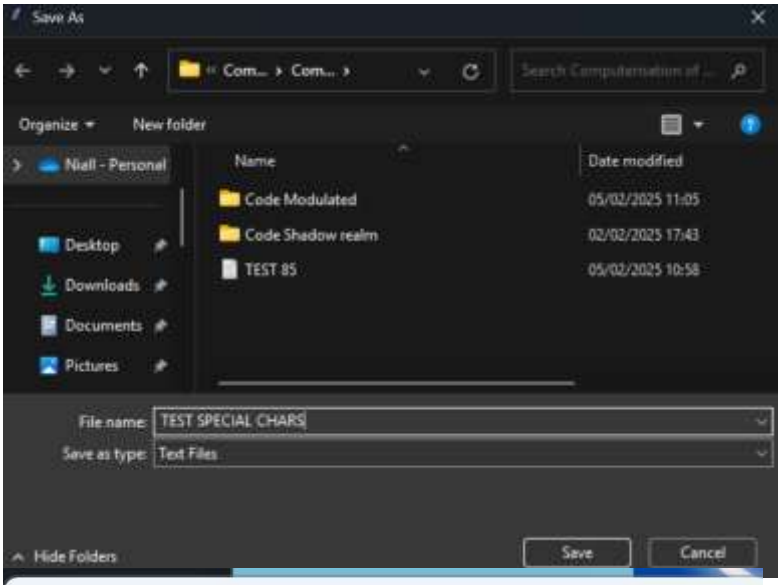
TEST 86

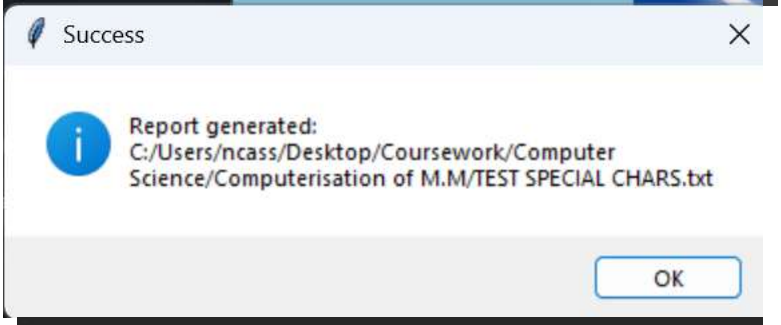


The cancel button will return the user to the Generate Report screen



TEST 87



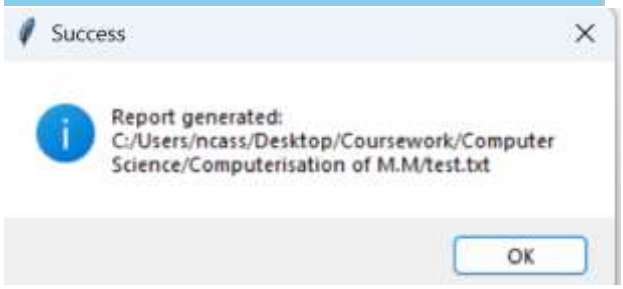


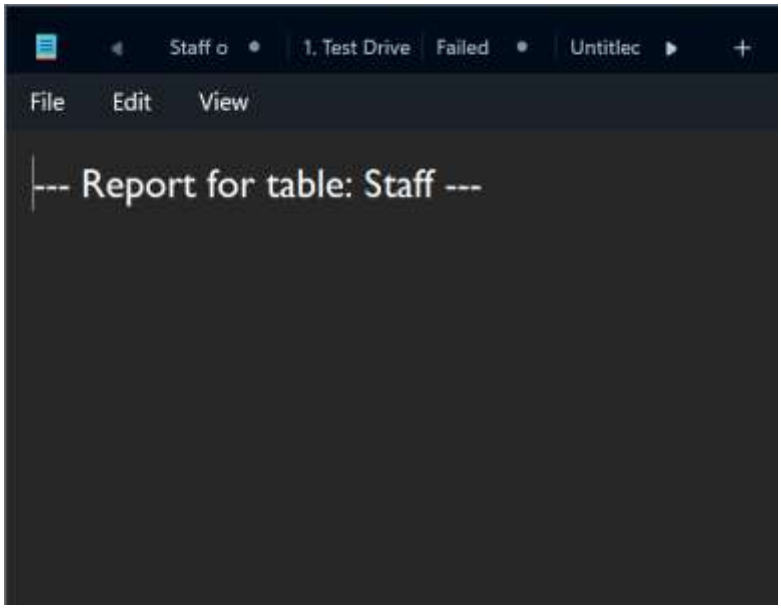


Special Characters in a text file can be handled as seen in the last row

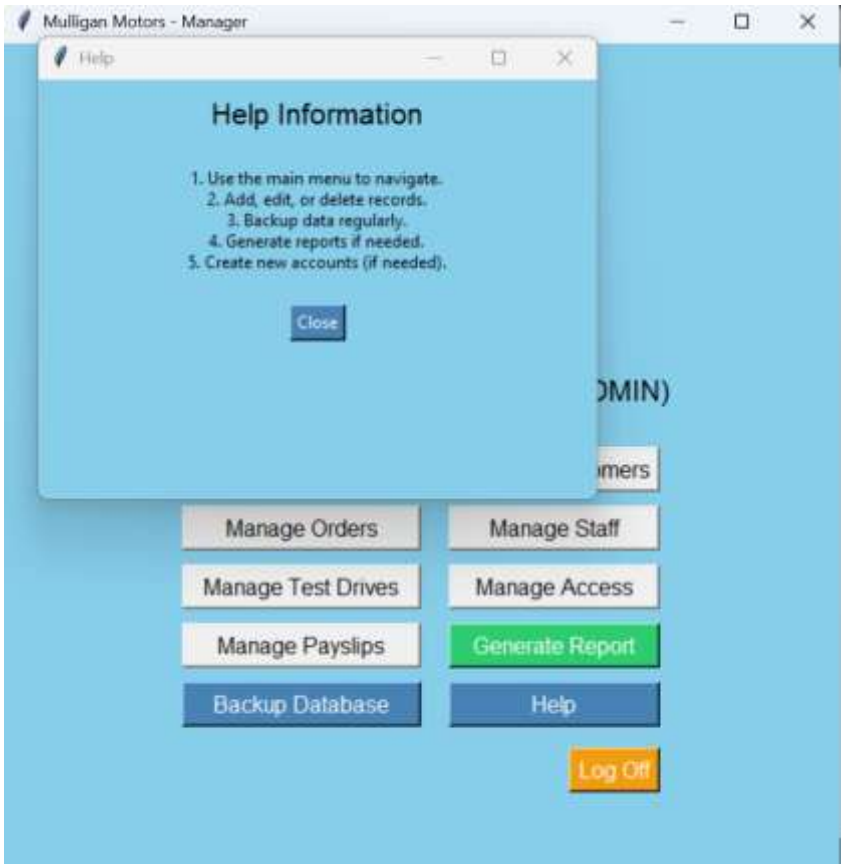
TEST 88


Staff Records


Success
Report generated:
C:/Users/ncass/Desktop/Coursework/Computer Science/Computerisation of M.M/test.txt
OK

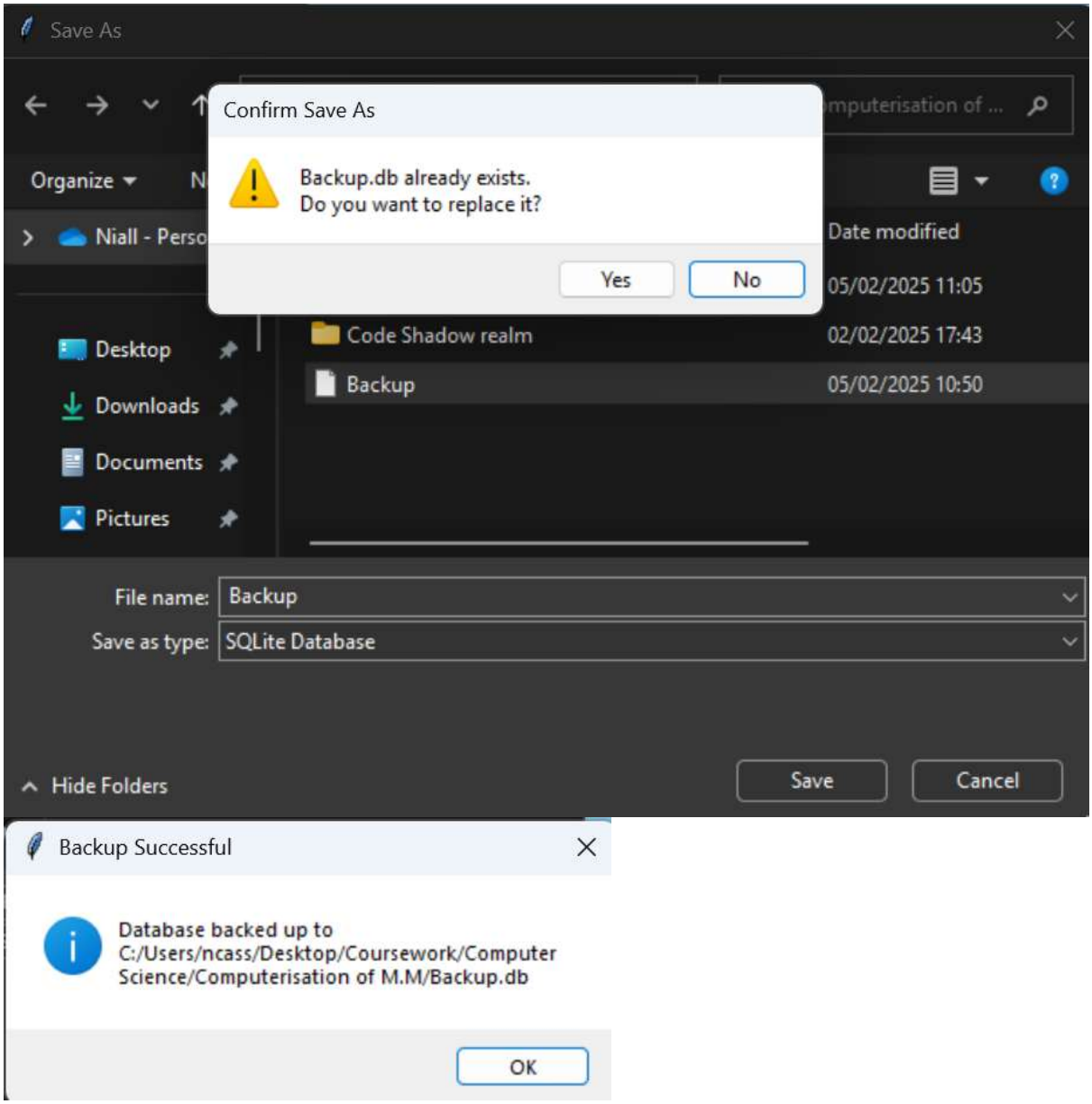

Staff o
1. Test Drive
Failed
Untitlec
File
Edit
View
--- Report for table: Staff ---

TEST 89


Mulligan Motors - Manager
Help
Help Information
1. Use the main menu to navigate.
2. Add, edit, or delete records.
3. Backup data regularly.
4. Generate reports if needed.
5. Create new accounts (if needed).
Close
Manage Orders
Manage Staff
Manage Test Drives
Manage Access
Manage Payslips
Generate Report
Backup Database
Help
Log Off


M
Mulligan Motors Management (ADMIN)
Manage Vehicle Stock
Manage Customers
Manage Orders
Manage Staff
Manage Test Drives
Manage Access
Manage Payslips
Generate Report
Backup Database
Help
Log Off

TEST 90


Save As
Confirm Save As
Backup.db already exists.
Do you want to replace it?
Yes
No
Organize
Niall - Perso
Desktop
Downloads
Documents
Pictures
Date modified
05/02/2025 11:05
Code Shadow realm
02/02/2025 17:43
Backup
05/02/2025 10:50
File name: Backup
Save as type: SQLite Database
Hide Folders
Save
Cancel

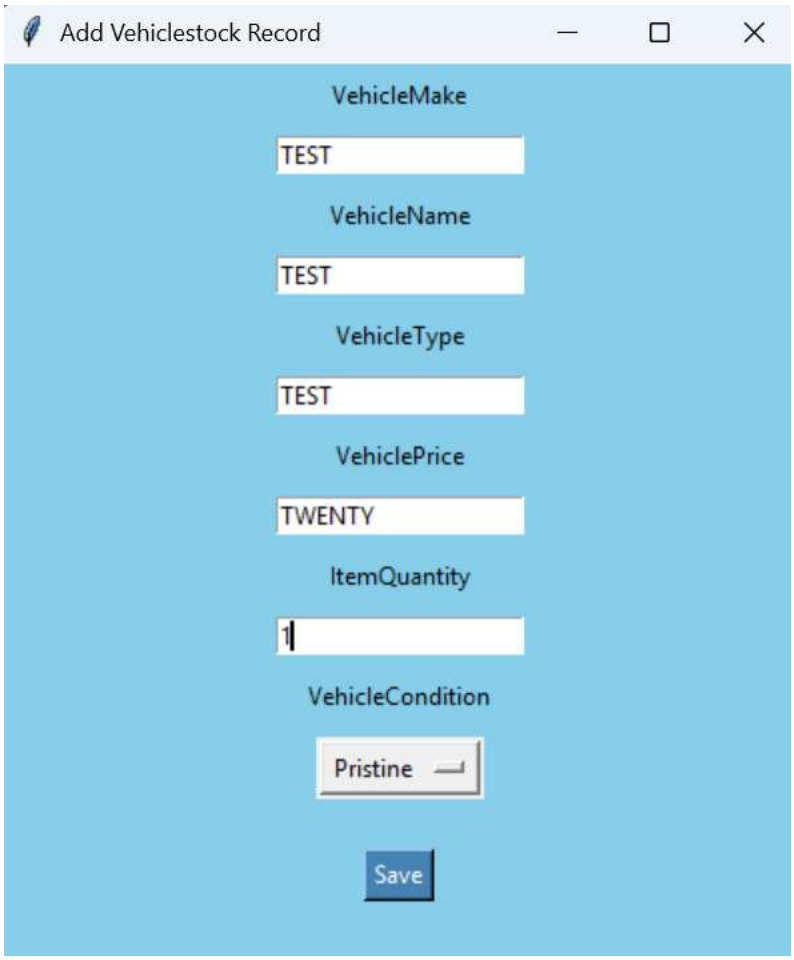
Backup Successful
Database backed up to
C:/Users/ncass/Desktop/Coursework/Computer Science/Computerisation of M.M/Backup.db
OK

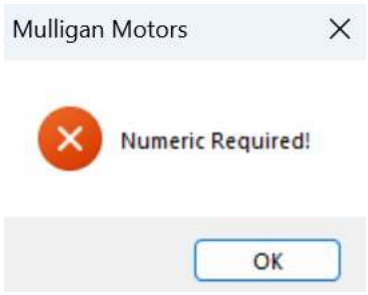
10. Input Validation (10 Tests)

File: InputValidation.py

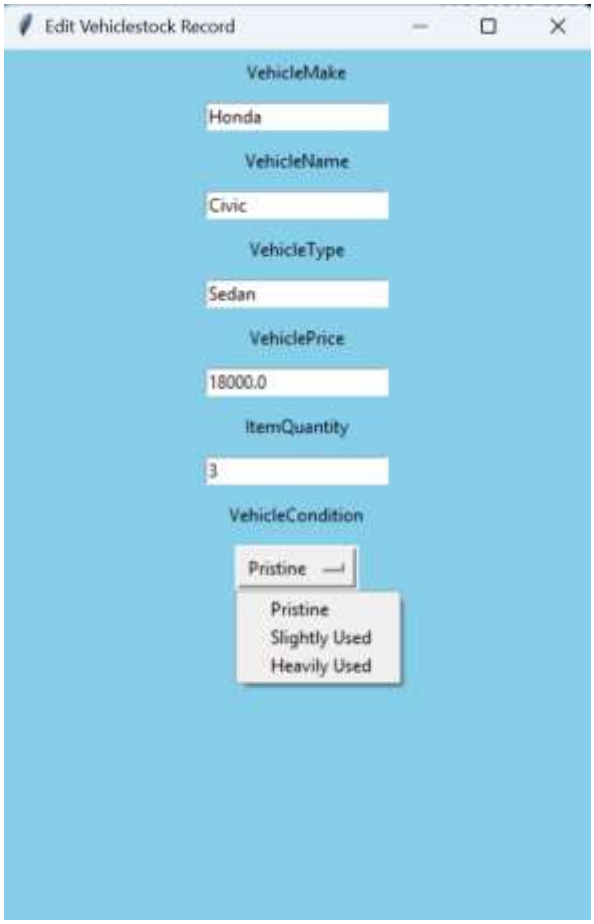
TEST#	DESCRIPTION	TESTING INPUTS (Table=VehicleStock)	TEST TYPE	DATA	EXPECTED RESULT	SUCCESS/FAIL
91	VehiclePrice Non-Numeric	Price=ten	Negative	I	Error: "Numeric required."	SUCCESS
92	VehicleCondition Invalid	Condition=Broken	Negative	I	The dropdown box should limit the user to only valid choices.	SUCCESS
93	Customer FirstName Empty	FirstName=""	Negative	I	Error: "FirstName required."	SUCCESS
94	Orders DatePlaced Invalid	DatePlaced=2025-13-32	Negative	I	Error: Date format.	SUCCESS
95	Staff PhoneNumber Letters	Phone=abc	Negative	I	Error, phone cannot be letters	SUCCESS
96	TestDrives Duration Empty	Duration=""	Functional	T	Error, duration must not be empty.	SUCCESS
97	AccessLevel Invalid	AccessLevel=guest	Negative	I	The dropdown box should limit the user to only valid choices.	SUCCESS
98	Payslips HoursWorked Negative	Hours=-5	Negative	I	Error: Invalid hours.	SUCCESS
99	Customer Email Missing "@"	Email=johndoe.com	Negative	I	Error: Invalid email.	SUCCESS
100	VehicleStock Quantity Non-Integer	Quantity=5.5	Negative	I	Error: Integer required.	SUCCESS

TEST 91


Add Vehiclestock Record
VehicleMake
TEST
VehicleName
TEST
VehicleType
TEST
VehiclePrice
TWENTY
ItemQuantity
1
VehicleCondition
Pristine
Save

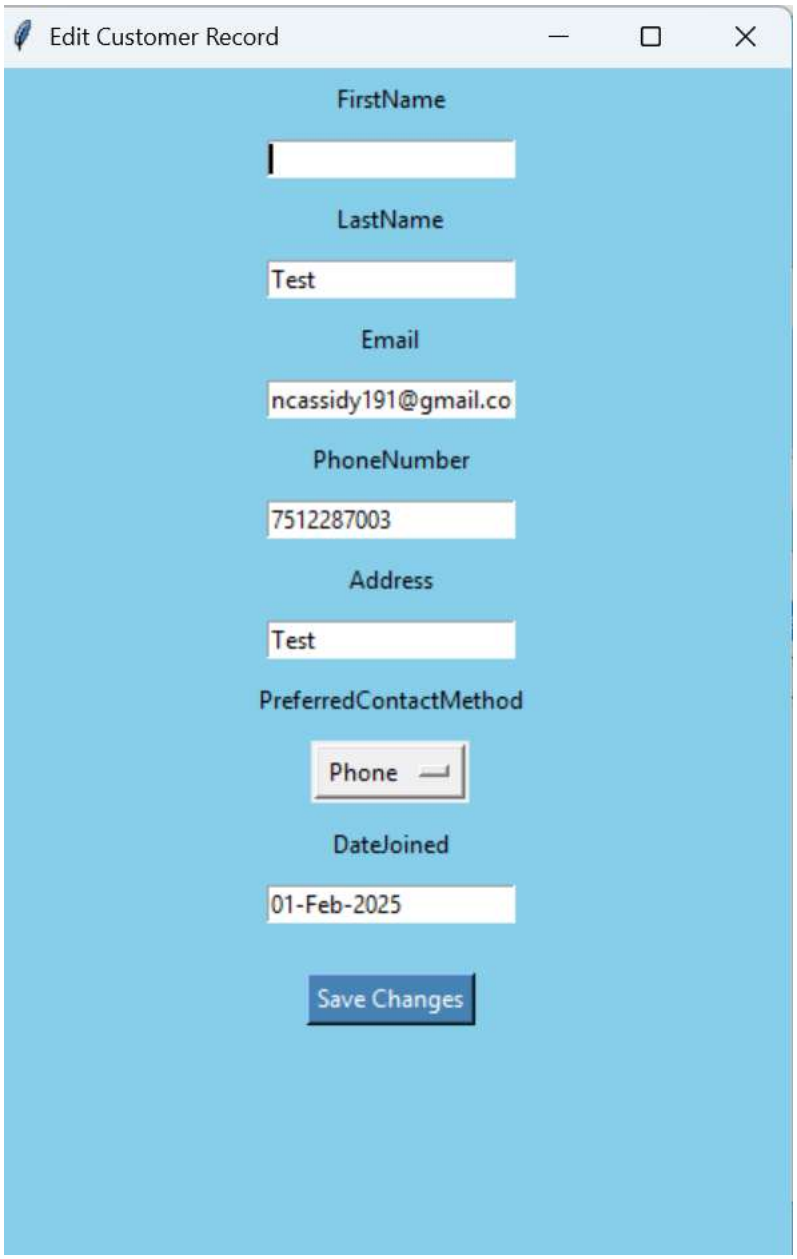

Mulligan Motors
Numeric Required!
OK

TEST 92



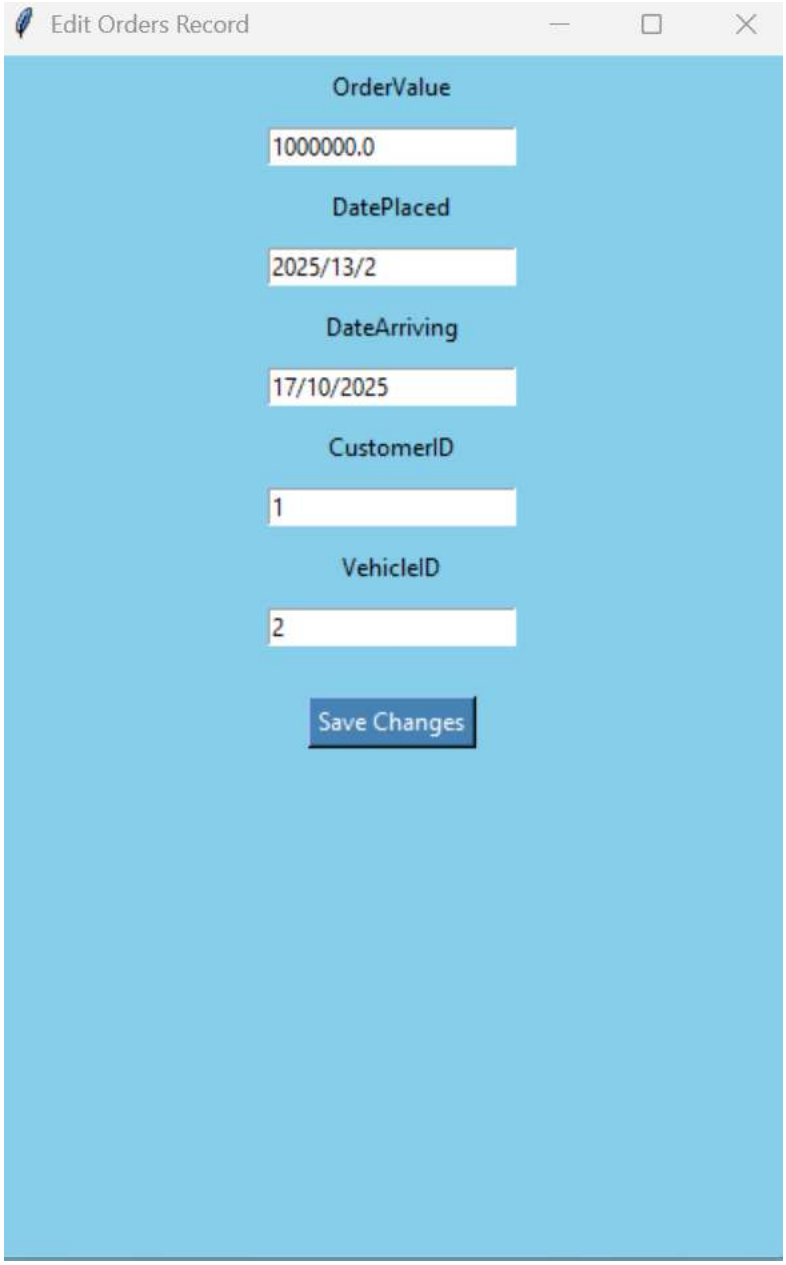
The user cannot select an option other than the pre-set choices, making it impossible to set an invalid value.

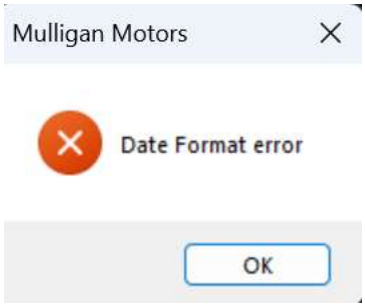
TEST 93


Edit Customer Record
FirstName
LastName
Test
Email
ncassidy191@gmail.co
PhoneNumber
7512287003
Address
Test
PreferredContactMethod
Phone
DateJoined
01-Feb-2025
Save Changes

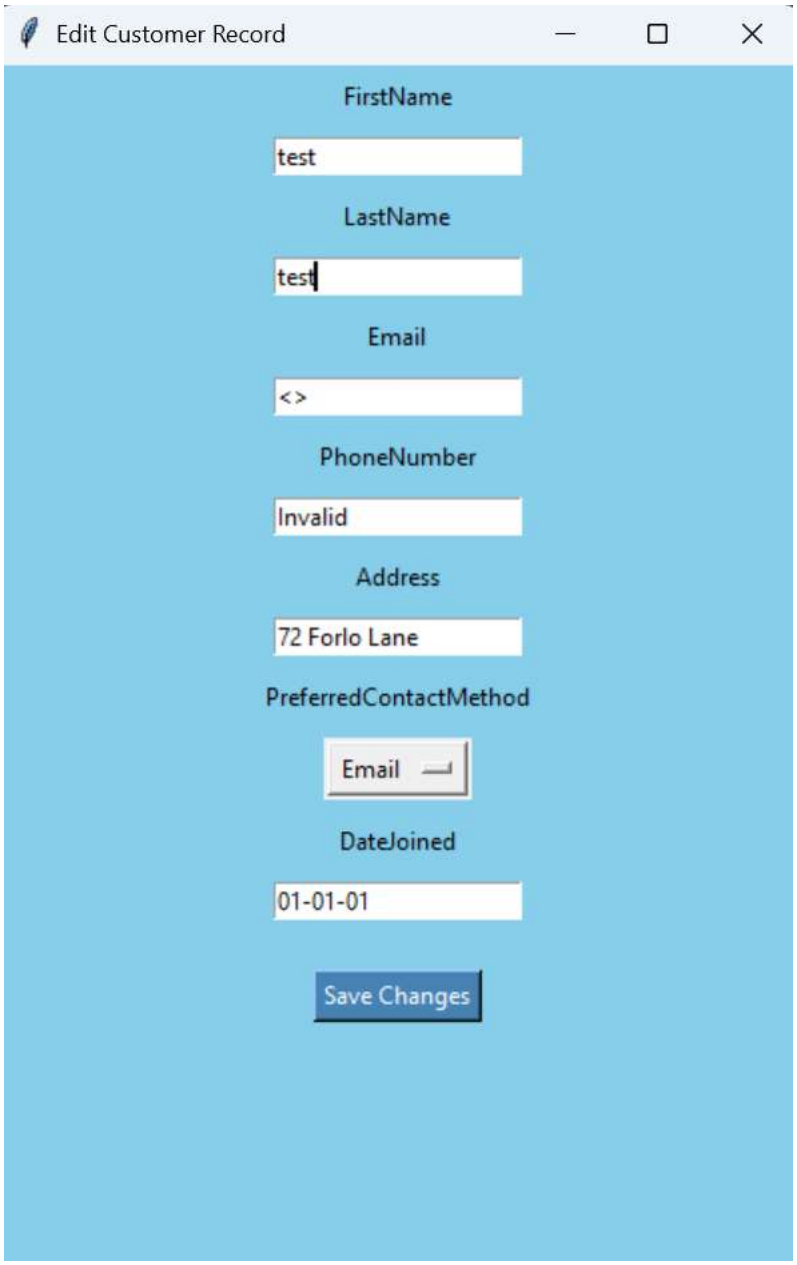

Mulligan Motors
FirstName required
OK

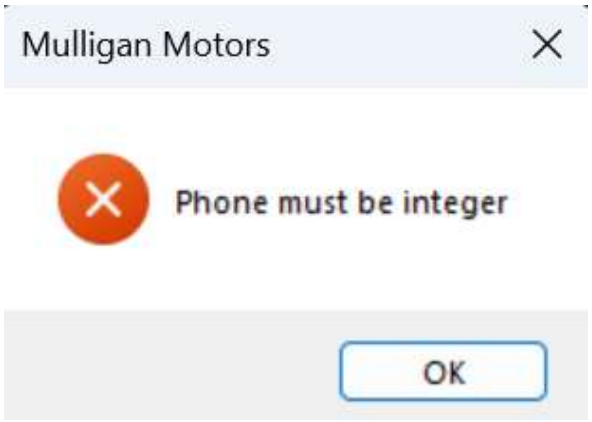
TEST 94


Edit Orders Record
OrderValue
1000000.0
DatePlaced
2025/13/2
DateArriving
17/10/2025
CustomerID
1
VehicleID
2
Save Changes

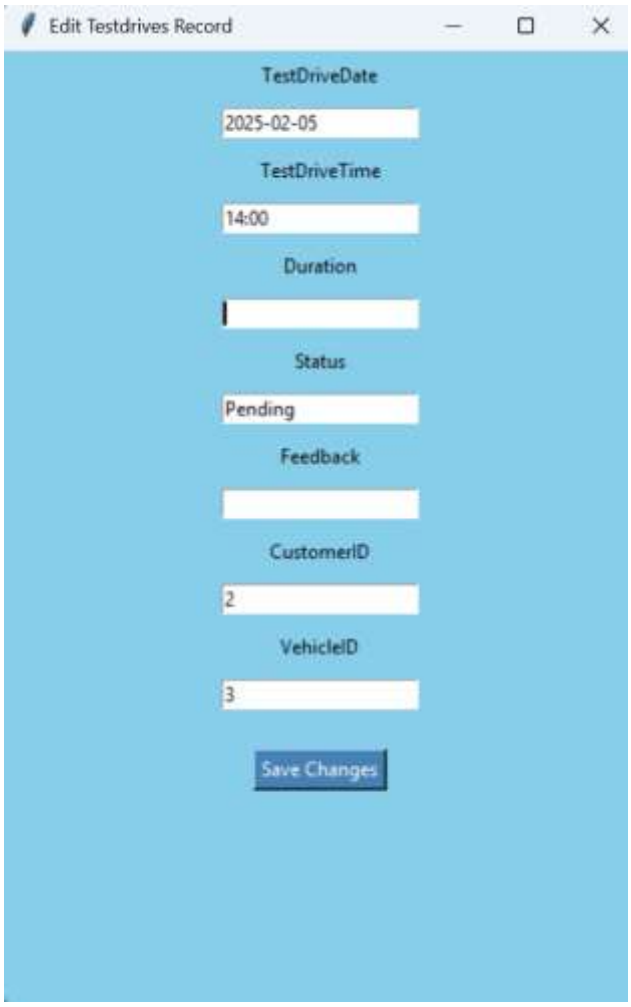

Mulligan Motors
Date Format error
OK

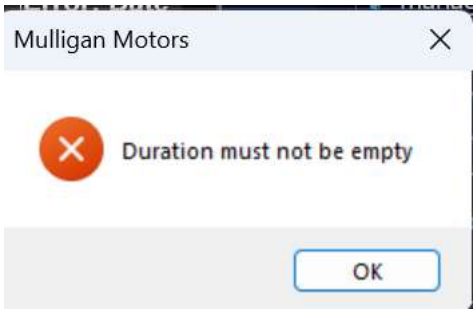
TEST 95


Edit Customer Record
FirstName
test
LastName
test
Email
<>
PhoneNumber
Invalid
Address
72 Forlo Lane
PreferredContactMethod
Email
DateJoined
01-01-01
Save Changes

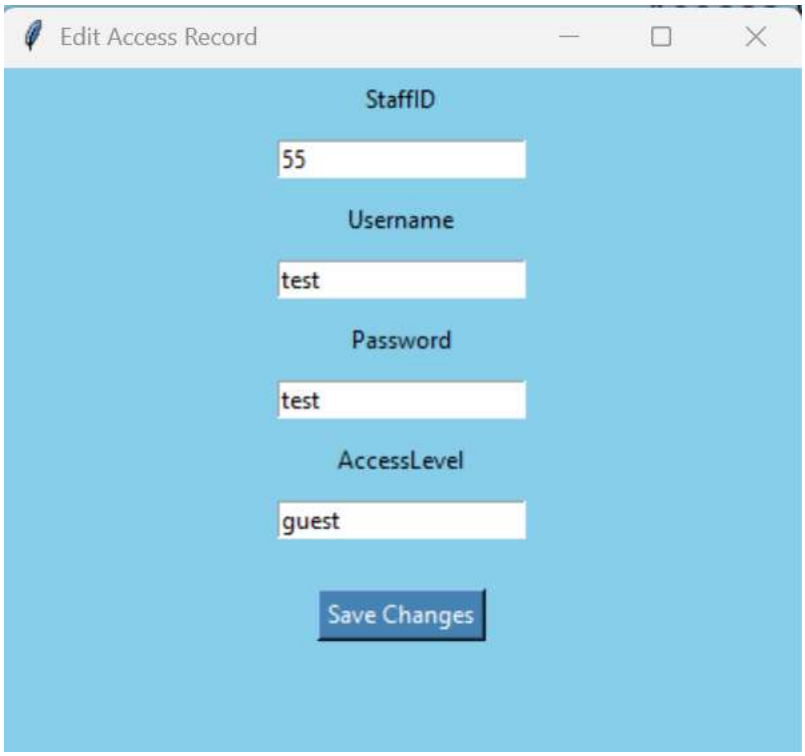

Mulligan Motors
Phone must be integer
OK

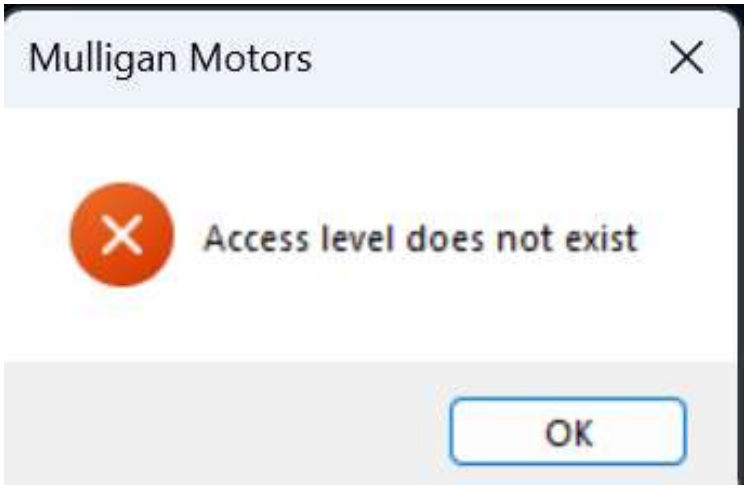
TEST 96


Edit Testdrives Record
TestDriveDate
2025-02-05
TestDriveTime
14:00
Duration
Status
Pending
Feedback
CustomerID
2
VehicleID
3
Save Changes


Mulligan Motors
Duration must not be empty
OK

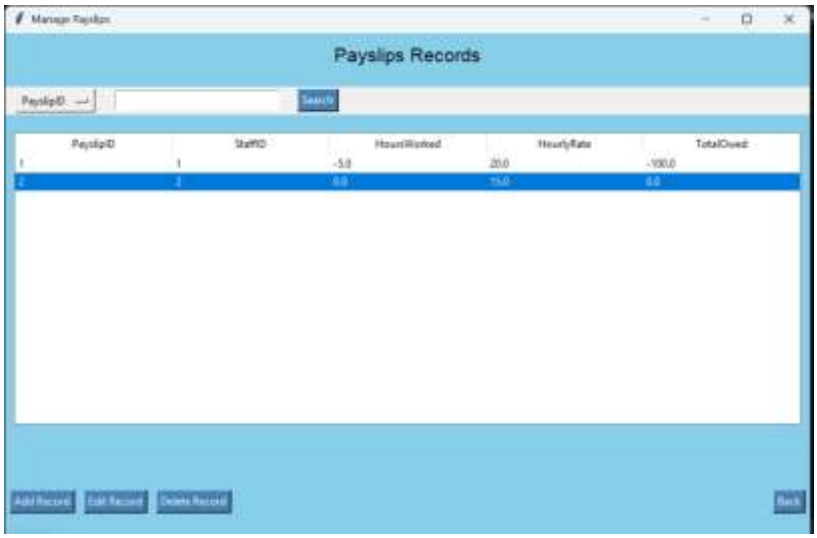
TEST 97


Edit Access Record
StaffID
55
Username
test
Password
test
AccessLevel
guest
Save Changes


Mulligan Motors
Access level does not exist
OK

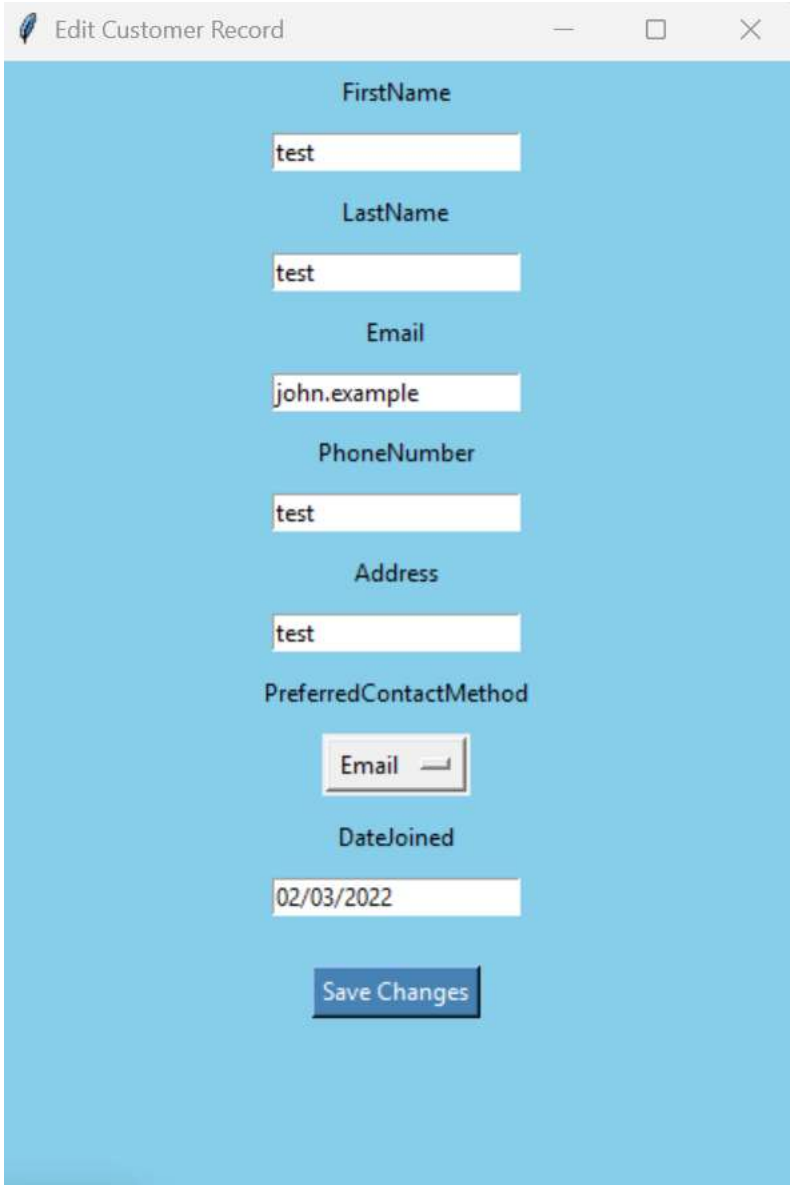
TEST 98

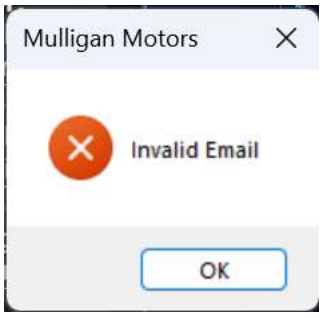




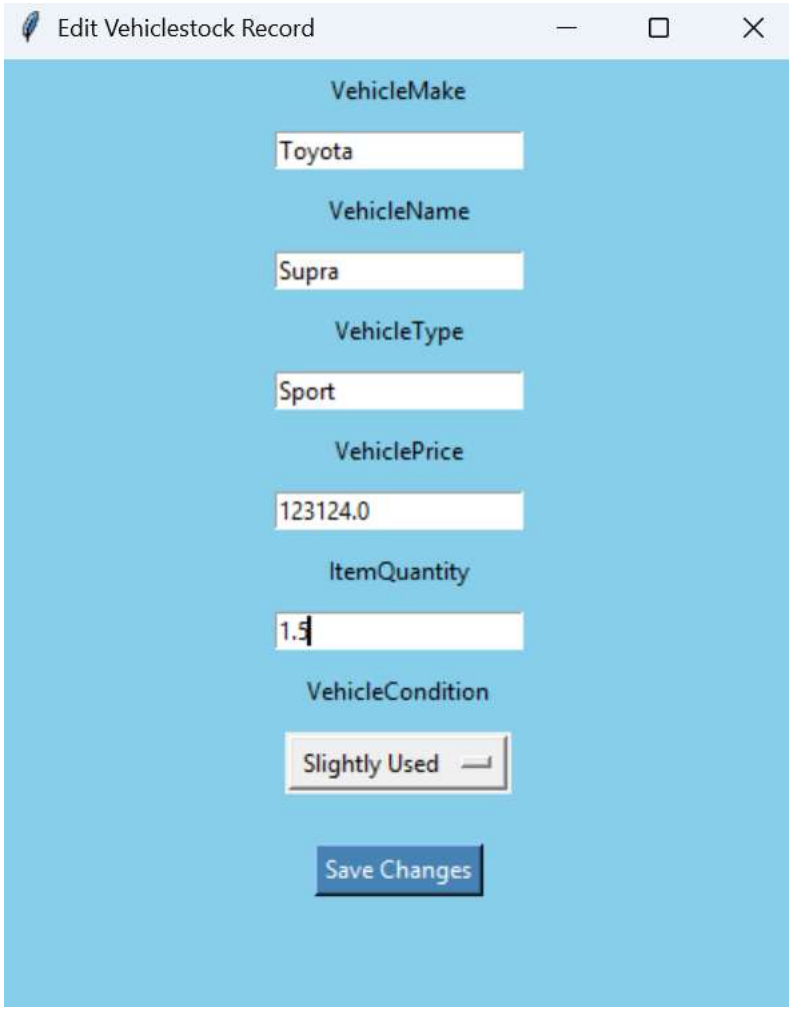
The ‘Hours worked’ attribute defaulted to 0.00 in case of an invalid value.

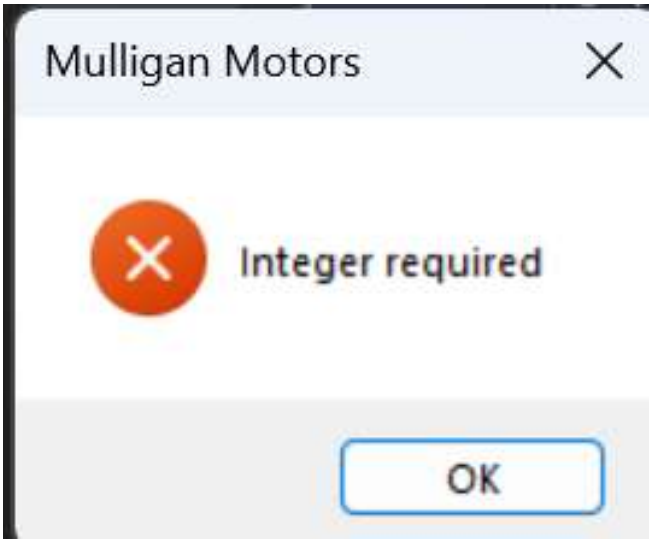
TEST 99


Edit Customer Record
FirstName
test
LastName
test
Email
john.example
PhoneNumber
test
Address
test
PreferredContactMethod
Email
DateJoined
02/03/2022
Save Changes


Mulligan Motors
Invalid Email
OK

TEST 100


Edit Vehiclestock Record
VehicleMake
Toyota
VehicleName
Supra
VehicleType
Sport
VehiclePrice
123124.0
ItemQuantity
1.5
VehicleCondition
Slightly Used
Save Changes


Mulligan Motors
Integer required
OK

8.2 End User Testing & Feedback

After completing white-box testing, the system was deployed to Mulligan Motors staff for a 5-day trial. Feedback collected from employees revealed a predominantly positive reception, with several strengths highlighted alongside constructive critiques. The system's user-friendly interface and intuitive design were frequently praised, as employees found it streamlined daily tasks such as inventory checks and customer interactions. Staff also commended its stability, noting that it operated smoothly on the dealership's existing computers without causing slowdowns or crashes, even during peak usage.

The search and reporting features emerged as standout tools. Sales teams highlighted the ability to filter cars by criteria like make, model, year, or availability, which allowed them to respond quickly to customer inquiries. Management, meanwhile, appreciated administrative features such as login permissions and payslip management, which simplified oversight tasks. Many employees emphasized that the system reduced time spent on manual processes compared to the dealership's outdated legacy system, boosting overall efficiency.

Despite these strengths, some areas for improvement were identified. A recurring criticism was the lack of range-based search functionality, which limited scenarios like filtering cars within a specific price range or mileage bracket, a critical need for a second-hand car dealership. The case sensitivity of the search feature also caused frustration, particularly when searching for customer names or car models with inconsistent capitalization. Additionally, the GUI's bland, monochrome design was noted as uninspiring, with suggestions to incorporate colour or modern design elements to enhance visual appeal. A handful of staff also expressed a desire for more customization, such as the ability to save frequently used filters or personalize dashboard layouts, and some newer users requested additional training resources to fully leverage the system's capabilities as the current help page seemed lackluster.

Overall, feedback underscored the system's reliability and effectiveness. No major technical issues, such as crashes or critical bugs, were reported, and most employees expressed enthusiasm for adopting the system permanently. Management noted that the tool had already improved transparency in inventory tracking and streamlined administrative workflows.

Moving forward, Mulligan Motors plans to prioritize updates to address key critiques. This includes implementing range**-**based **search** options, adjusting the search function to be case-insensitive, and redesigning the GUI to be more visually engaging. Customization features and expanded training materials will also be explored to maximize user adoption and satisfaction. The trial period provided valuable insights, confirming the system's potential to become a cornerstone of the dealership's operations while highlighting actionable refinements to better align with the unique needs of a second-hand car sales environment.